

Scratch Coding Class

LESSON 1

Hello!

Computer



Program=

- ▶ Instructions
- ▶ Tell computer
what to do





Code=

▶ How to write
instructions



Coding Language=

▶ Code
computer can
read

Code + Code +
Code = Program

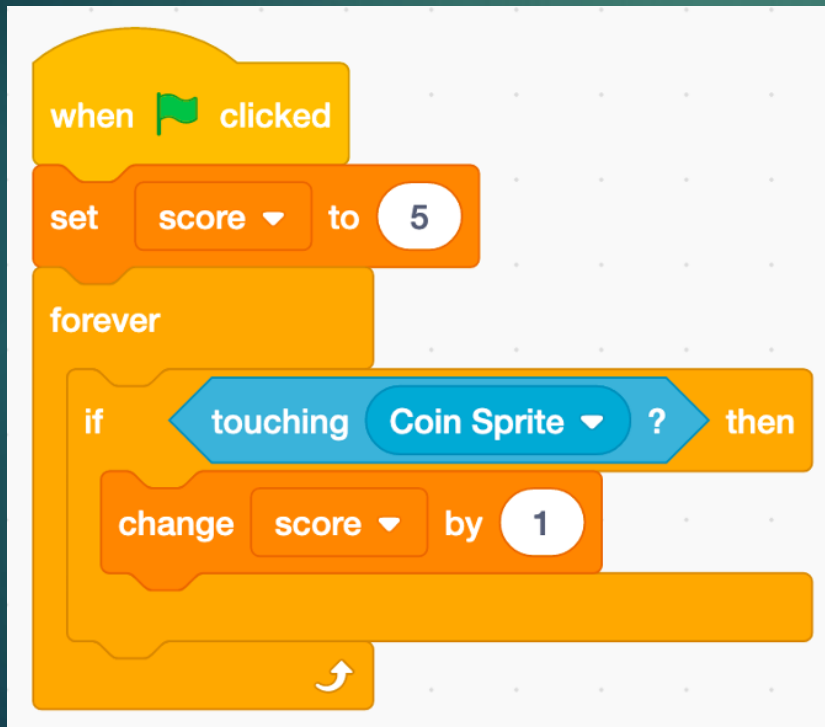


Program tells
computer what to do

Scratch =
Coding
Language



Scratch



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Scratch

- ▶ Read Code
and Programs
- ▶ Make computer
understand
stories



We will teach
you how to
use Scratch



Goodbye!



Scratch Coding Class

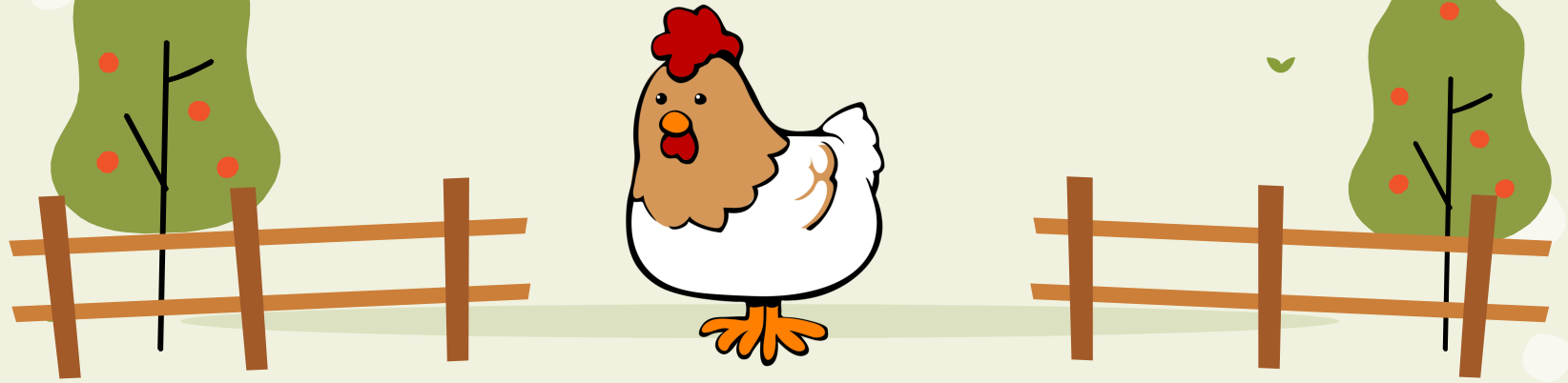
LESSON 2



Hello! Story
Time!

There once
was a
chicken
named Billy.





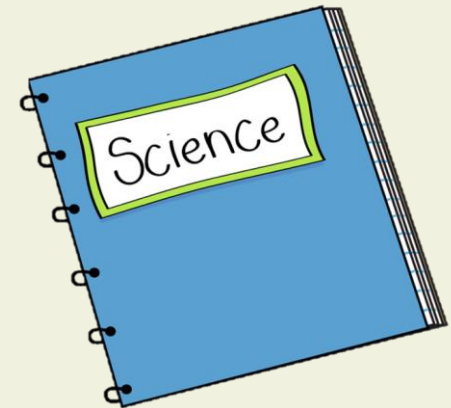
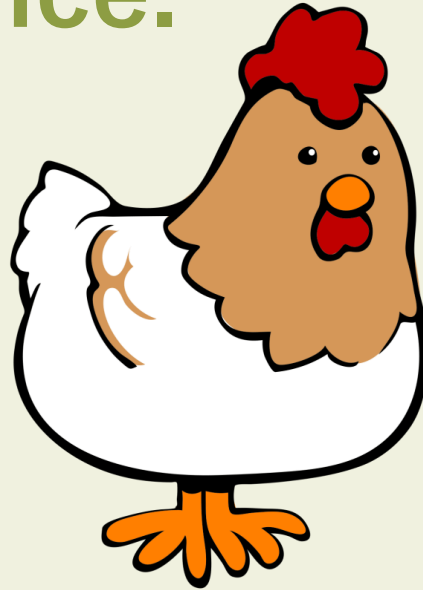
Billy The Chicken!



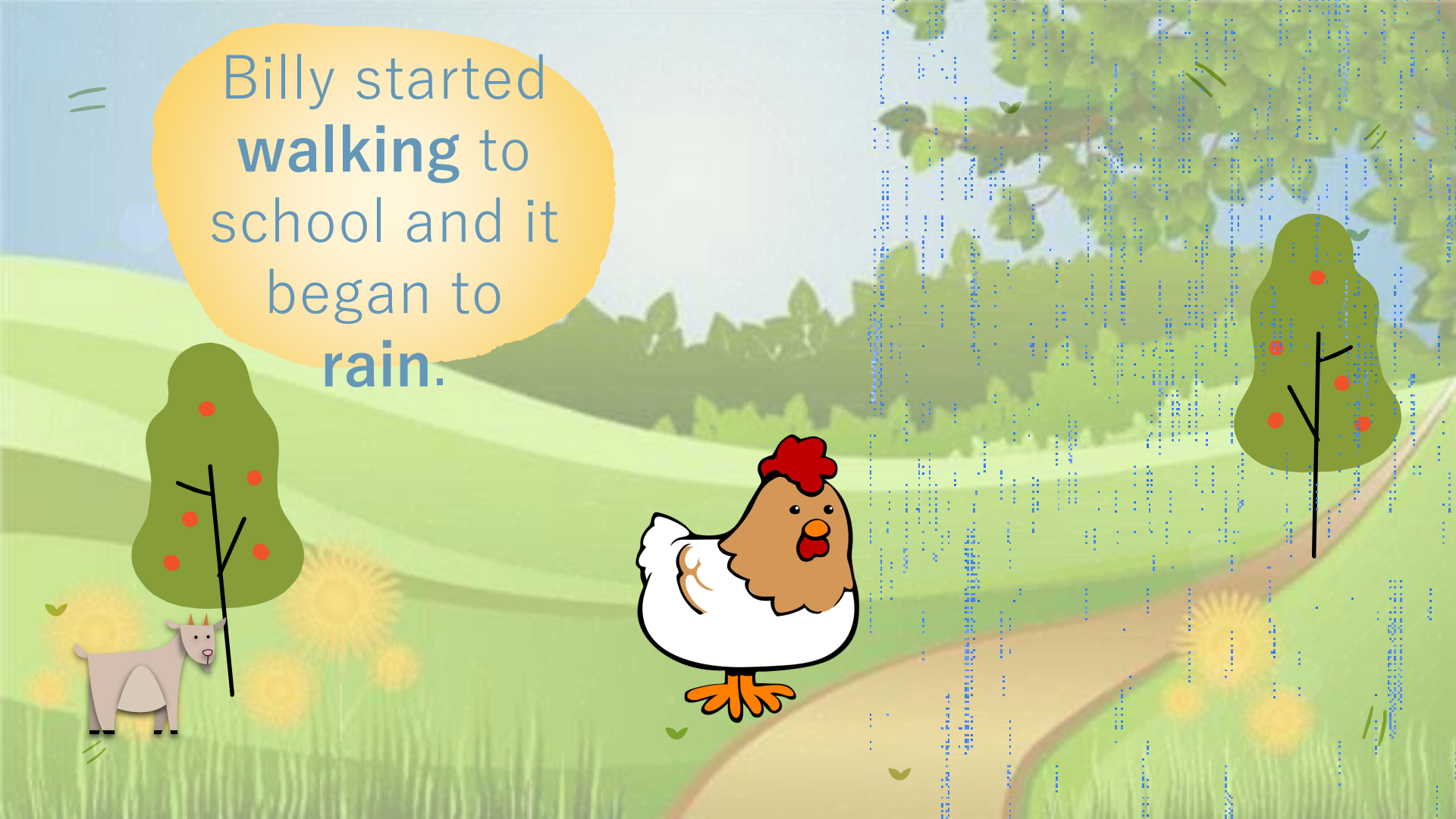
Billy

The chicken

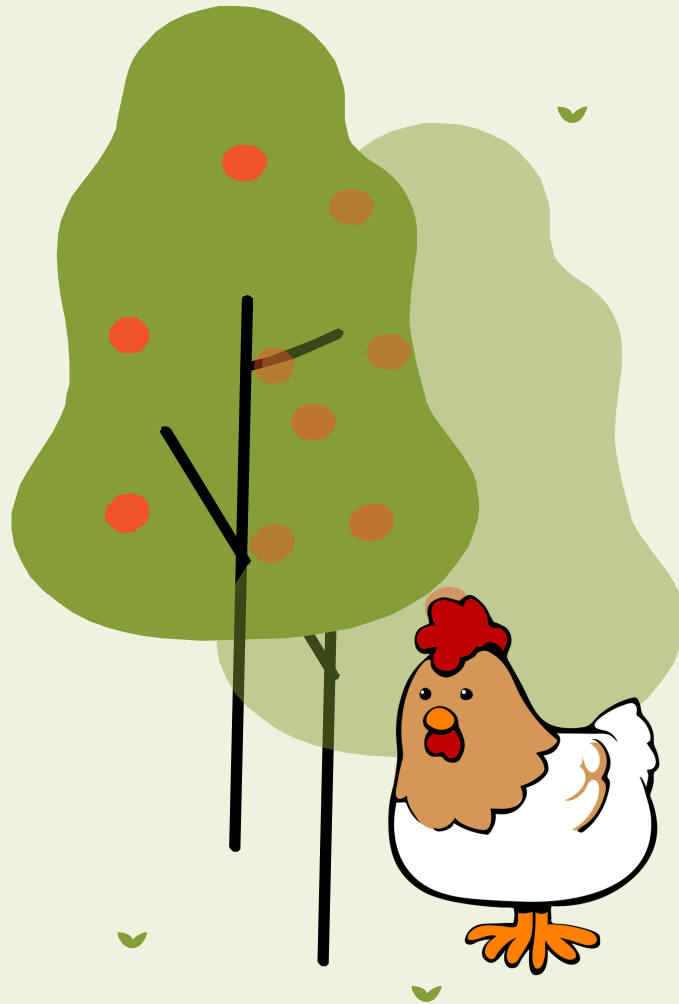
Billy had to go to
school to learn about
science.



Billy started
walking to
school and it
began to
rain.



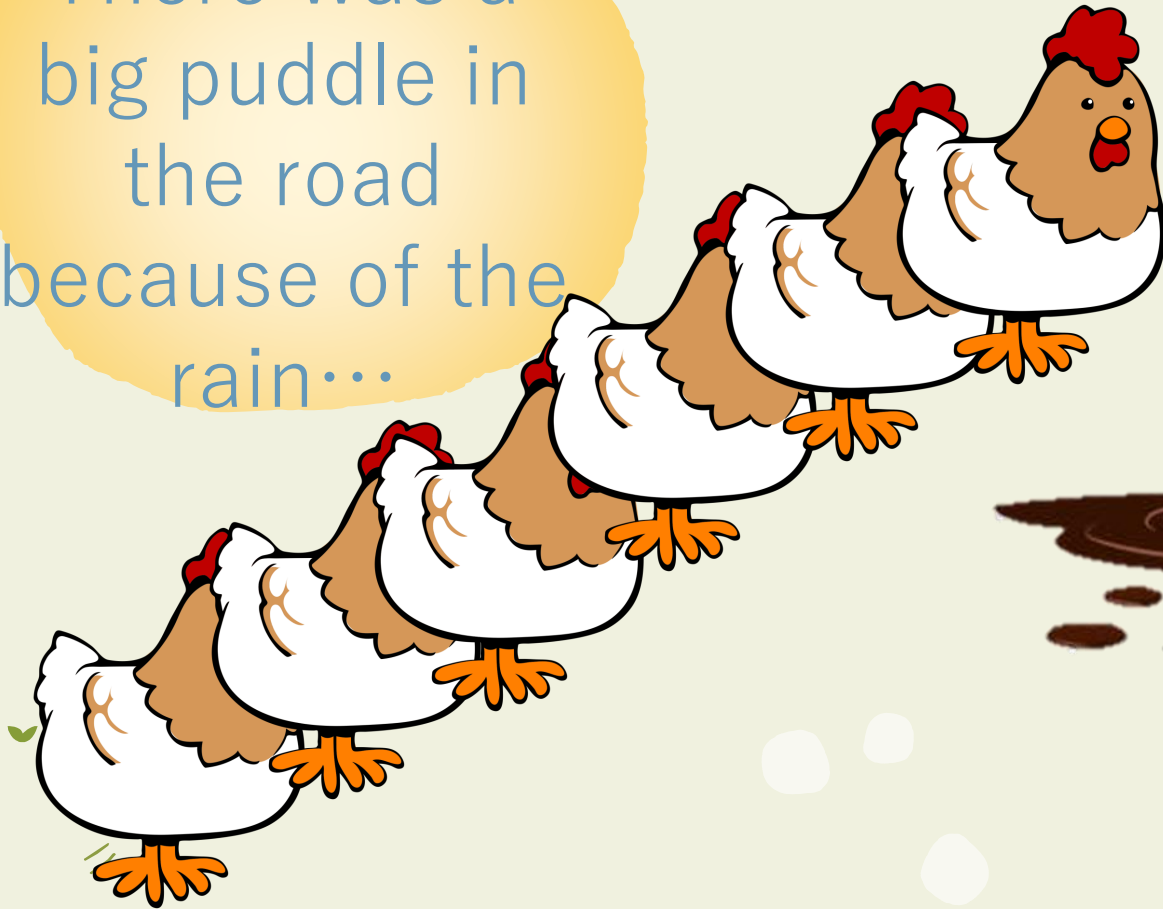
Billy went
under a **tree**
take **shelter**
from the rain.



Billy **turned
around** after
the rain
stopped and
kept going to
school.



There was a
big puddle in
the road
because of the
rain...



Billy
walked
around
the
puddle.

Billy then
walked
straight to
school.





Let's
understand
story.



Character?

Place?

Action?



Character? Billy.
Place? Grass Forest.
Action? Standing.



Let's make
computer
understand
story.



File

Edit



Tutorials

Scratch Project



Code

Backdrops

Sounds



Motion



Looks



Sound



Events



Control



Sensing



Operators



Variables



My Blocks

Motion

Stage selected: no motion blocks

Looks

switch backdrop to backdrop1 ▾

switch backdrop to backdrop1 ▾ and wait

next backdrop

change color ▾ effect by 25

set color ▾ effect to 0

clear graphic effects

☐ backdrop number ▾

Sound

play sound pop ▾ until done

start sound pop ▾

stop all sounds

change pitch ▾ effect by 10



Sprite

Name

↔ x

x

↕ y

y

Show



Size

Direction

Stage

Backdrops

1



Code

Backdrops

Sounds



Motion



Looks



Sound



Events



Control



Sensing



Operators



Variables



My Blocks

Motion

Stage selected: no motion blocks

Looks

switch backdrop to backdrop1 ▾

switch backdrop to backdrop1 ▾ and wait

next backdrop

change color ▾ effect by 25

set color ▾ effect to 0

clear graphic effects

☐ backdrop number ▾

Sound

play sound pop ▾ until done

start sound pop ▾

stop all sounds

change pitch ▾ effect by 10



Sprite

Name

↔ x

x

↕ y

y

Show



Size

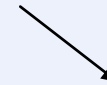
Direction

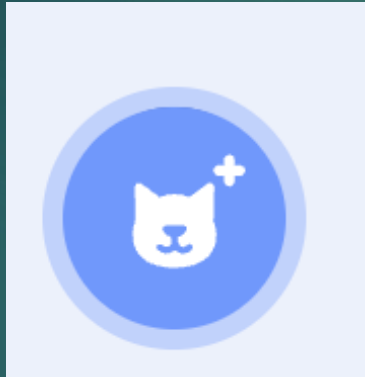
Stage

Backdrops

1

Let's
add
Billy





Use finger to
move
mouse over
character
button. Right
button to
click.

[← Back](#)

Choose a Sprite

All

Animals

People

Fantasy

Dance

Music

Sports

Food

Fashion

Letters



Abby



Amon



Andie



Anina Dance



Apple



Arrow1



Avery



Avery Walk...



Ball



Ballerina



Balloon1



Bananas



Baseball



Basketball



Bat



Batter



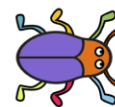
Beachball



Bear



Bear-walking



Beetle



Bell



Ben



Bowl



Bowtie



Bread



Broom



Buildings

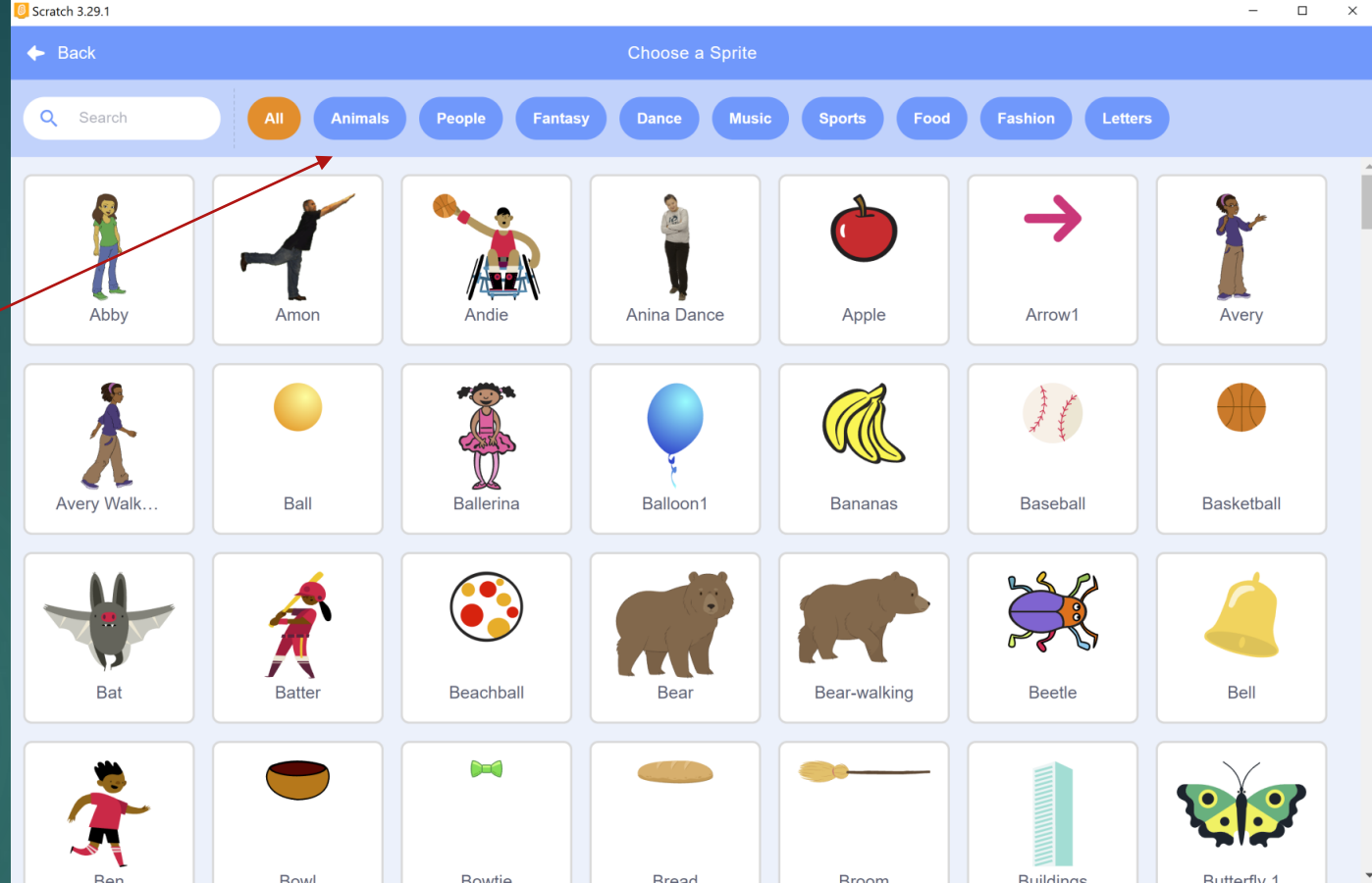


Butterfly 1



Billy the
Chicken =
Animal

Let's right
click
Animal





Bat



Bear



Bear-walking



Beetle



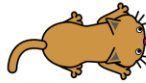
Butterfly 1



Butterfly 2



Cat



Cat 2



Cat Flying



Chick



Crab



Dinosaur1



Dinosaur2



Dinosaur3



Dinosaur4



Dinosaur5



Dog1



Dog2



Dot



Dove



Dragon



Dragonfly



Duck



Elephant



Fish



Fishbowl



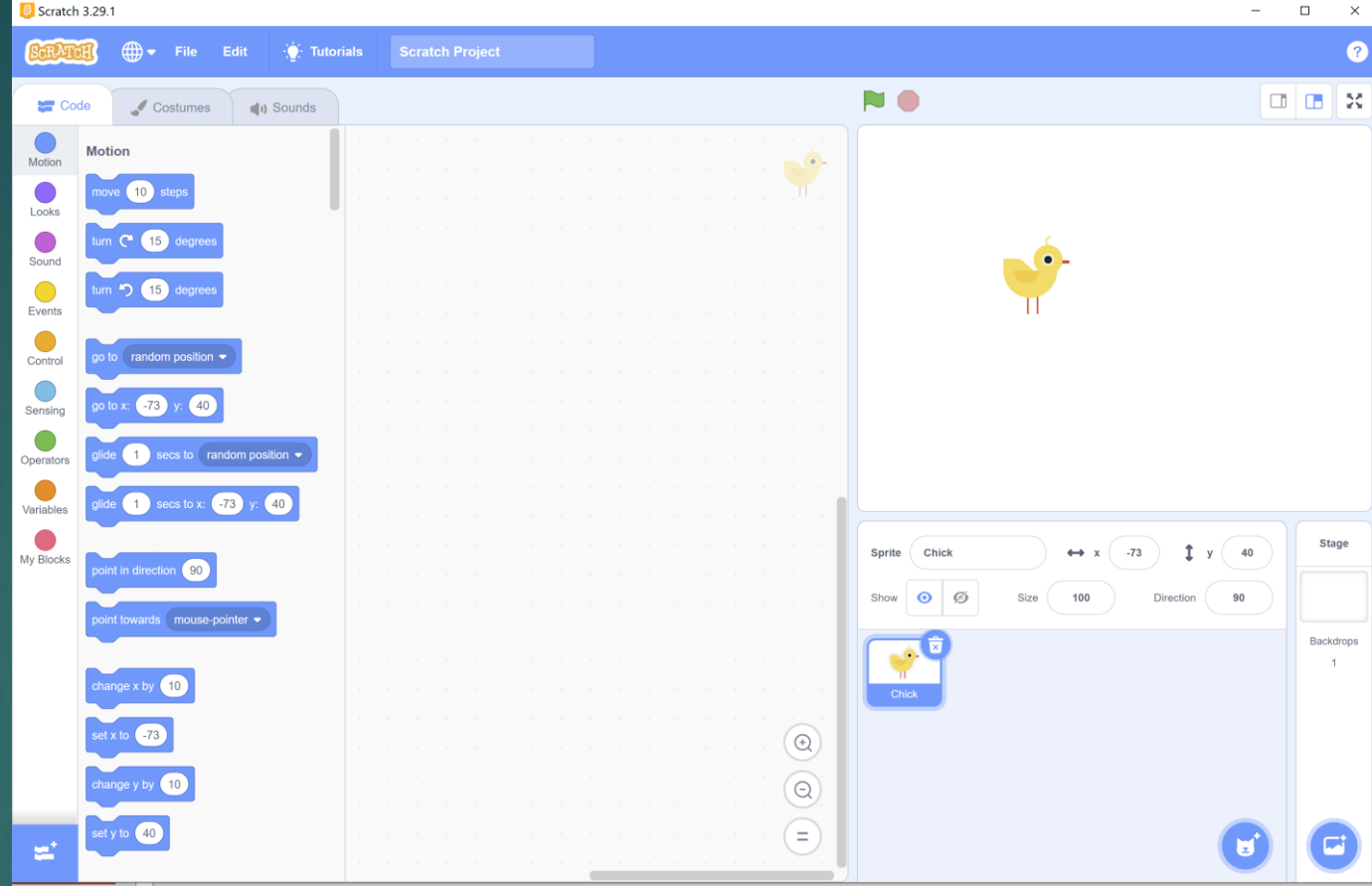
Fox



Frog

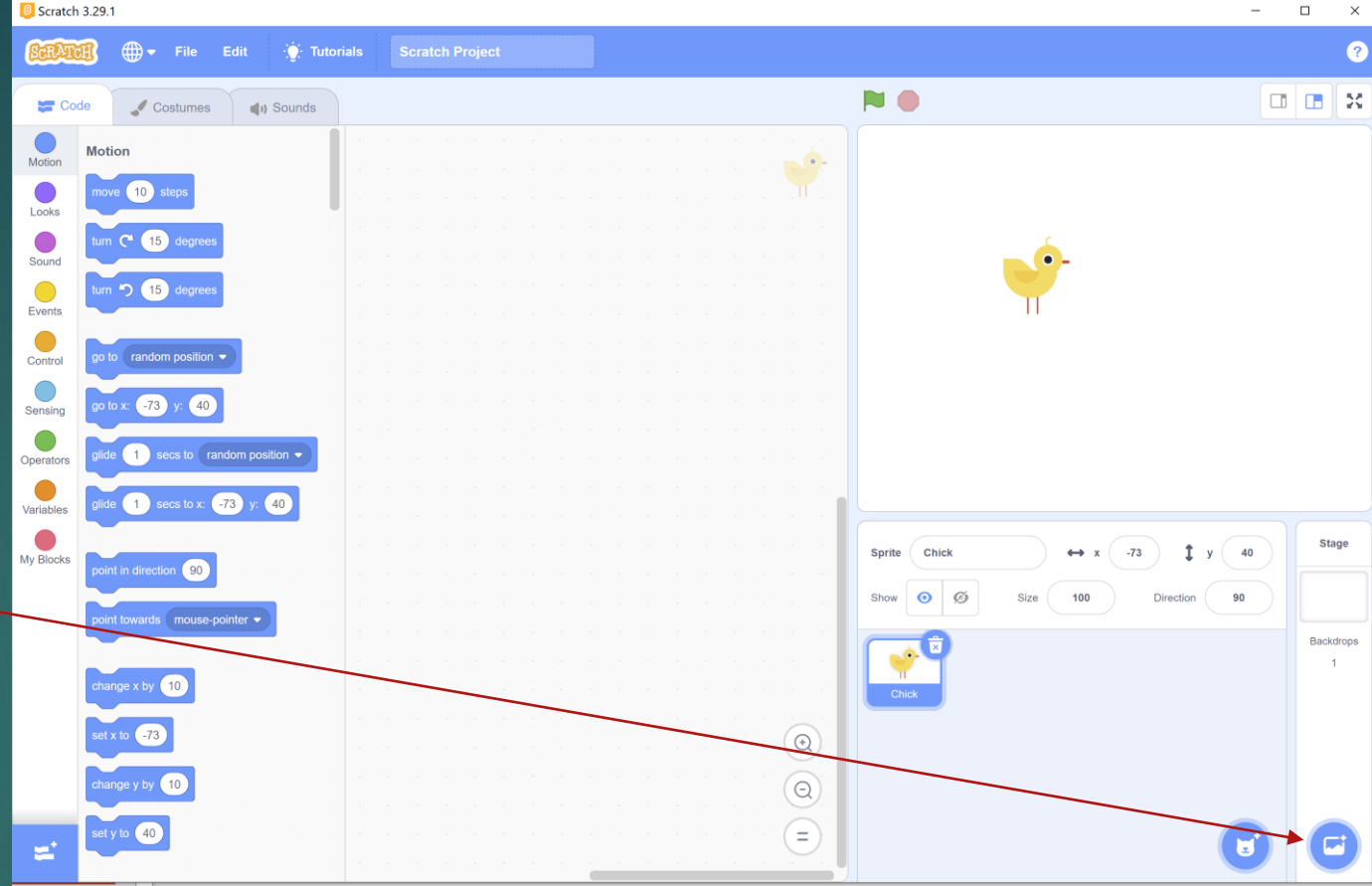
Let's
click
Chick

Computer
now sees
Billy!



Let's add Place

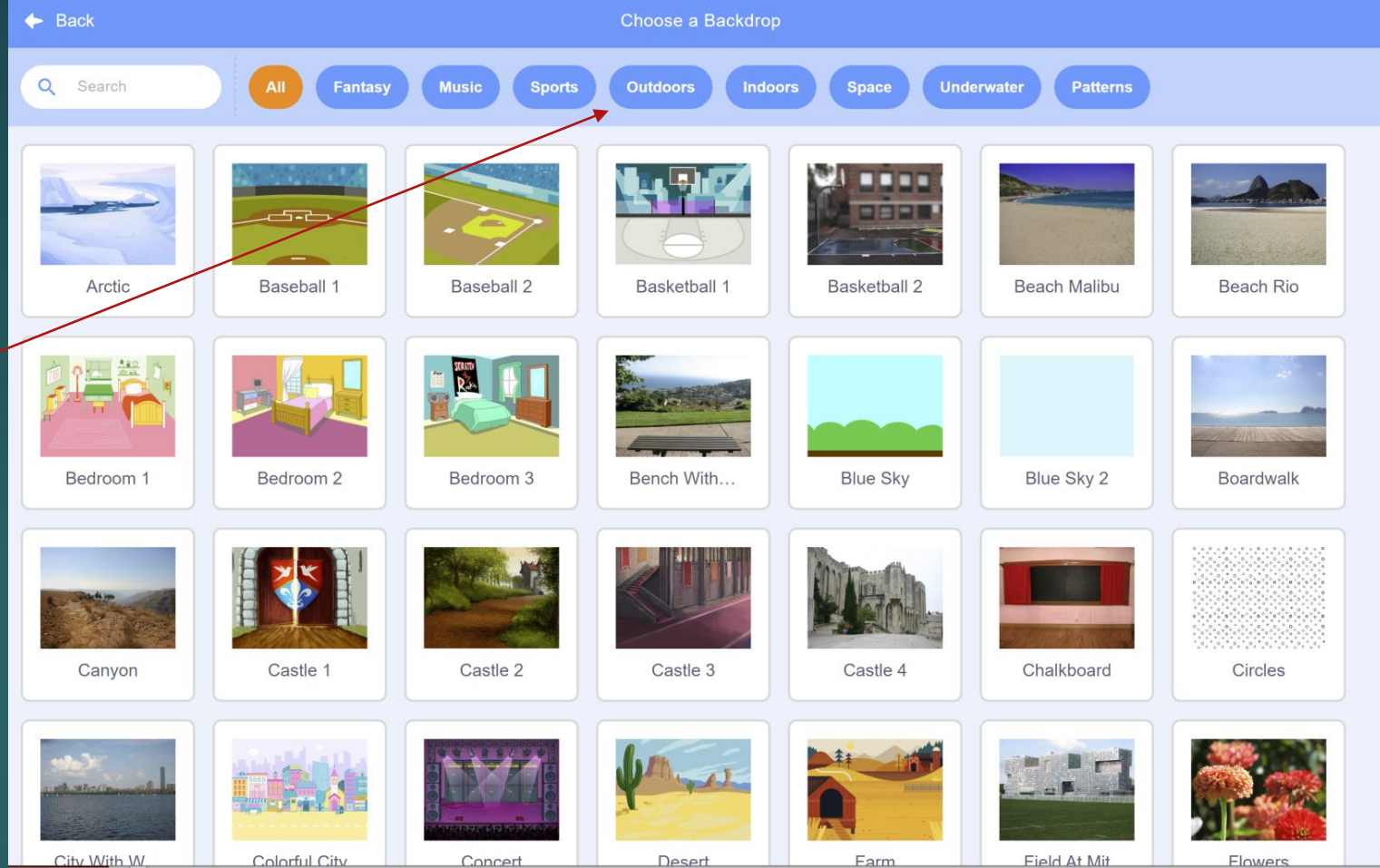
Place
Button

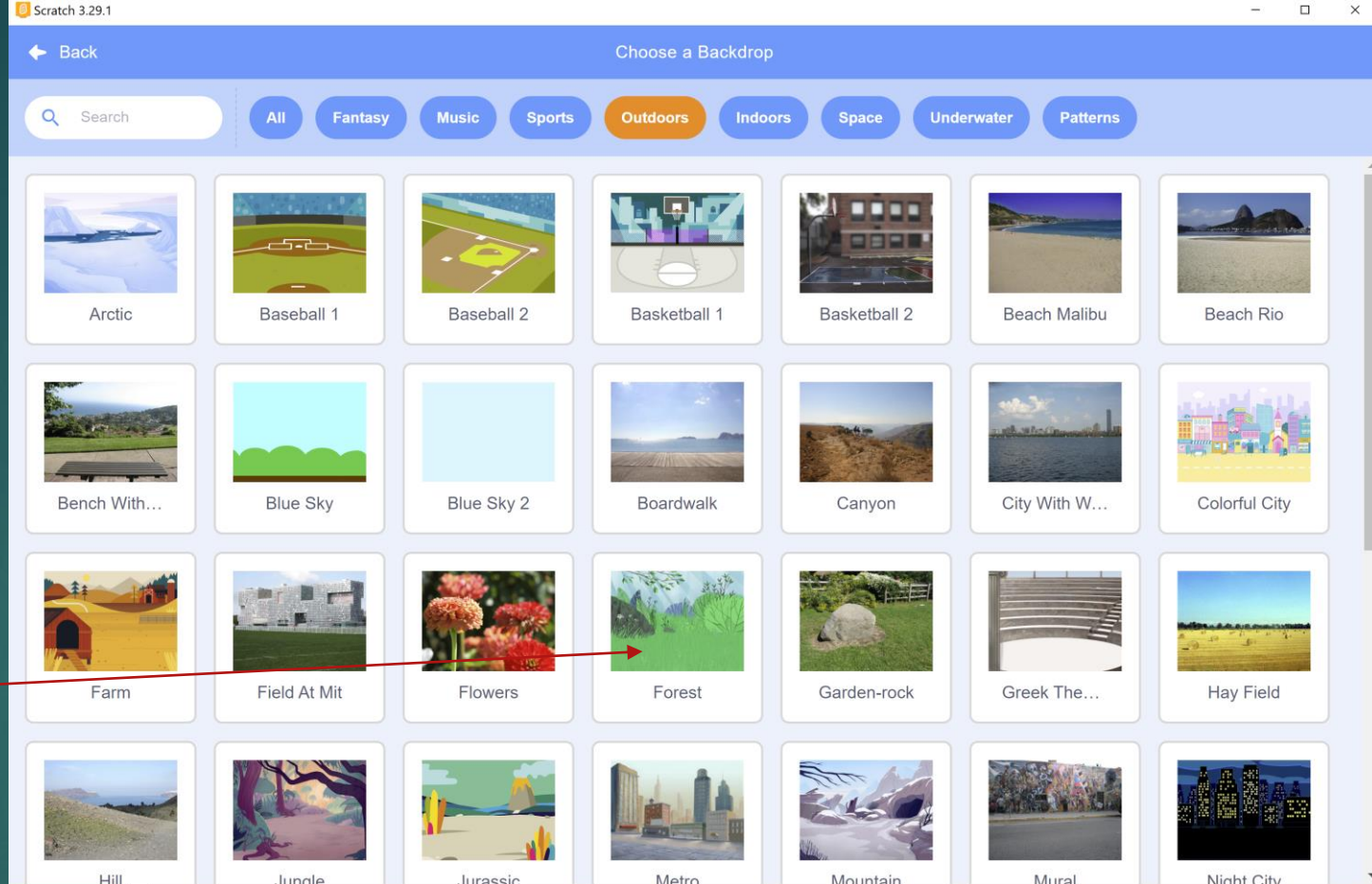




Use finger to
move
mouse over
place
button. Right
button to
click.

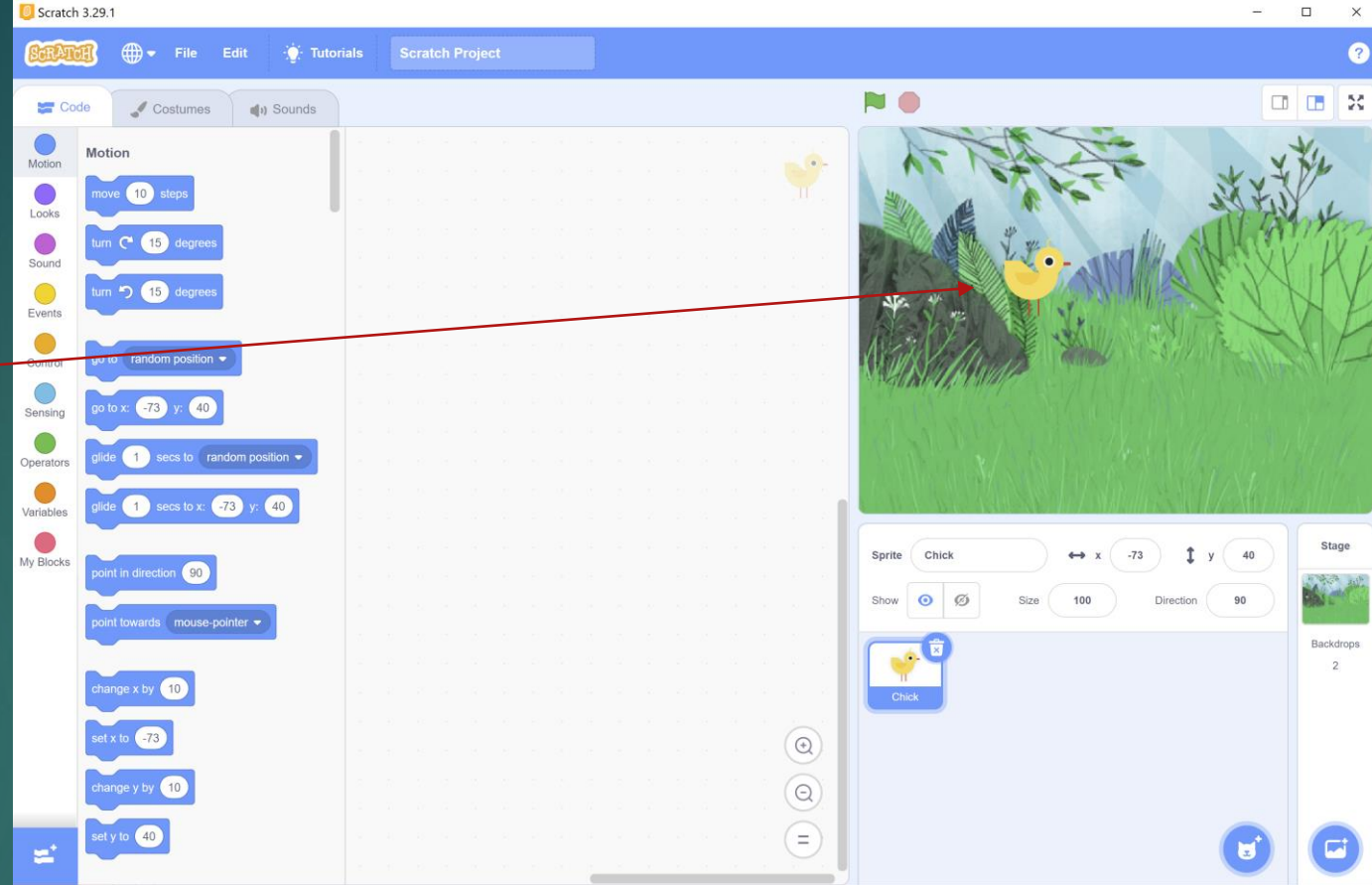
Billy is outside.
Let's
press
outdoors
button





Let's
press
forest

Click
Billy
and
hold to
move
him





Motion

move 10 steps

turn  15 degrees

turn  15 degrees

go to random position ▼

go to x: -4 y: -54

glide 1 secs to random position ▼

glide 1 secs to x: -4 y: -54

point in direction 90

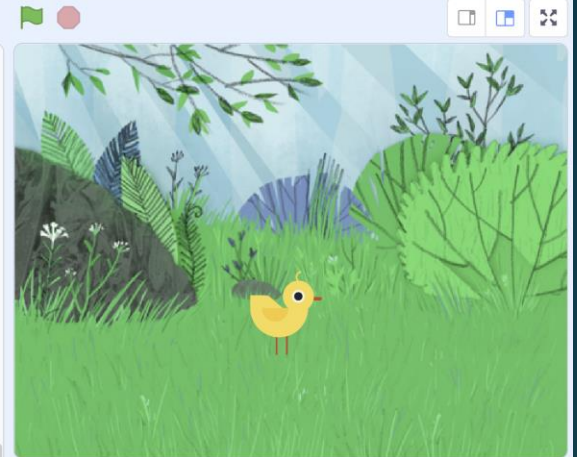
point towards mouse-pointer ▼

change x by 10

set x to -4

change y by 10

set y to -54



Sprite

4

$y = -54$

Show  

Size 100

Direction 90



Chick

Click

Stage



Backdrops

2





Goodbye!



Scratch Coding

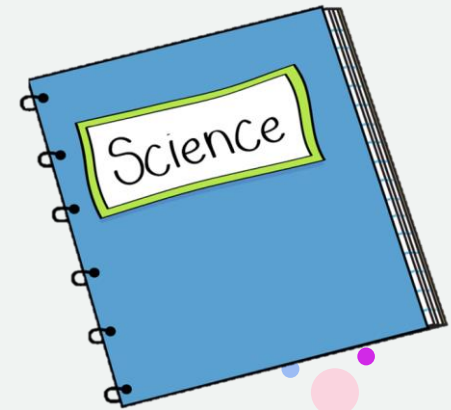
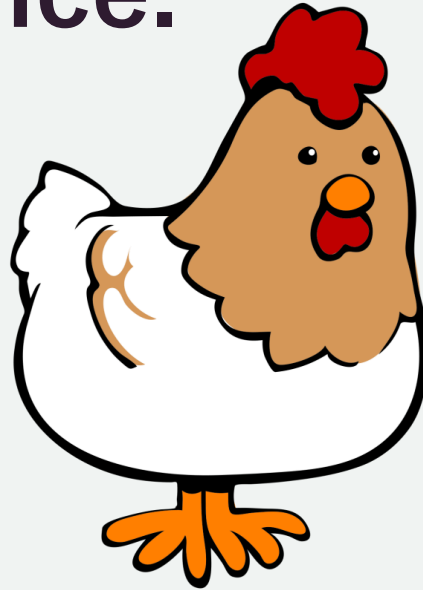
Lesson 3

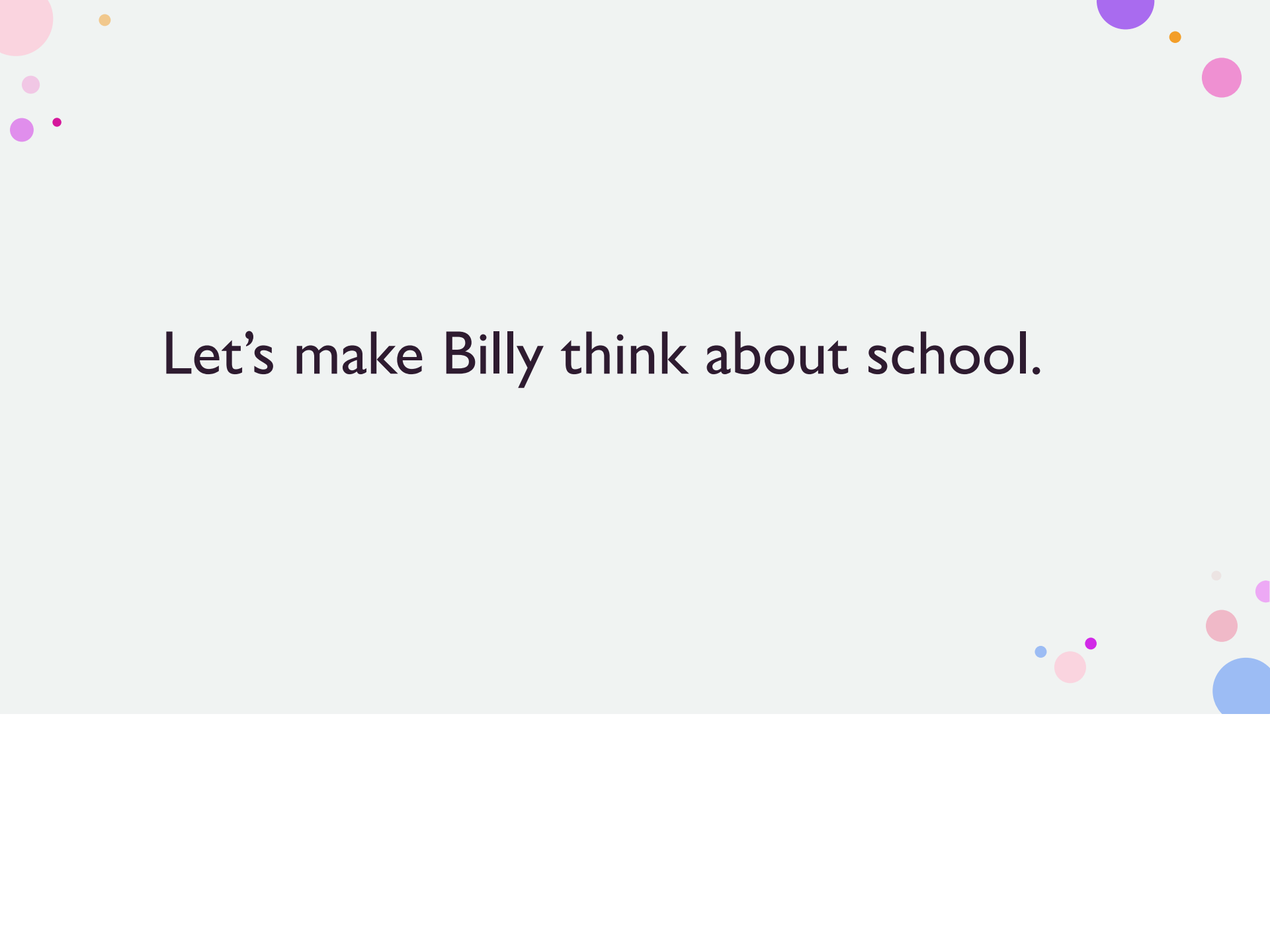


Billy

The chicken

Billy had to go to
school to learn about
science.

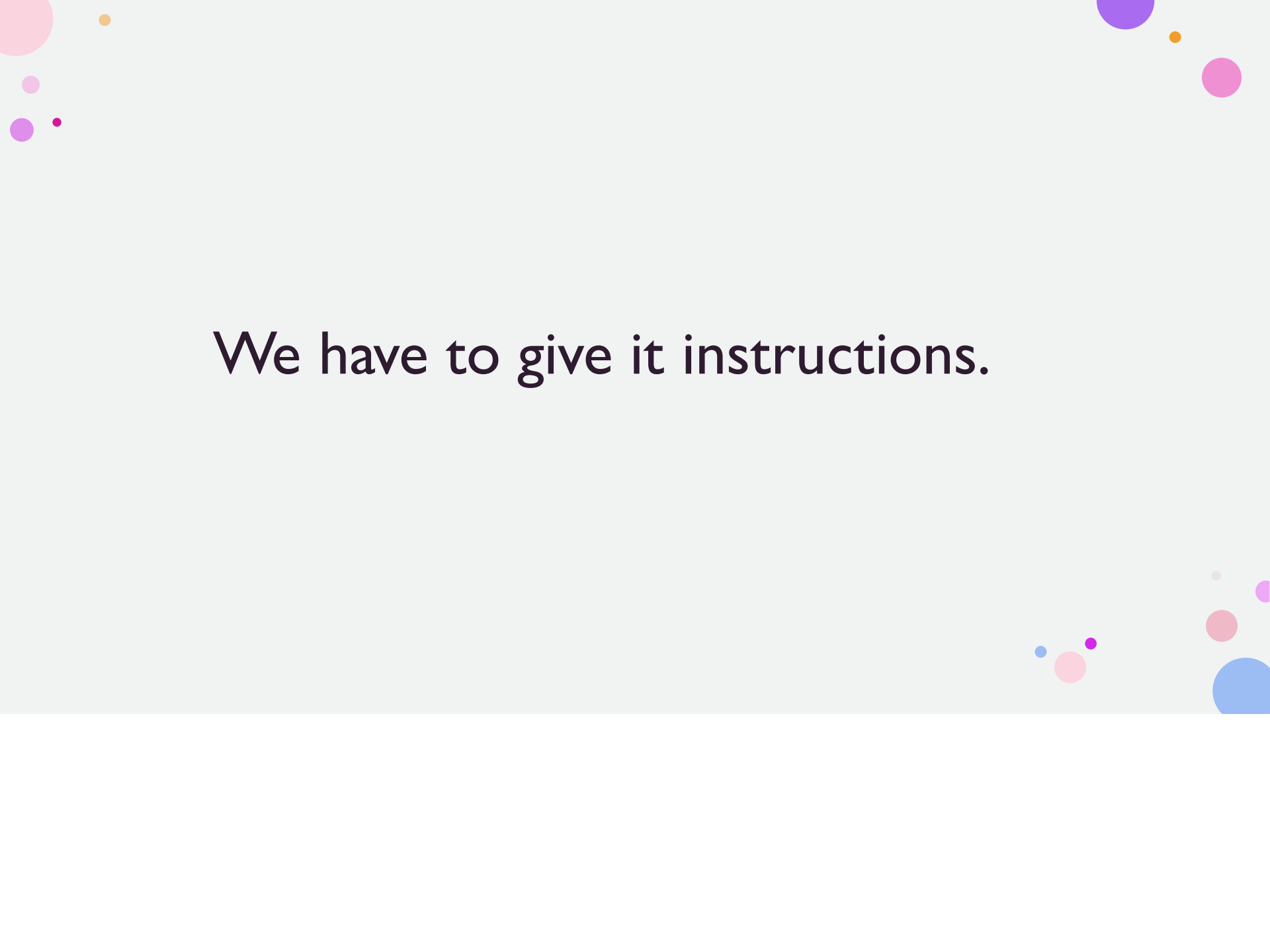


The image features a light gray background with several colorful circles of various sizes in the corners. In the top-left corner, there are pink, purple, and orange circles. In the top-right corner, there are purple, orange, and pink circles. In the bottom-right corner, there are blue, pink, and purple circles. The text is centered in the middle of the page.

Let's make Billy think about school.

The slide features a light gray background with decorative elements in the corners. The top-left corner contains several overlapping circles in shades of pink, purple, and orange. The top-right corner has a large purple circle, a smaller pink circle, and a tiny orange dot. The bottom-right corner is decorated with a large blue circle, a medium pink circle, a small purple dot, and a tiny light blue dot.

How does computer know how to read
story?

The slide features a light gray background with decorative elements in the corners. The top-left corner contains several small circles in shades of pink, purple, and orange. The top-right corner features a large purple circle, a smaller pink circle, and a tiny orange dot. The bottom-right corner is decorated with a cluster of circles in blue, pink, purple, and light gray. Centered on the slide is the text "We have to give it instructions." in a dark purple, sans-serif font.

We have to give it instructions.

Scratch 3.2.1

File Edit Tutorials Scratch Project

Code Costumes Sounds

Motion

- Motion
 - move 10 steps
- Looks
 - turn 15 degrees
- Sound
 - turn 15 degrees
- Events
- Control
 - go to random position
- Sensing
 - go to x: -24 y: -24
- Operators
 - glide 1 secs to random position
 - glide 1 secs to x: -24 y: -24
- Variables
- My Blocks

point in direction 90

point towards mouse-pointer

change x by 10

set x to -24

change y by 10

set y to -24

We want to see Billy think. This would be how Billy **Looks**

Stage

Sprite Chick x: -24 y: -24

Show Size 100 Direction 75

Backdrops 2

Code

Costumes

Sounds

Motion
Looks
Sound
Events
Control
Sensing
Operators
Variables
My Blocks

Looks

say Hello! for 2 seconds

say Hello!

think Hmm... for 2 seconds

think Hmm...

switch costume to chick-c

next costume

switch backdrop to Forest

next backdrop

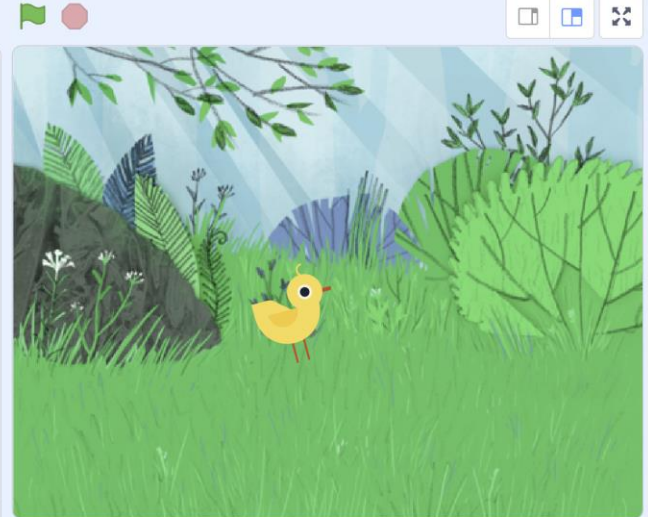
change size by 10

set size to 100 %

change color effect by 25

set color effect to 0

clear graphic effects



Sprite Chick

x -24

y -24

Show

Size

100

Direction

75



Chick

Stage



Backdrops

2

Billy needs to think. Right click and hold to drag to this empty space.

Looks

say Hello! for 2 seconds

say Hello!

think Hmm... for 2 seconds

think Hmm...

switch costume to chick-c

next costume

switch backdrop to Forest

next backdrop

change size by 10

set size to 100 %

change color effect by 25

set color effect to 0

clear graphic effects

think Hmm... for 2 seconds

We want Billy to think about school and science. Click the first bubble and use keyboard to change what he says.



Sprite Chick

x -24 y -24

Show

Size 100

Direction 75



Chick

Stage

Backdrops
2



Code

Costumes

Sounds



Motion



Looks



Sound



Events



Control



Sensing



Operators



Variables



My Blocks

Looks

say Hello! for 2 seconds

say Hello!

think Hmm... for 2 seconds

think Hmm...

switch costume to chick-c

next costume

switch backdrop to Forest

next backdrop

change size by 10

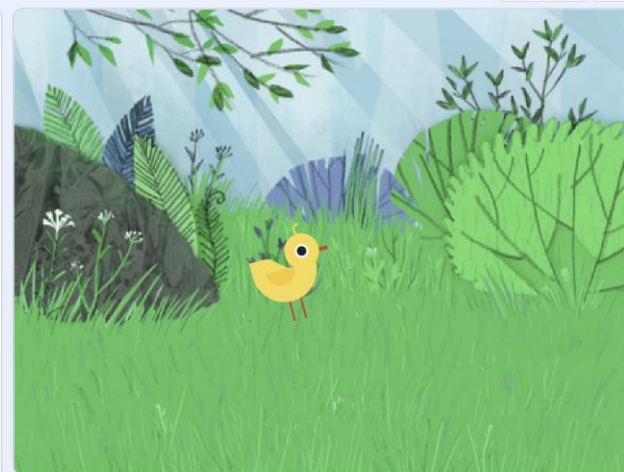
set size to 100 %

change color effect by 25

set color effect to 0

clear graphic effects

think I go to school to learn about Science. for 2 seconds



Sprite Chick

x -24

y -24

Show



Size

100

Direction

75



Chick

Stage

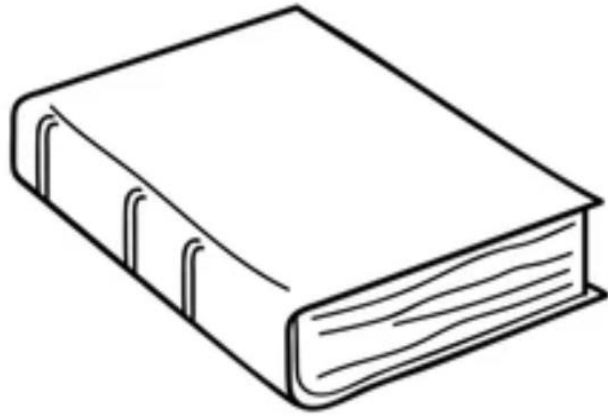


Backdrops

2

The slide features a light gray background with decorative elements in the corners. The top-left corner contains several overlapping circles in shades of pink, purple, and orange. The top-right corner has a large purple circle, a smaller pink circle, and a tiny orange dot. The bottom-right corner is decorated with a large blue circle, a pink circle, a small purple dot, and a tiny light blue dot.

How do we make the computer read
instructions?



What computer sees right now. Instructions in book but it can't see them.



Computer needs
to open the book
to read
instructions. How
do this in Scratch?

The image shows the Scratch 3.29.1 interface. The top bar includes the Scratch logo, a dropdown menu, and tabs for File, Edit, Tutorials, and a Scratch Project button. Below the top bar are tabs for Code, Costumes, and Sounds. The left sidebar contains a 'My Blocks' section and a 'Looks' section with various blocks for Motion, Looks, Sound, Events, Control, Sensing, Operators, Variables, and My Blocks. The main workspace is a grid where a 'think' block is placed, containing the text 'I go to school to learn about Science.' for 2 seconds. A red arrow points from the text 'Green flag. Tell computer to go and open book to read instructions' to a green flag icon in the top right corner of the workspace. The stage area on the right shows a chick sprite on a forest backdrop. The bottom right panel displays the 'Sprite' and 'Stage' sections, including a 'Backdrops' list with 2 items.

Scratch 3.29.1

File Edit Tutorials Scratch Project

Code Costumes Sounds

Looks

say Hello! for 2 seconds

say Hello!

think Hmm... for 2 seconds

think Hmm...

switch costume to chick-c

next costume

switch backdrop to Forest

next backdrop

change size by 10

set size to 100 %

change color effect by 25

set color effect to 0

clear graphic effects

think I go to school to learn about Science. for 2 seconds

Green flag. Tell computer to **go** and open book to read instructions

Sprite Chick x -24 y -24

Show Size 100 Direction 75

Stage

Backdrops 2

Events

when green flag clicked

when space key pressed

when this sprite clicked

when backdrop switches to backdrop1

when loudness > 10

when I receive message1

broadcast message1

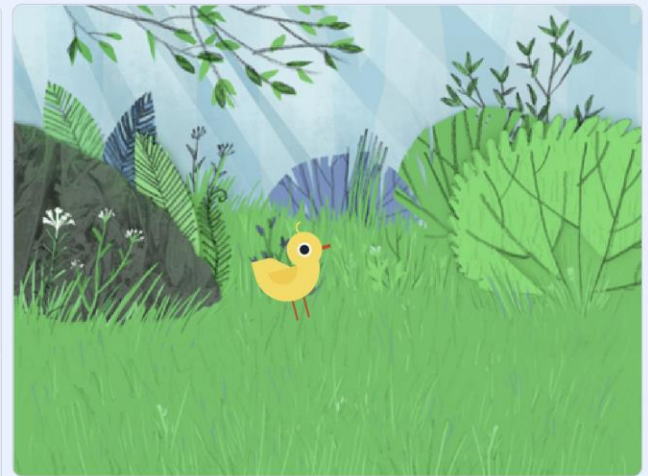
broadcast message1 and wait

Control

wait 1 seconds

repeat 10

think I go to school to learn about Science. for 2 seconds



Sprite Chick

x

y

-24 -24

Show

Size

100

Direction

75



Chick

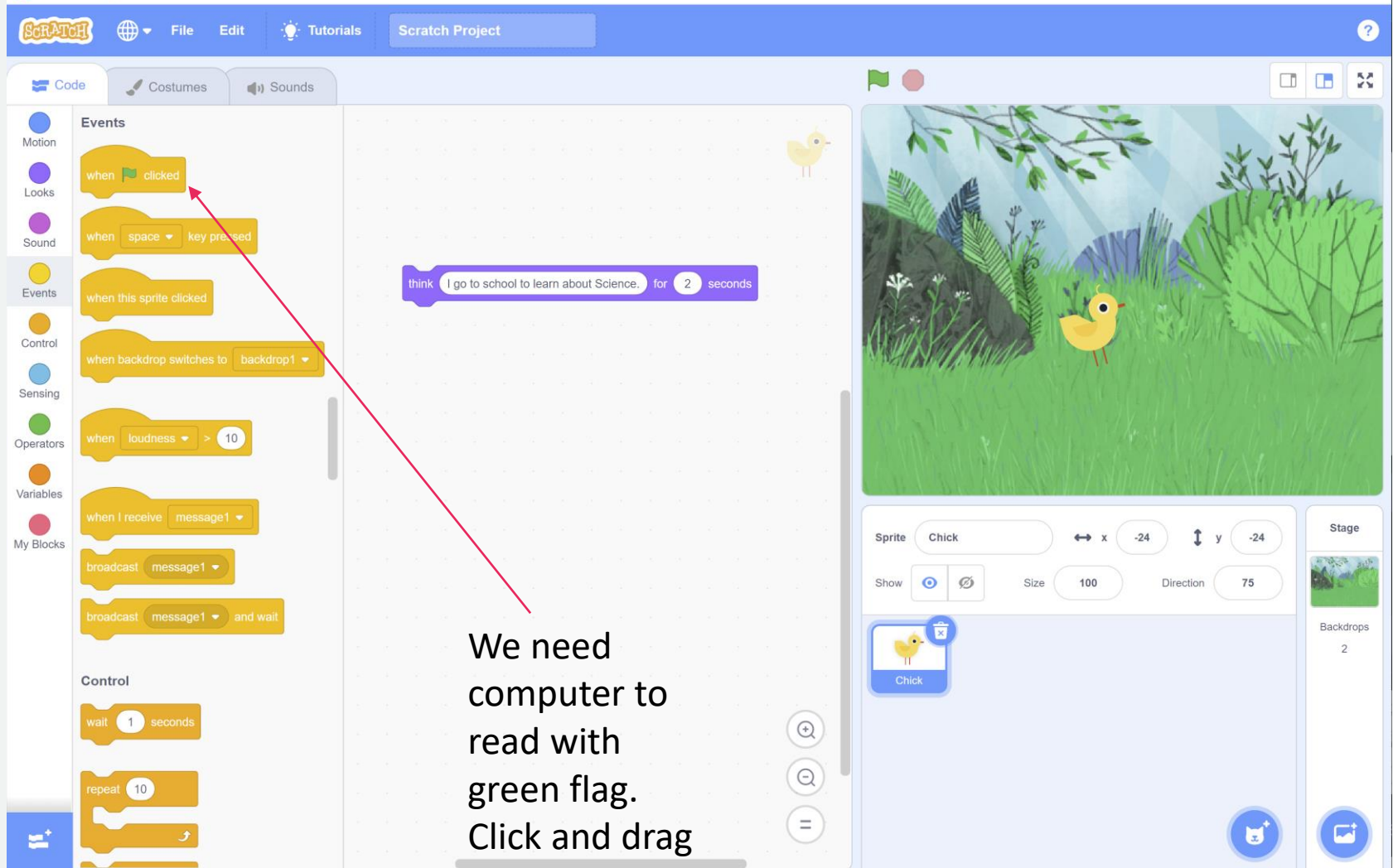
Stage



Backdrops

2

Opening a
book is an
event.
Clicking
green flag
is an
event.
Click
event.



The image shows the Scratch 3.29.1 interface. The top bar includes the Scratch logo, a dropdown menu, and tabs for File, Edit, Tutorials, and Scratch Project. The left sidebar contains categories: Code, Costumes, and Sounds. The 'Code' tab is active, showing a list of block categories: Motion, Looks, Sound, Events, Control, Sensing, Operators, Variables, and My Blocks. The 'Events' category is selected, displaying several blocks: 'when green flag clicked', 'when space key pressed', 'when this sprite clicked', 'when backdrop switches to', 'when loudness > 10', 'when I receive message1', 'broadcast message1', and 'broadcast message1 and wait'. The 'Control' category is also visible, showing 'wait 1 seconds' and 'repeat 10' blocks. The main workspace shows a yellow chick sprite on a green grassy field backdrop. A purple 'think' block is placed on the stage, containing the text 'I go to school to learn about Science.' and a duration of '2 seconds'. The right sidebar shows the 'Sprite' panel with 'Chick' selected, and the 'Stage' panel with 'Backdrops' and '2' listed. A red arrow points from the 'when green flag clicked' block to the text 'We need computer to read with green flag. Click and drag to space.'

Scratch 3.29.1

File Edit Tutorials Scratch Project

Code Costumes Sounds

Events

- when green flag clicked
- when space key pressed
- when this sprite clicked
- when backdrop switches to backdrop1
- when loudness > 10
- when I receive message1
- broadcast message1
- broadcast message1 and wait

Control

- wait 1 seconds
- repeat 10

think I go to school to learn about Science. for 2 seconds

Sprite Chick x -24 y -24

Show Size 100 Direction 75

Backdrops 2

We need computer to read with green flag. Click and drag to space.

Scratch 3.29.1

File Edit Tutorials Scratch Project

Code Costumes Sounds

Events

- Motion
- Looks
- Sound
- Events
- Control
- Sensing
- Operators
- Variables
- My Blocks

when green flag clicked

when space key pressed

when this sprite clicked

when backdrop switches to backdrop1

when loudness > 10

when I receive message1

broadcast message1

broadcast message1 and wait

Control

wait 1 seconds

repeat 10

when green flag clicked

think I go to school to learn about Science. for 2 seconds

We need to connect instructions. So computer knows when green flag click, to make Billy think.

Sprite Chick x: -24 y: -24

Show Size 100 Direction 75

Stage

Backdrops 2

Code

Costumes

Sounds

Motion

Looks

Sound

Events

Control

Sensing

Operators

Variables

My Blocks

Events

when green flag clicked

when space key pressed

when this sprite clicked

when backdrop switches to backdrop1

when loudness > 10

when I receive message1

broadcast message1

broadcast message1 and wait

Control

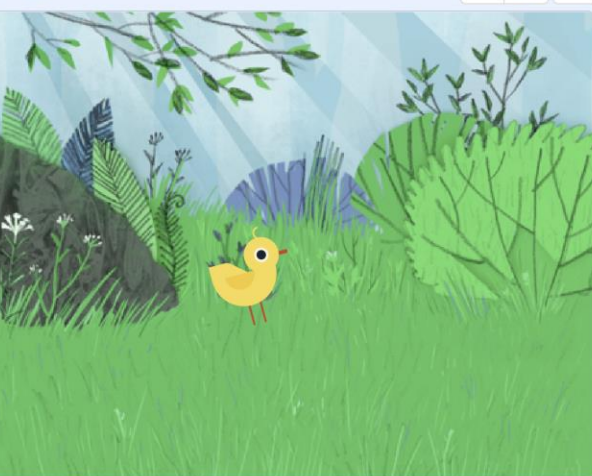
wait 1 seconds

repeat 10

when green flag clicked

think I go to school to learn about Science. for 2 seconds

Now, click
green flag!



Sprite Chick

x

y

-24

-24

Show

Size

100

Direction

75



Chick

Stage

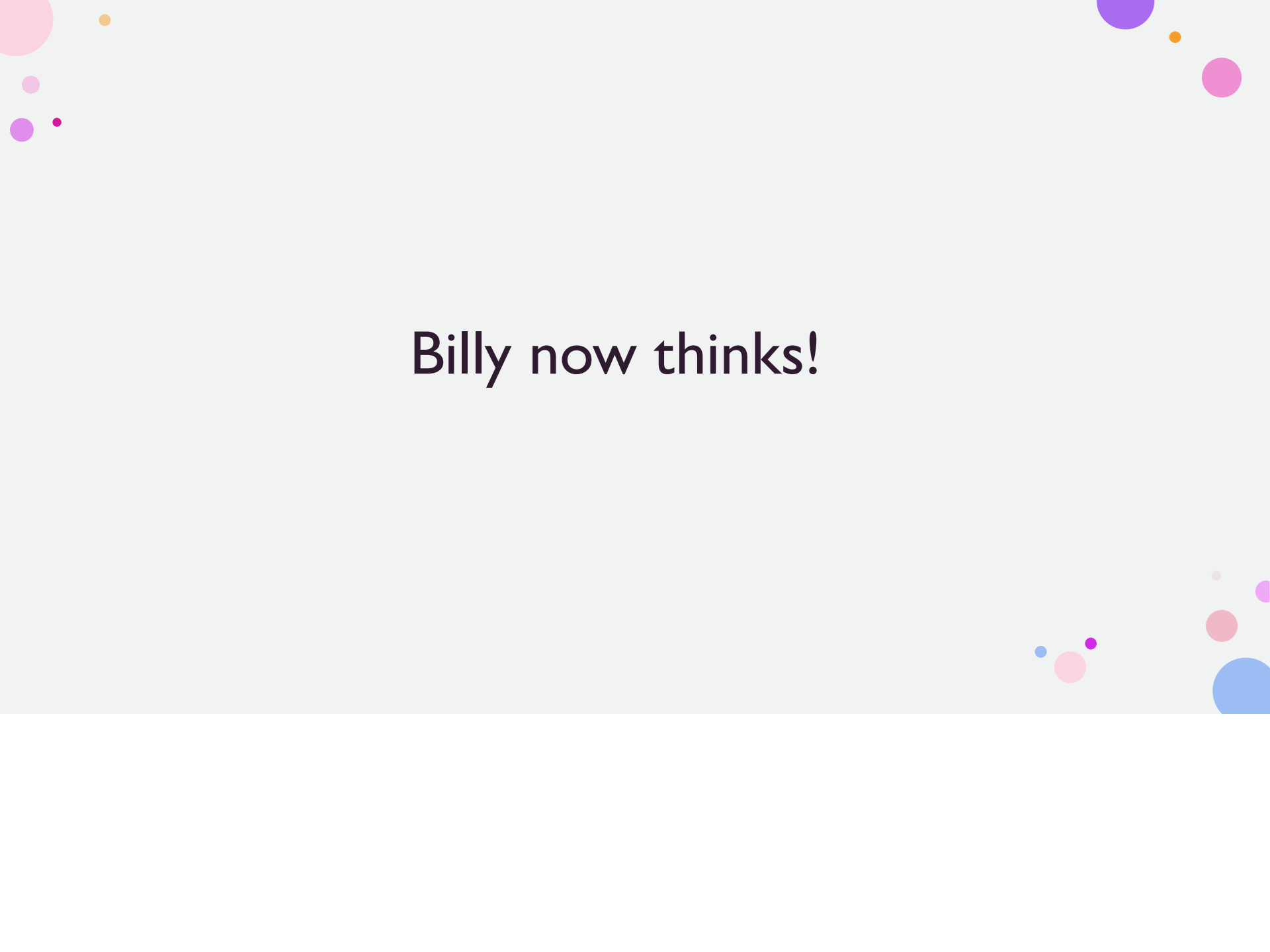


Backdrops

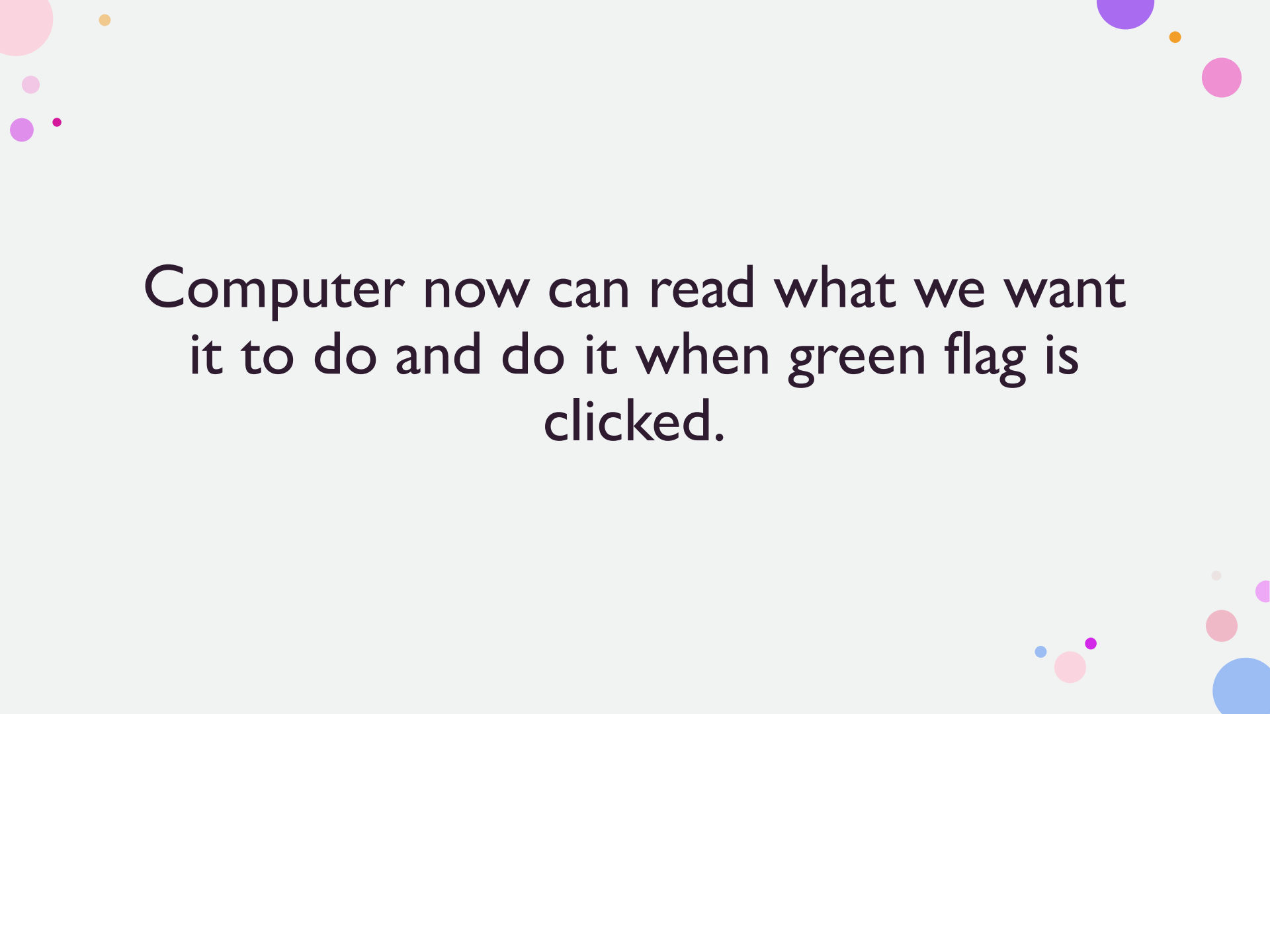
2



I go to school to learn
about science.

The slide features a light gray background with decorative elements in the corners. The top-left corner contains a large pink circle, a small orange circle, a small light pink circle, a small purple circle, and a tiny dark pink dot. The top-right corner features a large purple circle, a small orange circle, and a medium pink circle. The bottom-right corner includes a small blue circle, a medium pink circle, a tiny purple circle, a small light pink circle, a tiny light gray circle, a medium red circle, a small purple circle, and a large blue circle.

Billy now thinks!

The slide features a light gray background with decorative elements in the corners. The top-left corner has a cluster of pink, purple, and orange circles. The top-right corner has a purple circle, an orange circle, and a pink circle. The bottom-right corner has a blue circle, a pink circle, a purple circle, and a small blue circle.

Computer now can read what we want
it to do and do it when green flag is
clicked.

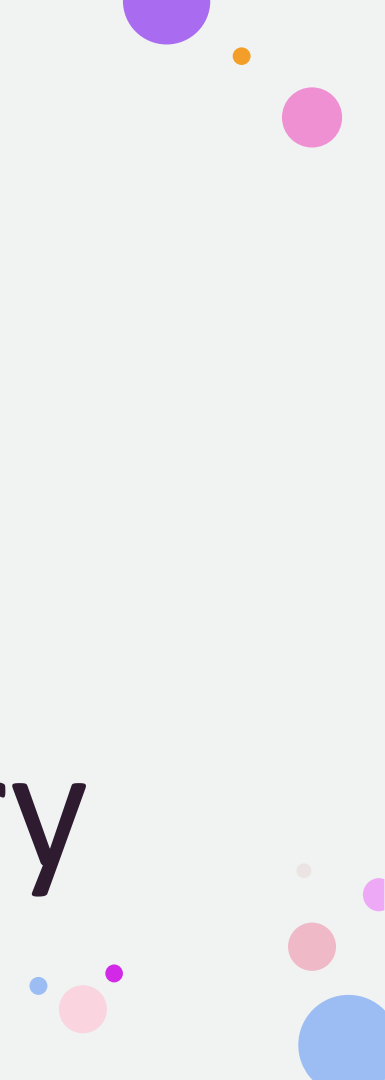


What did we learn?

- Computer needs instructions to read story
- 



What did we learn?

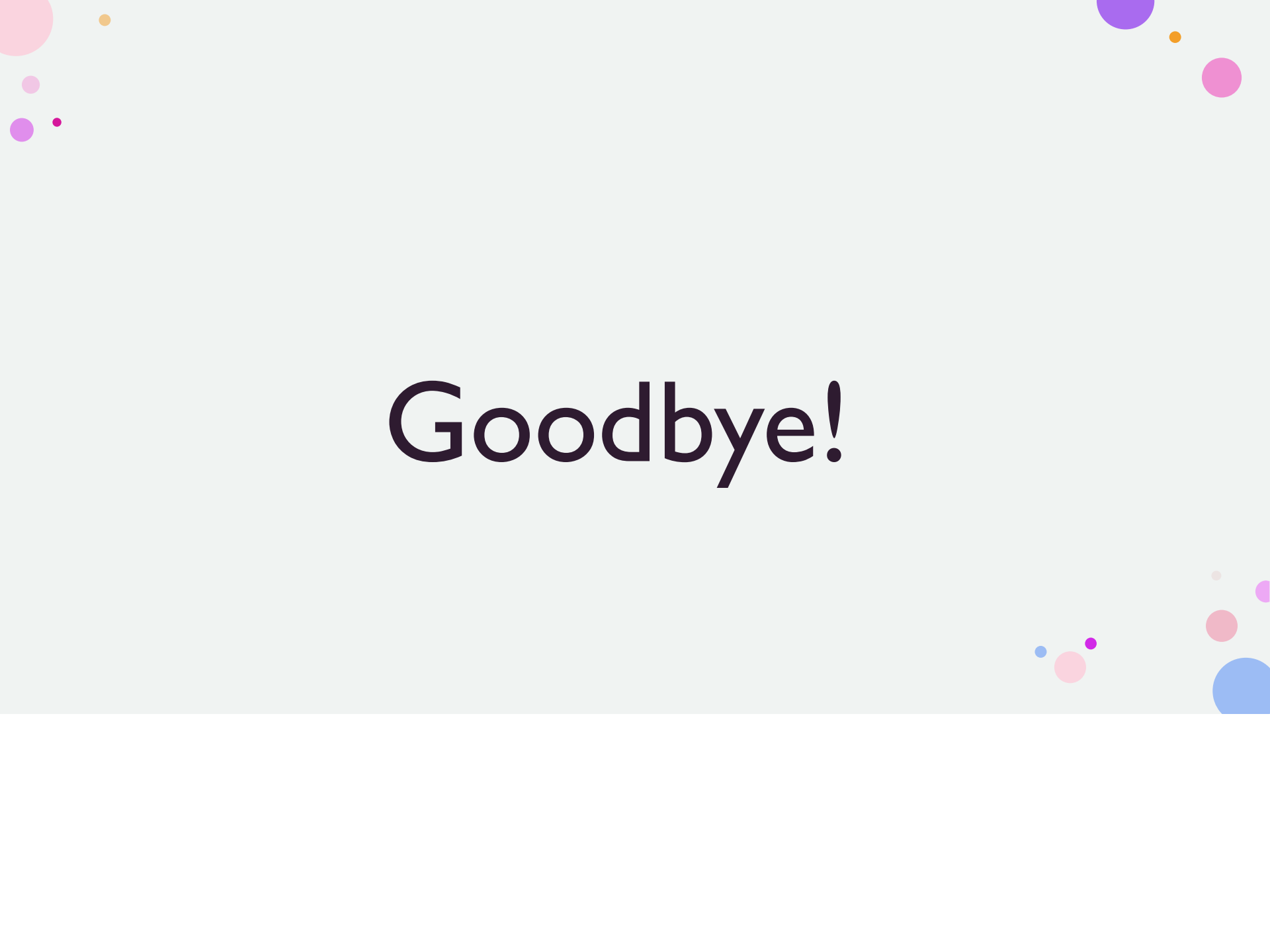
- Computer needs green flag to be clicked to read story
- 

What did we learn?



- For computer green flag and instructions is like opening a book and reading



The image features a light gray background with the word "Goodbye!" centered in a dark purple, sans-serif font. The corners of the image are decorated with clusters of small, semi-transparent circles in various colors including pink, purple, blue, and orange.

Goodbye!

The background of the slide is a complex, abstract geometric pattern. It consists of numerous overlapping triangles and polygons in various colors, including shades of pink, orange, yellow, teal, and purple. The pattern is dense and covers the entire upper half of the slide. The text is centered over this pattern.

Scratch Coding Class

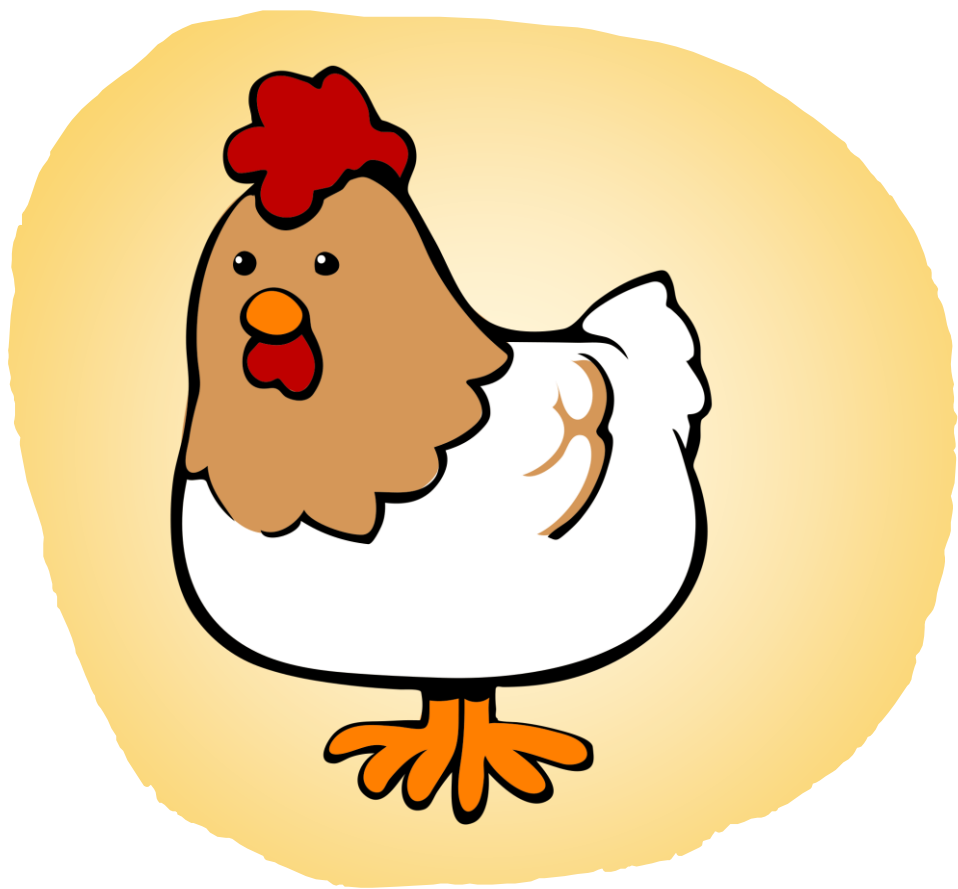
Lesson 4

The top of the image features a decorative header with a repeating pattern of semi-circles. Each semi-circle is filled with a different geometric pattern: concentric lines, a grid of dots, or radiating dashes. The colors are in shades of brown and tan.

Hello!

The top of the image features a decorative header with a repeating pattern of semi-circles. Each semi-circle is filled with a different geometric pattern: concentric lines, a grid of dots, or radiating lines. The colors are in shades of brown and tan.

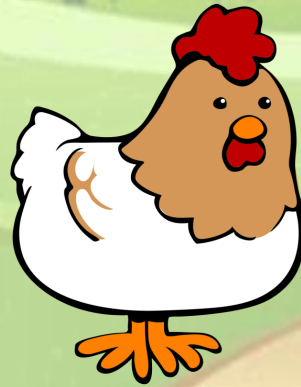
**Today we
will continue
our story.**



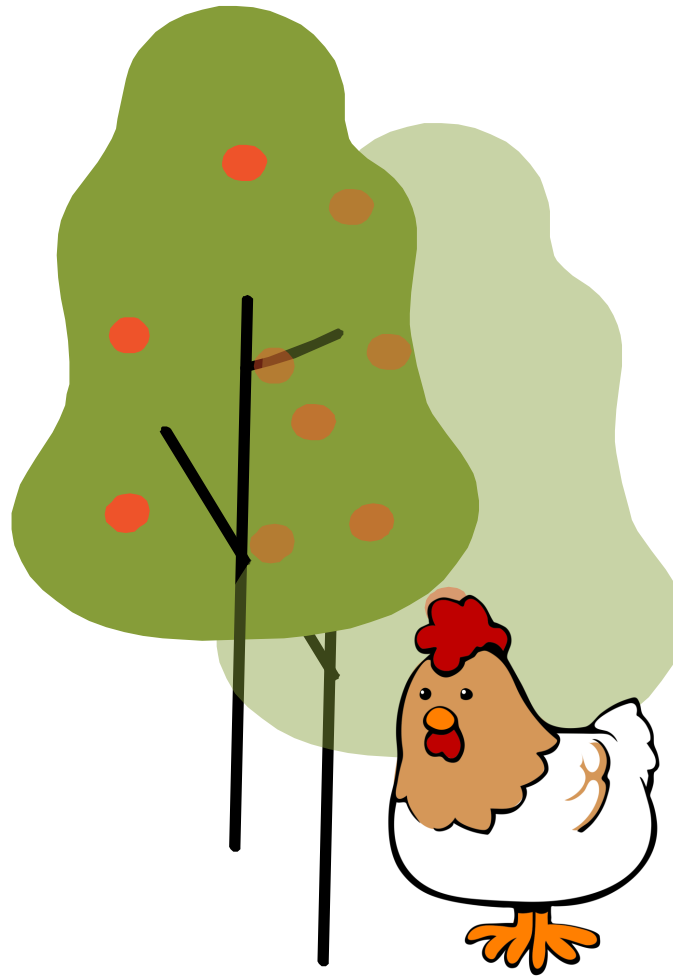
Billy

The chicken

Billy started
walking to
school and it
began to
rain.



Billy went
under a **tree**
take **shelter**
from the rain.



The top of the image features a decorative header with a repeating pattern of semi-circles and concentric arcs in a dark red color. Below this, the main text is centered on a solid dark red background.

**We want Billy to glide from start
to end**

Scratch 3.29.1

File Edit Tutorials Scratch Project

Code Costumes Sounds

Motion

- move 10 steps
- turn 15 degrees
- turn 15 degrees
- go to random position
- go to x: -185 y: -65
- glide 1 secs to random position
- glide 1 secs to x: -185 y: -65
- point in direction 90
- point towards mouse-pointer
- change x by 10
- set x to -185
- change y by 10
- set y to -65

We want Billy to be in motion

when clicked
think 1 secs
dance for 2 seconds

Use your finger to right click *Motion*

Sprite: Chick x: -185 y: -65

Show: Size: 100 Direction: 90

Stage: Backdrops: 2

Scratch 3.29.1

File Edit Tutorials Scratch Project

Code Costumes Sounds

Motion

- move 10 steps
- turn 15 degrees
- turn 15 degrees
- go to random position
- go to x: 165 y: -80
- glide 1 secs to random position
- glide 1 secs to x: 165 y: -80
- point in direction 90
- point towards mouse-pointer
- change x by 10
- set x to 165
- change y by 10
- set y to -80

when clicked

think I go to school to learn science for 2 seconds

glide 1 secs to x: -185 y: -65

We want Billy to glide to a certain position

Sprite: Chick x: 165 y: -80

Show: Size: 100 Direction: 90

Backdrops: 2

The top of the slide features a decorative header with a repeating pattern of semi-circles and concentric arcs in a dark red color. The main body of the slide is a solid, medium-brown color.

**How do we make computer
understand where Billy should
go?**

The top of the slide features a decorative header with a repeating pattern of semi-circles. Each semi-circle is filled with a different geometric pattern: concentric lines, a grid of dots, or radiating lines. The colors are in shades of brown and tan.

X and Y position



X Position: How far side to side
Y position: How far up and down



Scratch Project

Code Costumes Sounds

Motion

- move 10 steps
- turn 15 degrees
- turn 15 degrees
- go to random position
- go to x: -182 y: -62
- glide 1 secs to random position
- glide 1 secs to x: -182 y: -62
- point in direction 90
- point towards mouse-pointer
- change x by 10
- set x to -182
- change y by 10
- set y to -62

when clicked

think I go to school to learn science for 2 seconds

glide 3 secs to x: 165 y: -80

Where Billy is right now.

Sprite: Chick

x: -182 y: -62

Size: 100 Direction: 90

Stage

Backdrops: 2

Scratch 3.29.1

File Edit Tutorials Scratch Project

Code Costumes Sounds

Motion

- move 10 steps
- turn 15 degrees
- turn 15 degrees
- go to random position
- go to x: 165 y: -80
- glide 1 secs to random position
- glide 1 secs to x: 165 y: -80
- point in direction 90
- point towards mouse-pointer
- change x by 10
- set x to 165
- change y by 10
- set y to -80

when clicked

think I go to school to learn science for 2 seconds

glide 1 secs to x: -185 y: -65

Right click and move Billy where we would like him to move.

Sprite: Chick

Size: 100

Direction: 90

Stage

Backdrops: 2

The image shows the Scratch 3.29.1 interface. The top bar includes the Scratch logo, a menu (File, Edit, Tutorials), and a "Scratch Project" button. The left sidebar contains tabs for Code, Costumes, and Sounds. The "Code" tab is active, showing a "Motion" category with various blocks. The main workspace displays a script starting with "when green flag clicked", followed by "think I go to school to learn science for 2 seconds", and "glide 1 secs to x: -185 y: -65". A yellow chick sprite is visible in the top right corner of the workspace. The right sidebar shows the "Stage" area with a preview of the stage and a "Backdrops" section. Below the stage, there are buttons for "Sprite" and "Stage". A text overlay in the center of the workspace reads: "The X (side to side) and Y position (up and down) of where we want him to move". A blue arrow points from the text to the "x" coordinate field in the "Sprite" panel, which is set to 165. The "y" coordinate field is set to -80. The "Sprite" panel also shows "Chick" as the selected sprite, "Size" as 100, and "Direction" as 90.

Scratch 3.29.1

File Edit Tutorials Scratch Project

Code Costumes Sounds

Motion

move 10 steps

turn 15 degrees

turn 15 degrees

go to random position

go to x: 165 y: -80

glide 1 secs to random position

glide 1 secs to x: 165 y: -80

point in direction 90

point towards mouse-pointer

change x by 10

set x to 165

change y by 10

set y to -80

when green flag clicked

think I go to school to learn science for 2 seconds

glide 1 secs to x: -185 y: -65

The X (side to side) and Y position (up and down) of where we want him to move

Sprite Chick x 165 y -80

Show Size 100 Direction 90

Chick

Stage

Backdrops 2

Scratch Project

Code Costumes Sounds

Motion

- move 10 steps
- turn 15 degrees
- turn 15 degrees
- go to random position
- go to x: -182 y: -62
- glide 1 secs to random position
- glide 1 secs to x: -182 y: -62
- point in direction 90
- point towards mouse-pointer
- change x by 10
- set x to -182
- change y by 10
- set y to -62

when green flag clicked

think I go to school to learn science for 2 seconds

glide 3 secs to x: 165 y: -80

Let's copy the numbers

Sprite: Chick

x: -182 y: -62

Show: [] Size: 100 Direction: 90

Stage

Backdrops: 2

Scratch Project

Code Costumes Sounds

Motion

- move 10 steps
- turn 15 degrees
- turn 15 degrees
- go to random position
- go to x: -182 y: -62
- glide 1 secs to random position
- glide 1 secs to x: -182 y: -62
- point in direction 90
- point towards mouse-pointer
- change x by 10
- set x to -182
- change y by 10
- set y to -62

when green flag clicked

think I go to school to learn science for 2 seconds

glide 3 secs to x: 165 y: -80

And let's pick how long it will take for Billy to move

Sprite: Chick

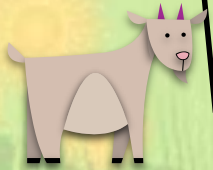
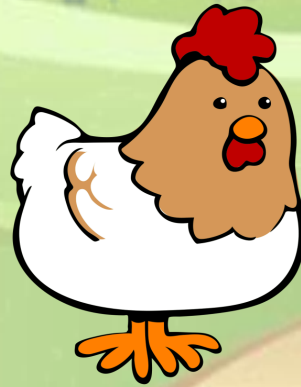
x: -182 y: -62

Show: [] Size: 100 Direction: 90

Stage

Backdrops: 2

Billy started
walking to
school and it
began to
rain.



The top of the image features a decorative header with a repeating pattern of semi-circles and concentric arcs in a dark red color. The main background is a solid, medium brown color.

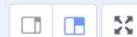
**Let's draw the rain Billy walks
through**

Backdrops

2

Paint





1

costume1
0 x 0

Costume costume1



Fill

Outline

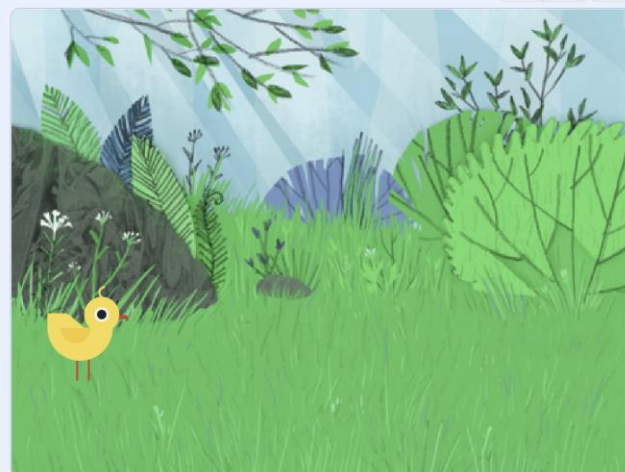


4



Choose a
color

Convert to Bitmap



Sprite Sprite1

x

y

36

28

Show



Size

100

Direction

90

Stage



Backdrops

2



Chick



Sprite1



Code

Costumes

Sounds

1

costume1
0 x 0

Costume

costume1

Fill

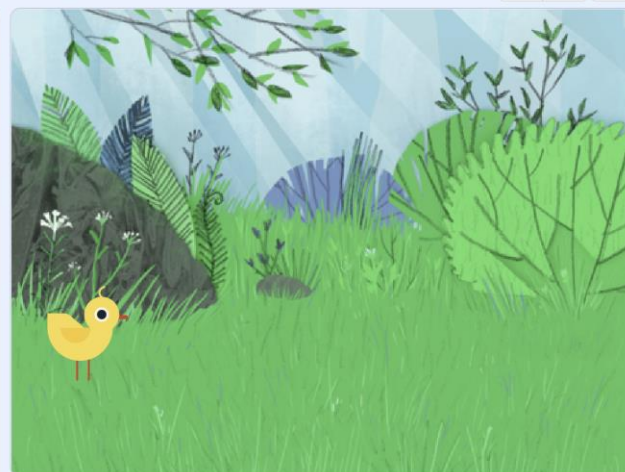
Outline

4



Use
paintbrush

Convert to Bitmap



Sprite

Sprite1

x

36

y

28

Show



Size

100

Direction

90

Stage



Backdrops

2

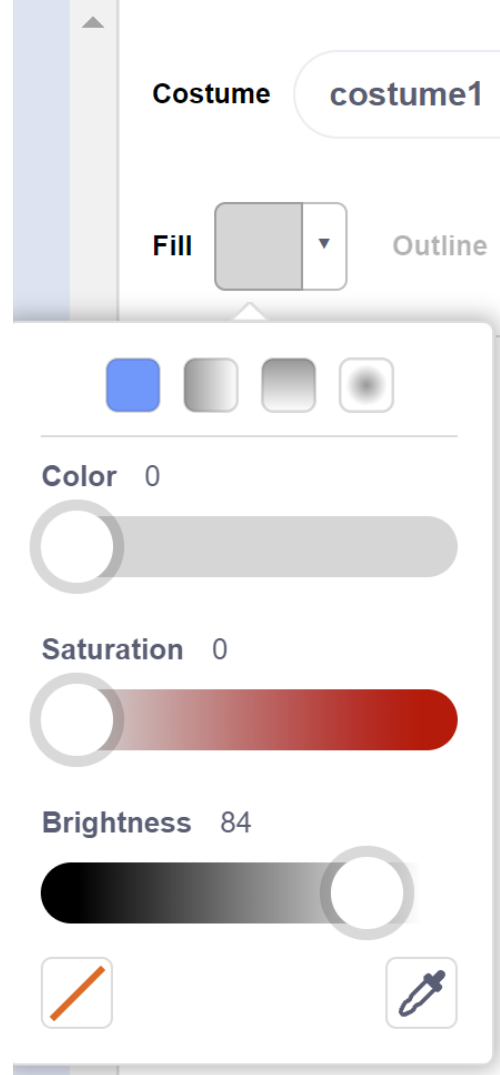


Chick



Sprite1





Scratch Project

Code Costumes Sounds

Costume: costume1

Fill:  Outline:  4

Color 0
Saturation 0
Brightness 84

Convert to Bitmap

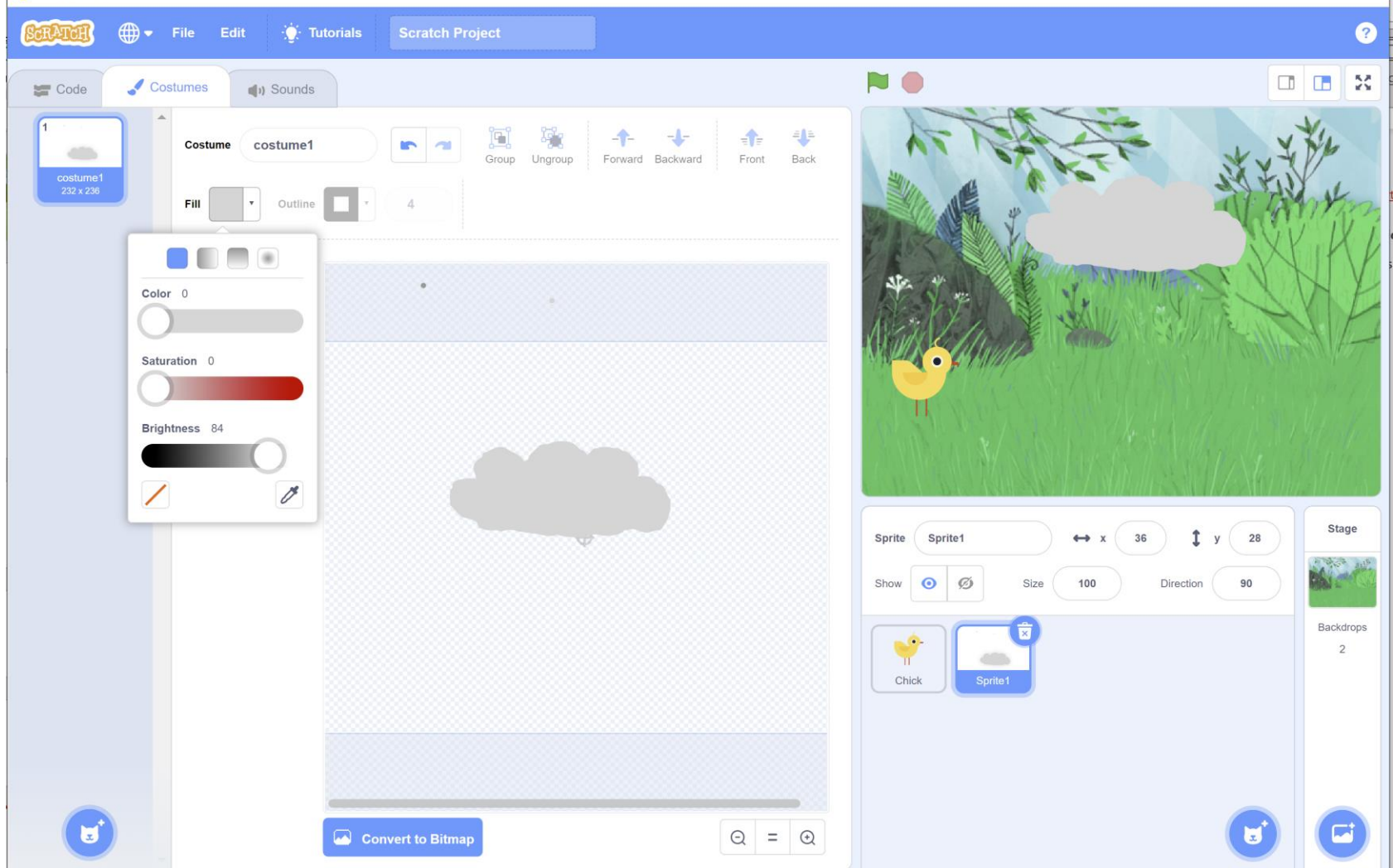
Sprite: Sprite1 x: 36 y: 28

Show:   Size: 100 Direction: 90

Chick Sprite1

Stage

Backdrops: 2



  File Edit  Tutorials Scratch Project 

Code Costumes Sounds

1 
costume1
232 x 398

Costume costume1  

Group Ungroup

Forward Backward

Front Back

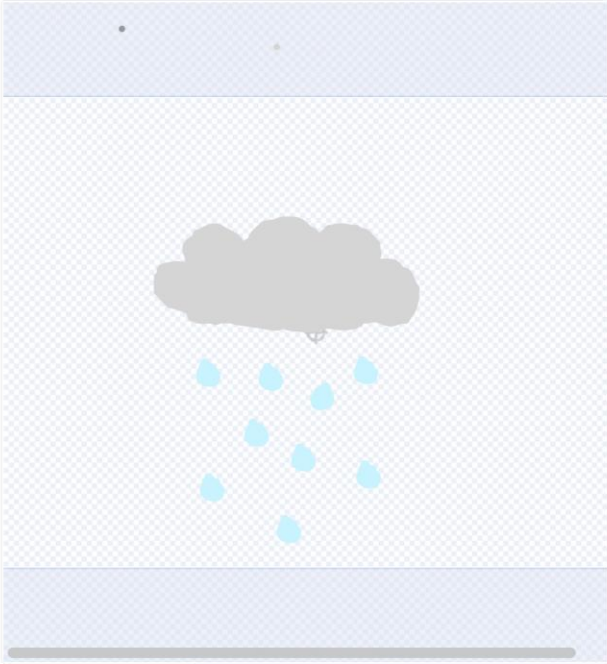
Copy Paste

Delete

Flip Horizontal Flip Vertical

Fill  Outline  0

 Convert to Bitmap  = 



Sprite Sprite1  x 36  y 28

Show  

Size 100

Direction 90

 Chick

 Sprite1

Stage


Backdrops
2

Scratch Project

Code Costumes Sounds

1
costume1
232 x 318

Costume: costume1

Fill: Outline: 0

Group Ungroup Forward Backward Front Back Copy Paste Delete Flip Horizontal Flip Vertical

Return to code when done

Convert to Bitmap

Sprite: Sprite1 x: 36 y: 28

Show: Size: 100 Direction: 90

Chick Sprite1

Stage

Backdrops: 2

The top of the slide features a decorative header with a repeating pattern of semi-circles and concentric arcs in a light brown color against a darker brown background.

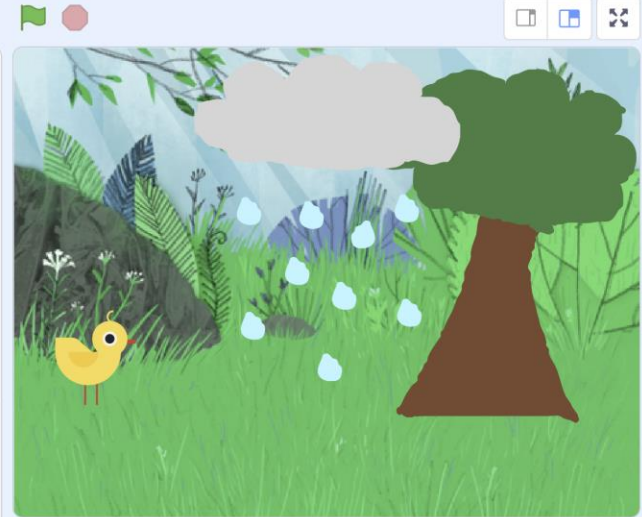
**You try: Draw the tree Billy
walks through**

 Sounds

My Block

clear graphic effects

glide 3 secs to x: 165 y: -80



Sprite2

2



The top of the slide features a decorative header with a repeating pattern of semi-circles. Each semi-circle is filled with a different geometric pattern: concentric lines, a grid of dots, or radiating dashes. The colors are in shades of brown and tan.

Press green flag!

SCRATCH CODING CLASS

Lesson 5

Billy **turned
around** after
the rain
stopped and
kept going to
school.



LET'S REVIEW OUR CODE

when  clicked

think I go to school to learn science for 2 seconds

glide 3 secs to x: 165 y: -80

when  clicked

When we click the
green flag

think I go to school to learn science for 2 seconds

glide 3 secs to x: 165 y: -80

when  clicked

When we click the
green flag

think I go to school to learn science for 2 seconds

Think about science
for 2 seconds

glide 3 secs to x: 165 y: -80

when  clicked

When we click the
green flag

think I go to school to learn science for 2 seconds

Think about science
for 2 seconds

glide 3 secs to x: 165 y: -80

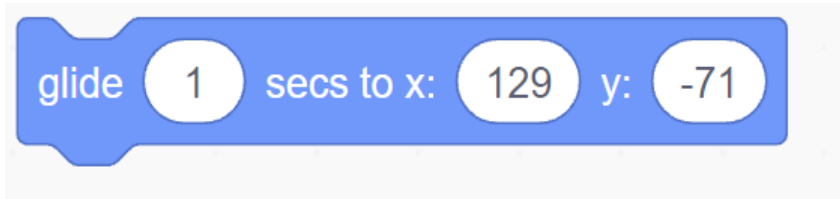
In 3 seconds glide to
the

Billy **turned
around** after
the rain
stopped and
kept going to
school.



YOUR TURN

How do we get Billy to move?



- Pick how many seconds we want it to take for Billy to move.
- Pick a location of where we want Billy to move.

Scratch 3.29.1

Scratch File Edit Tutorials CHIBCK

Code Costumes Sounds

Motion

- turn 15 degrees

Looks

- go to random position

Sound

- go to x: 165 y: -80

Events

- glide 1 secs to random position

Control

- glide 1 secs to x: 165 y: -80

Sensing

- point in direction 90

Operators

- point towards mouse-pointer

Variables

- change x by 10
- set x to 165
- change y by 10
- set y to -80

My Blocks

- if on edge, bounce
- set rotation style left-right

when clicked

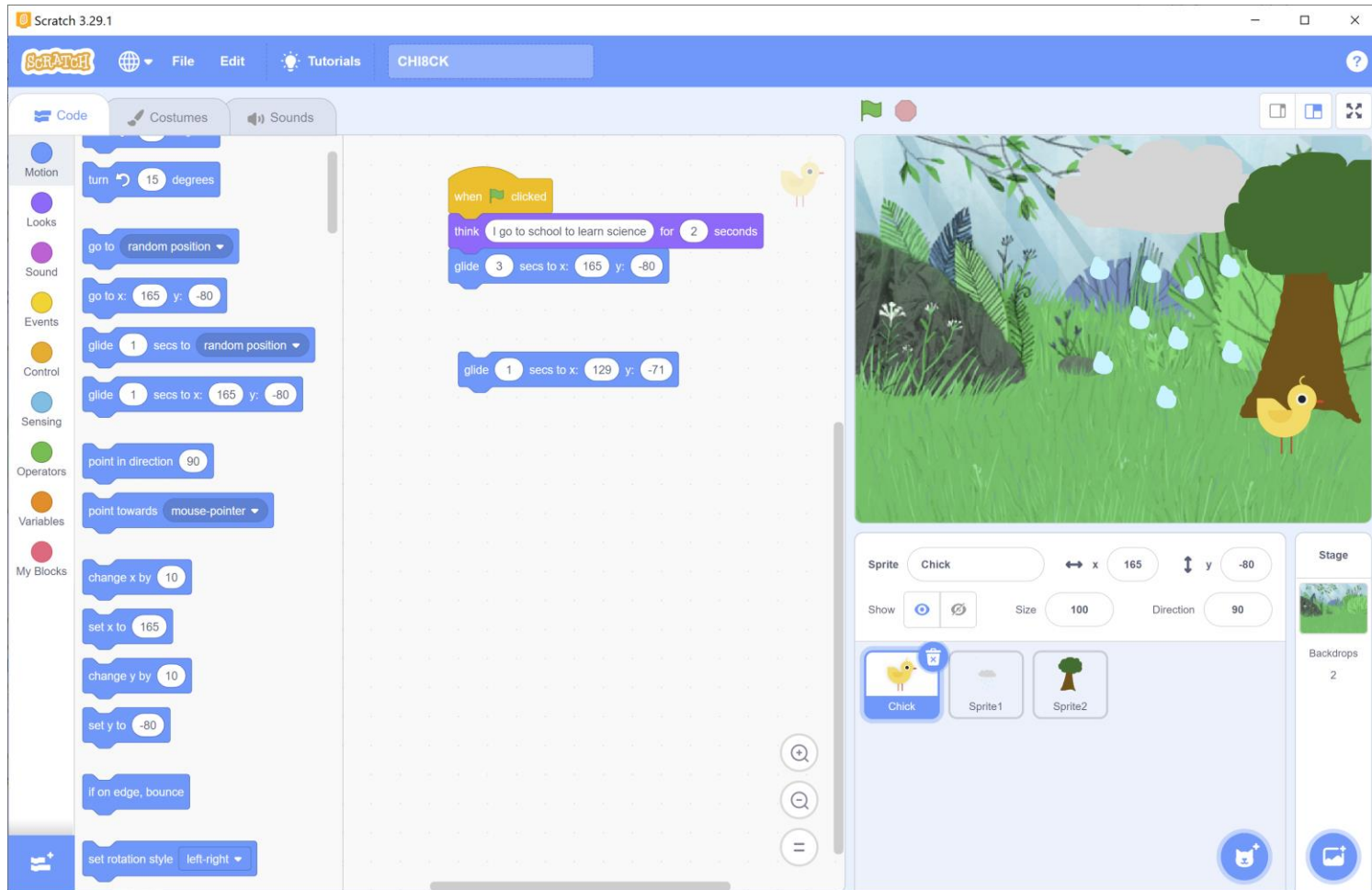
- think I go to school to learn science for 2 seconds
- glide 3 secs to x: 165 y: -80
- glide 1 secs to x: 129 y: -71

Sprite: Chick

Position: x: 165 y: -80

Size: 100 Direction: 90

Backdrops: 2



Scratch 3.29.1

CHI8CK

Code Costumes Sounds

Motion

- turn 15 degrees

Looks

- go to random position

Sound

- go to x: -170 y: -62

Events

- glide 1 secs to random position

Control

- glide 1 secs to x: -170 y: -62

Sensing

- point in direction 90

Operators

- point towards mouse-pointer

Variables

- change x by 10
- set x to -170
- change y by 10
- set y to -62

My Blocks

- if on edge, bounce
- set rotation style left-right

when clicked

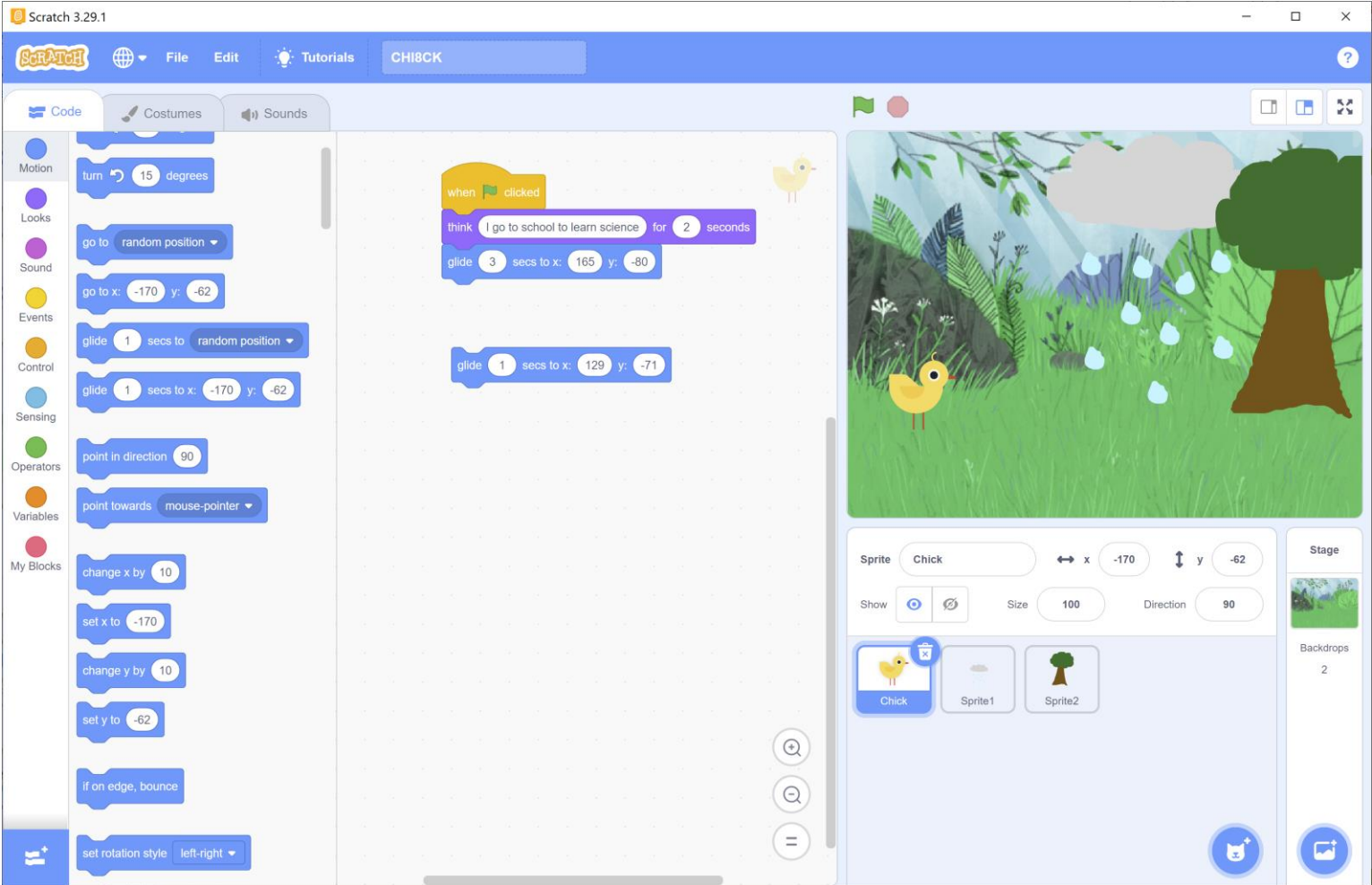
- think I go to school to learn science for 2 seconds
- glide 3 secs to x: 165 y: -80
- glide 1 secs to x: 129 y: -71

Sprite: Chick

Show: ☒ ☐

Size: 100 Direction: 90

Backdrops: 2



Sprite

Chick

↔ x

-170

↕ y

-62

Show

☒

☐

Size

100

Direction

90



Chick

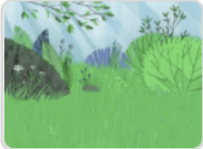


Sprite1



Sprite2

Stage



Backdrops

2

Scratch 3.29.1

CHI8CK

Code Costumes Sounds

Motion

- turn 15 degrees
- go to random position
- go to x: -170 y: -62
- glide 1 secs to random position
- glide 1 secs to x: -170 y: -62

Looks

Sound

Events

Control

Sensing

Operators

Variables

My Blocks

when clicked

- think I go to school to learn science for 2 seconds
- glide 3 secs to x: 165 y: -80
- glide 3 secs to x: -170 y: -62

point in direction 90

point towards mouse-pointer

change x by 10

set x to -170

change y by 10

set y to -62

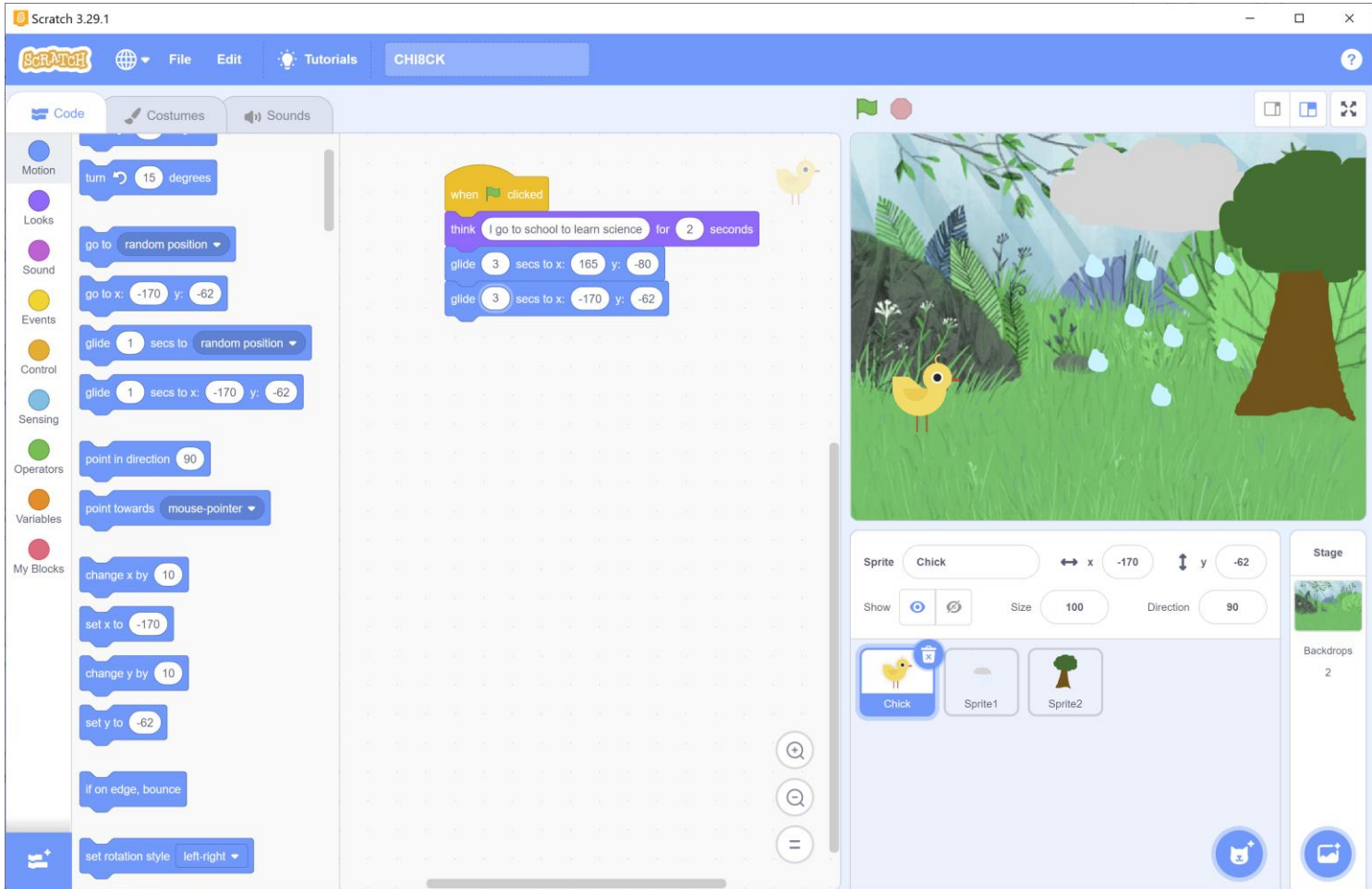
if on edge, bounce

set rotation style left-right

Sprite Chick

Stage

Backdrops 2

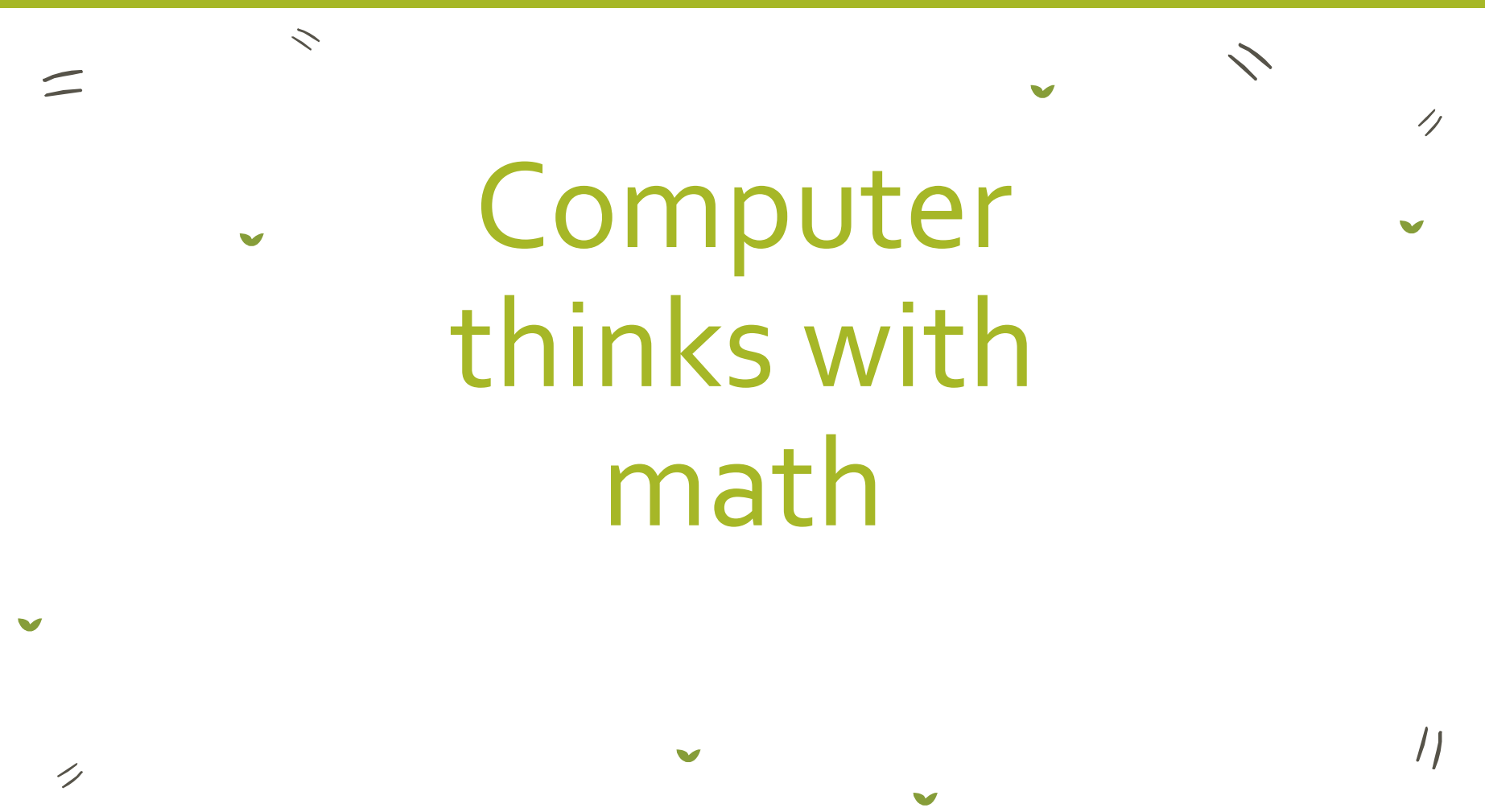


Billy **turned
around** after
the rain
stopped and
kept going to
school.





How do we
make the rain
go away?



Computer
thinks with
math

when  clicked

think I go to school to learn science for 2 seconds

glide 3 secs to x: 165 y: -80

glide 3 secs to x: -170 y: -62

when  clicked

think I go to school to learn science for 2 seconds

glide 3 secs to x: 165 y: -80

glide 3 secs to x: -170 y: -62

→

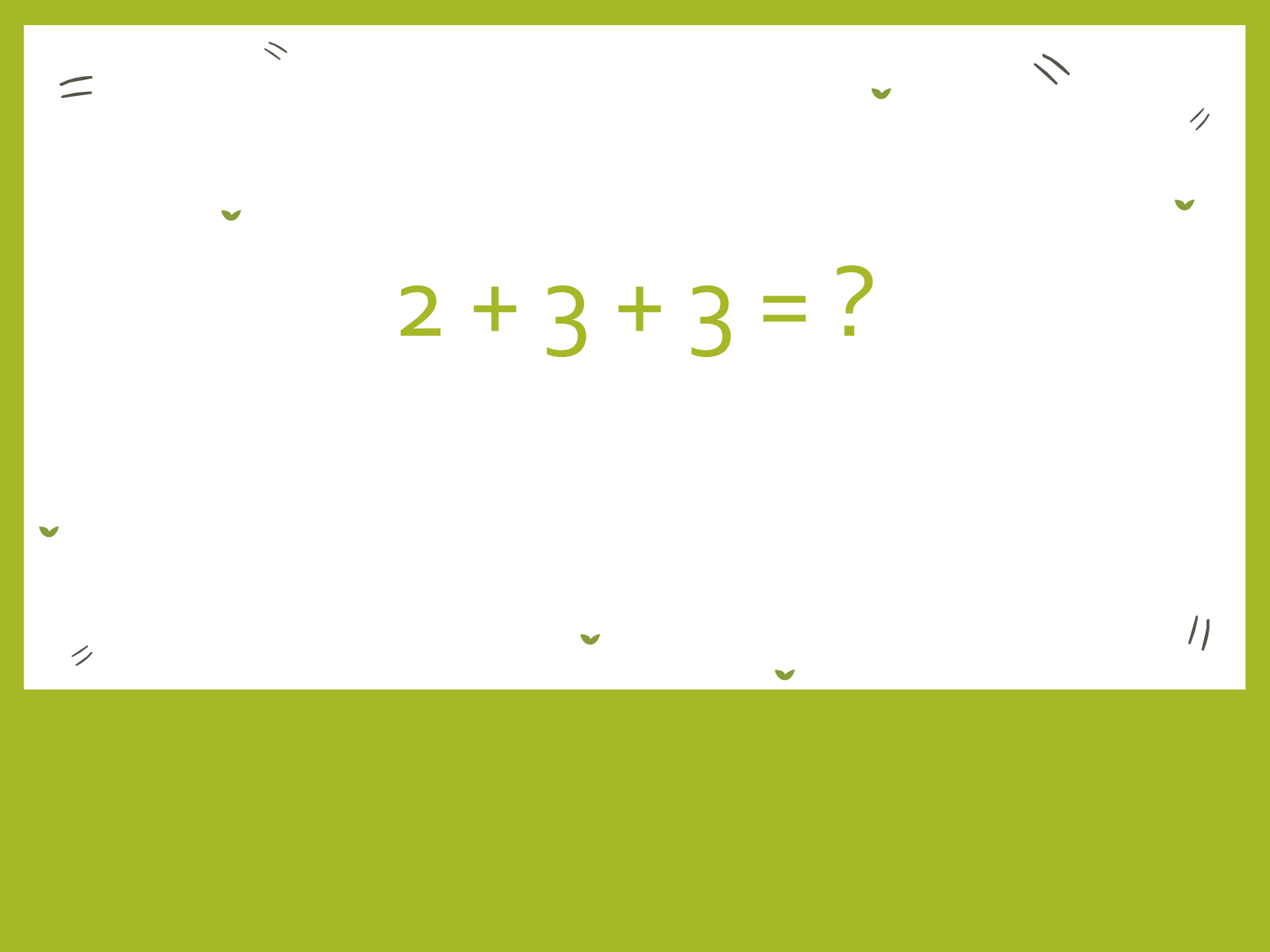
when  clicked

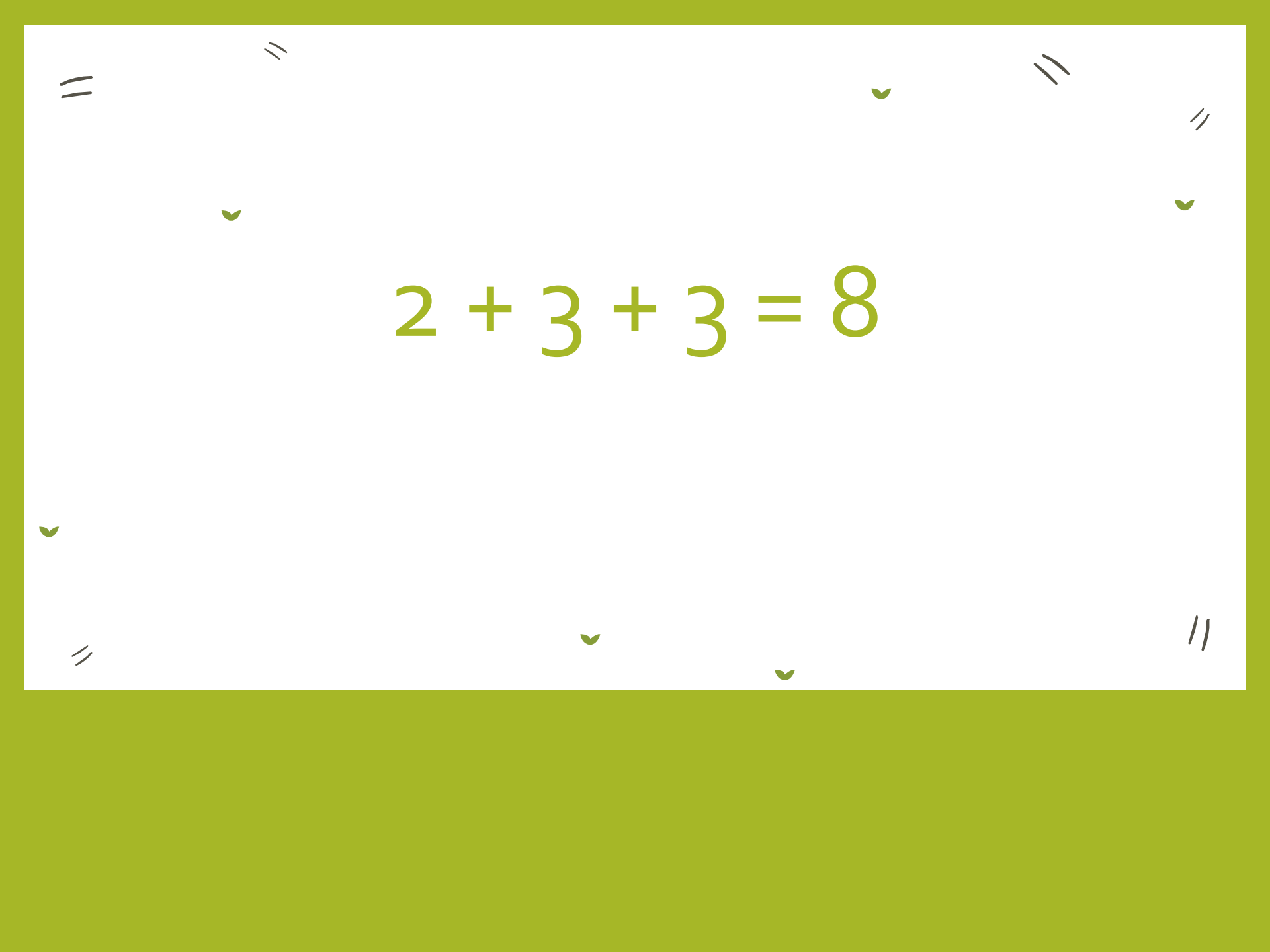
think I go to school to learn science for 2 seconds

glide 3 secs to x: 165 y: -80

glide 3 secs to x: -170 y: -62

$2 + 3 + 3$


$$2 + 3 + 3 = ?$$


$$2 + 3 + 3 = 8$$

Scratch 3.29.1

CHI8CK

Code Costumes Sounds

Motion

- turn 15 degrees

Looks

- go to random position

Sound

- go to x: -170 y: -62

Events

- glide 1 secs to random position

Control

- glide 1 secs to x: -170 y: -62

Sensing

- point in direction 90

Operators

- point towards mouse-pointer

Variables

- change x by 10
- set x to -170
- change y by 10
- set y to -62

My Blocks

- if on edge, bounce
- set rotation style left-right

when clicked

- think I go to school to learn science for 2 seconds
- glide 3 secs to x: 165 y: -80
- glide 3 secs to x: -170 y: -62

Press the rain

Sprite Chick

Stage

Backdrops 2

Scratch 3.29.1

CHI8CK

Code Costumes Sounds

Motion

- Change x by 10
- set x to 70
- change y by 10
- set y to 88
- if on edge, bounce
- set rotation style left-right

Sensing

- when green flag clicked

Operators

- x position
- y position
- direction

Variables

My Blocks

Looks

- say Hello! for 2 seconds
- say Hello!
- think Hmm... for 2 seconds
- think Hmm...
- switch costume to costume1

Add green flag so computer knows to read code

Stage

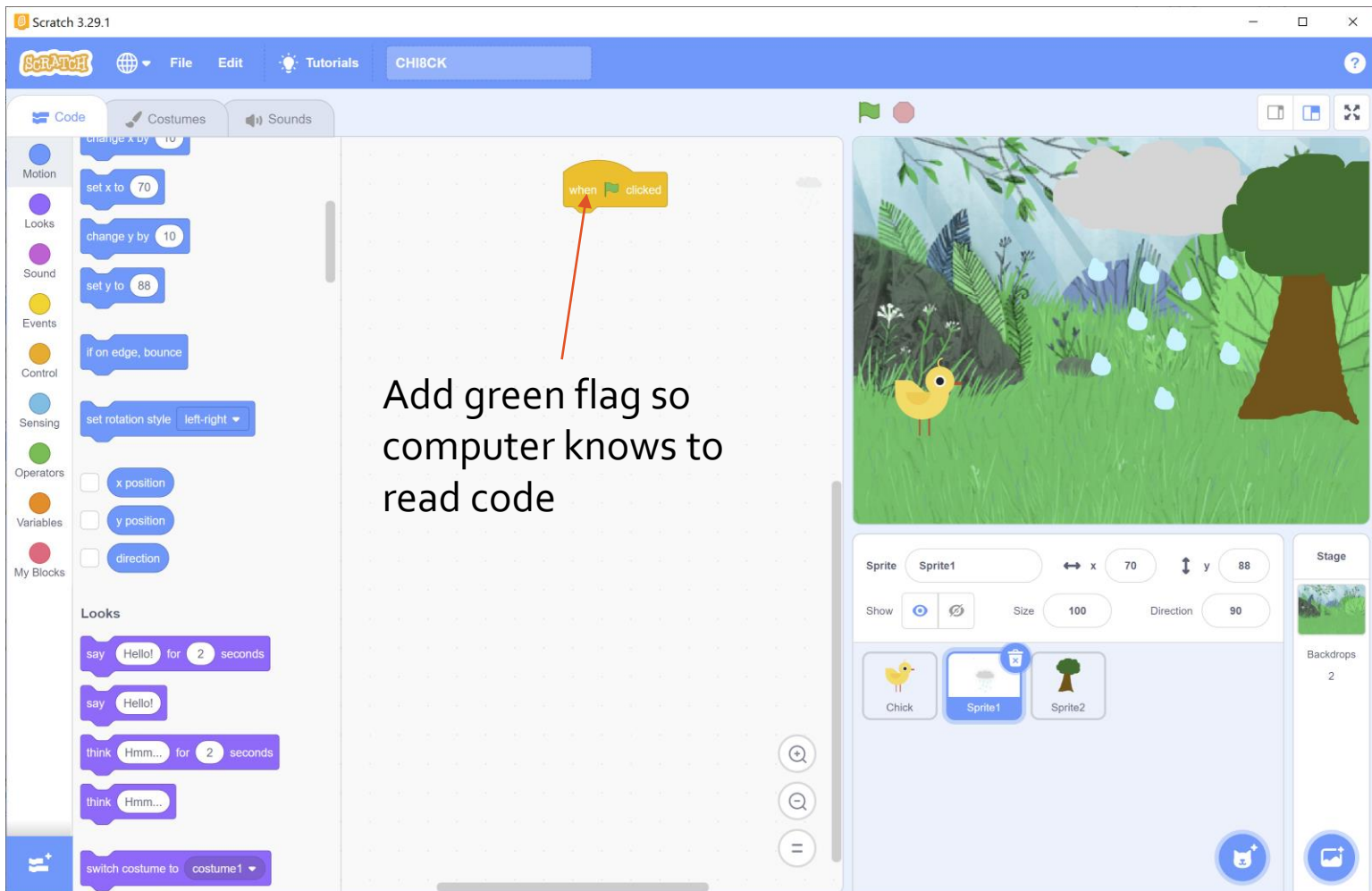
Sprite1

x 70 y 88

Show Size 100 Direction 90

Chick Sprite1 Sprite2

Backdrops 2



Scratch 3.29.1

Scratch File Edit Tutorials CH18CK

Code Costumes Sounds

Control

- Motion
- Looks
- Sound
- Events
- Control**
- Sensing
- Operators
- Variables
- My Blocks

when green flag clicked

wait 1 seconds

repeat 10

forever

if then

if then else

wait until

repeat until

stop all

We want to wait

Sprite Sprite1 x 70 y 88

Show Show Size 100 Direction 90

Chick Sprite1 Sprite2

Stage

Backdrops 2

The image shows the Scratch 3.29.1 interface. The 'Code' tab is selected, and the 'Control' category is active in the left sidebar. A 'when green flag clicked' block is on the script area, followed by a 'wait 1 seconds' block. A red arrow points to the 'wait 1 seconds' block, and the text 'We want to wait' is written next to it. The 'Sprite' panel on the right shows 'Sprite1' selected, with a 'Chick' sprite also visible. The 'Stage' panel shows two backdrops.

Scratch 3.29.1

Scratch File Edit Tutorials CHI8CK

Code Costumes Sounds

next backdrop

change size by 10

set size to 100 %

change color effect by 25

set color effect to 0

clear graphic effects

show

hide

go to front layer

go forward 1 layers

costume number

backdrop number

size

Sound

play sound pop until done

when clicked

wait 8 seconds

We want the rain to hide after 8 seconds

Sprite Sprite1 x 70 y 88

Show Size 100 Direction 90

Chick Sprite1 Sprite2

Stage

Backdrops 2

CLICK GREEN FLAG!

Scratch 3.29.1

Scratch File Edit Tutorials CHIBCK

Code Costumes Sounds

Motion

Looks

Sound

Events

Control

Sensing

Operators

Variables

My Blocks

next backdrop

change size by 10

set size to 100 %

change color effect by 25

set color effect to 0

clear graphic effects

show

hide

go to front layer

go forward 1 layers

costume: number

backdrop: number

size

Sound

play sound pop until done

when clicked

wait 8 seconds

hide

Sprite1 x 70 y 88

Show

Size 100

Direction 90

Chick

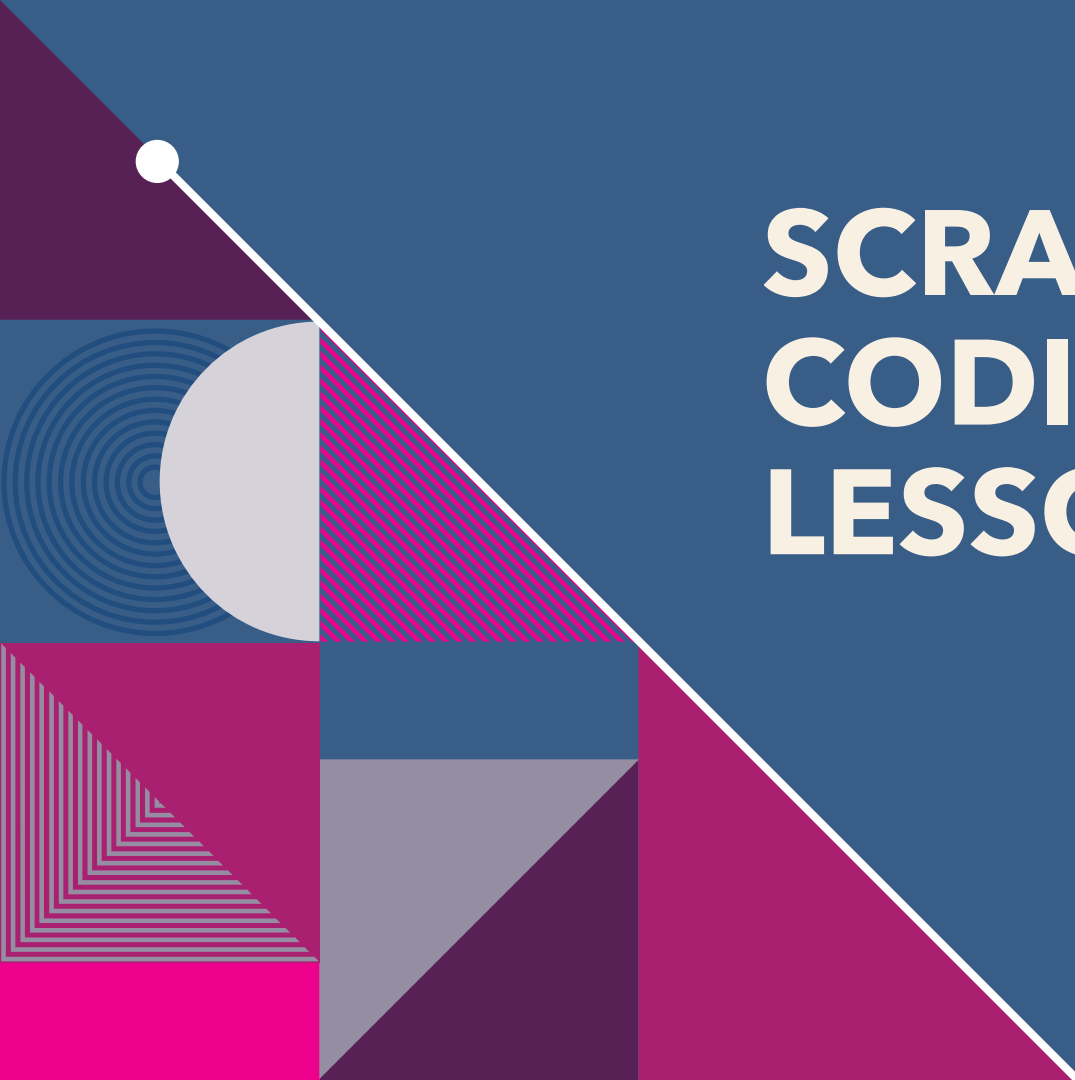
Sprite1

Sprite2

Stage

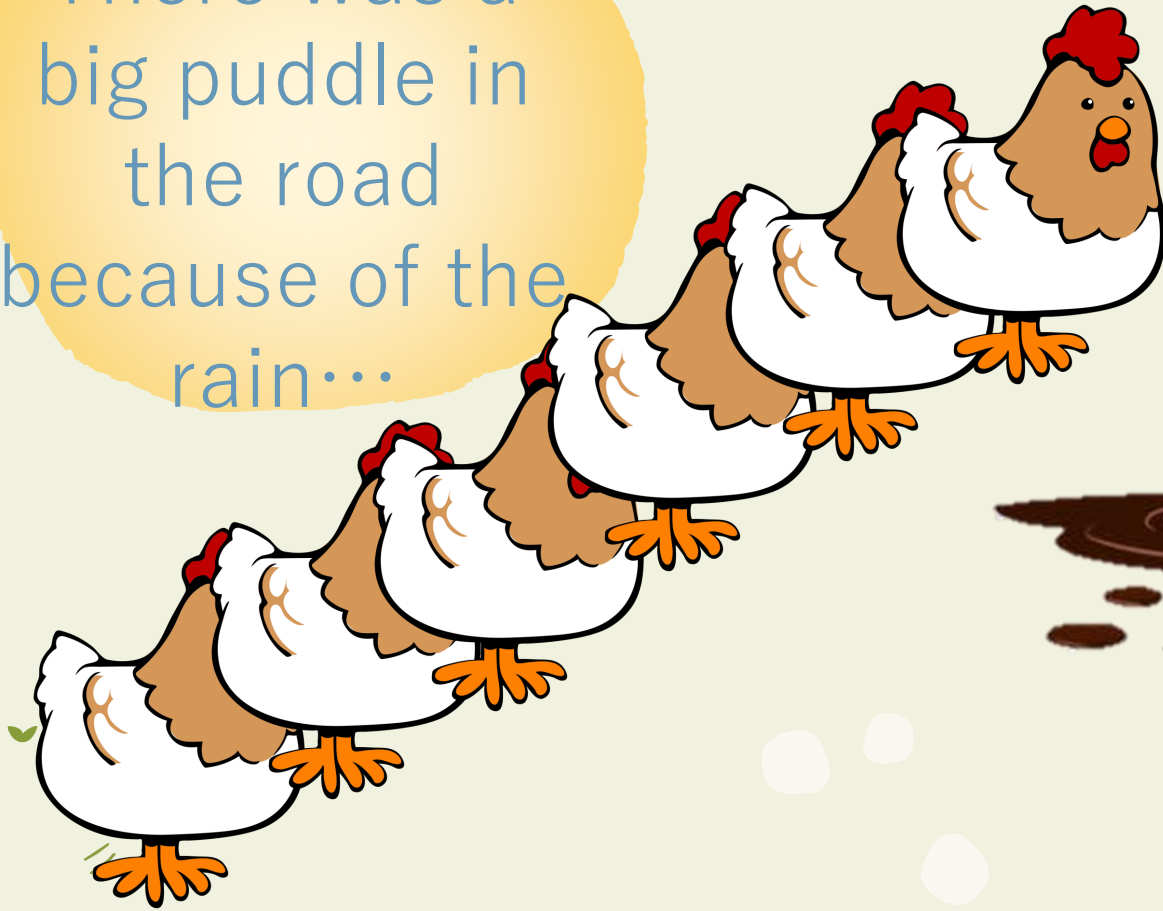
Backdrops 2

The image shows the Scratch 3.29.1 interface. The top bar includes the Scratch logo, a menu bar (File, Edit, Tutorials), and a username 'CHIBCK'. Below the menu bar are tabs for 'Code', 'Costumes', and 'Sounds'. The 'Code' tab is active, showing a script area with a grid. The script starts with a 'when clicked' event block, followed by a 'wait 8 seconds' block, and a 'hide' block. The 'Costumes' tab is also visible, showing a list of costumes: 'Chick', 'Sprite1', and 'Sprite2'. The 'Stage' area shows a forest backdrop with a yellow chick sprite. The 'Sprite' panel on the right shows the 'Sprite1' costume selected, with a size of 100 and a direction of 90. The 'Backdrops' panel shows two backdrops.

An abstract geometric design on the left side of the slide. It features a dark blue background with a white diagonal line running from the top-left corner towards the bottom-right. Along this line, there are several geometric shapes: a small white circle near the top, a light gray semi-circle, a magenta triangle with diagonal lines, a magenta square with concentric lines, a magenta square, a light gray triangle, and a magenta triangle. The text 'SCRATCH CODING CLASS LESSON 6' is written in white, bold, sans-serif font on the right side of the slide.

SCRATCH CODING CLASS LESSON 6

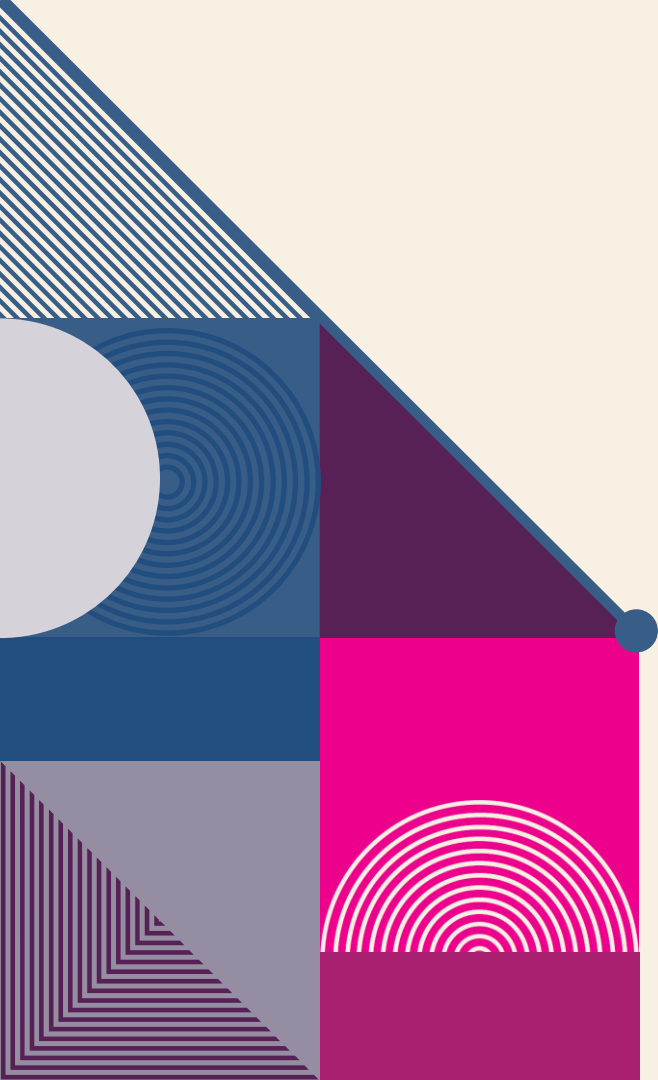
There was a
big puddle in
the road
because of the
rain...



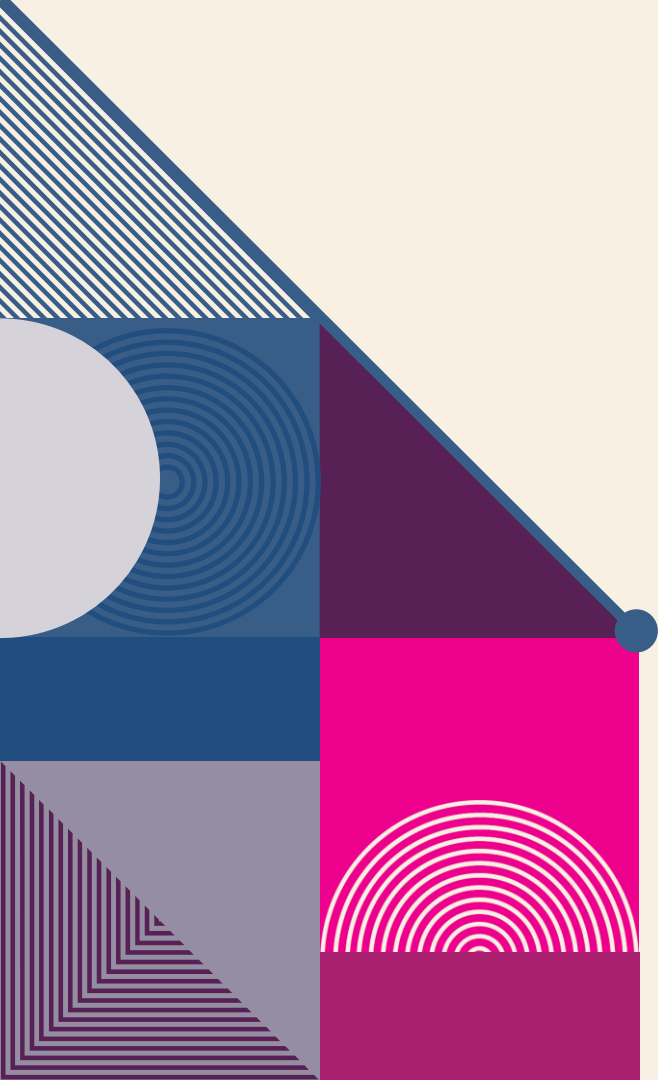
Billy
walked
around
the
puddle.

Billy then
walked
straight to
school.





**LET'S GET BILLY IN A NEW
LOCATION**



LET'S USE MATH.

From last time...

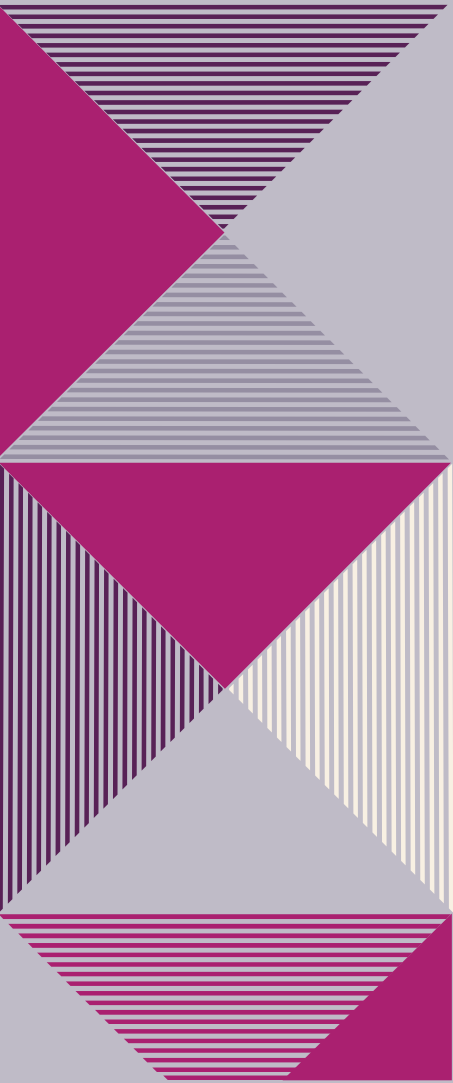
when  clicked

think I go to school to learn science for 2 seconds

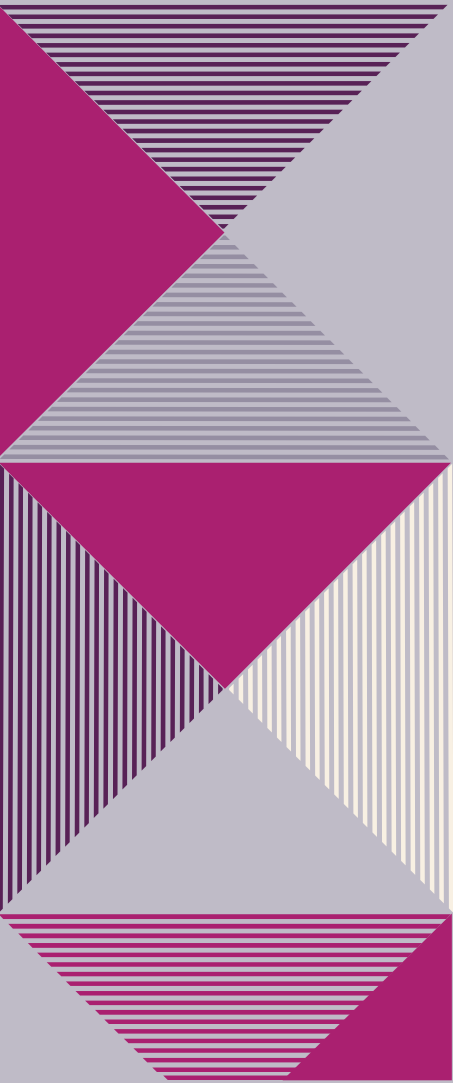
glide 3 secs to x: 165 y: -80

glide 3 secs to x: -170 y: -62

$2 + 3 + 3$



$$2 + 3 + 3 = ?$$



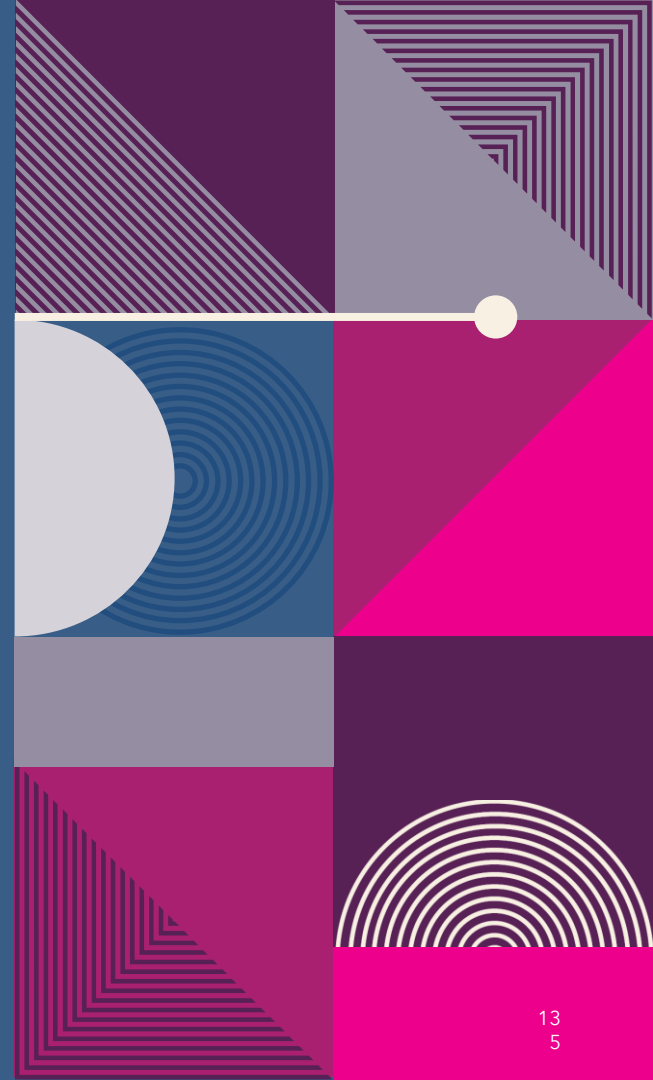
$$2 + 3 + 3 = 8$$

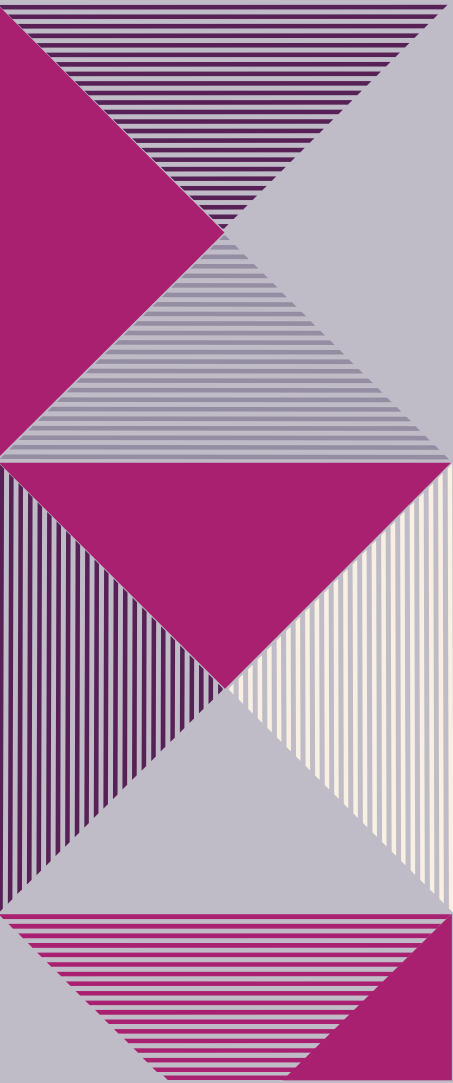




WHAT NEEDS TO LEAVE?

- Tree
- Old Place





**FOR TREE- COPY
CODE FROM
LAST TIME!**

Scratch 3.29.1

Scratch

FileEditTutorialsScratch Project

CodeCostumesSounds

MotionLooksSoundEventsControlSensingOperatorsVariablesMy Blocks

Looks

say Hello! for 2 seconds

say Hello!

think Hmm... for 2 seconds

think Hmm...

switch costume to costume1

next costume

switch backdrop to School

next backdrop

change size by 10

set size to 100 %

change color effect by 25

set color effect to 0

clear graphic effects

show

hide

when clicked

show

wait 8 seconds

hide

Stage

Sprite16461

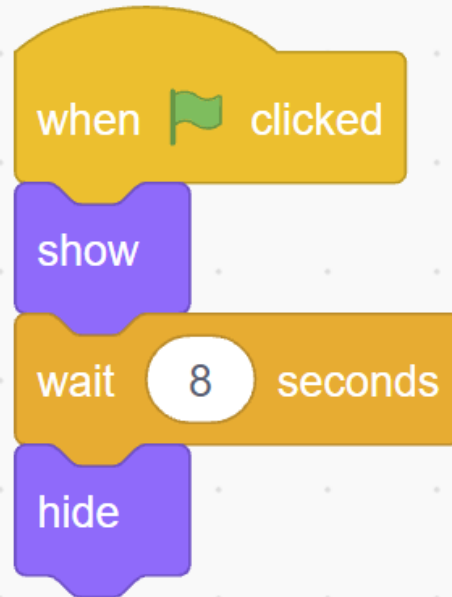
ShowSize100Direction90

Chick

Sprite1

Sprite2

Backdrops2

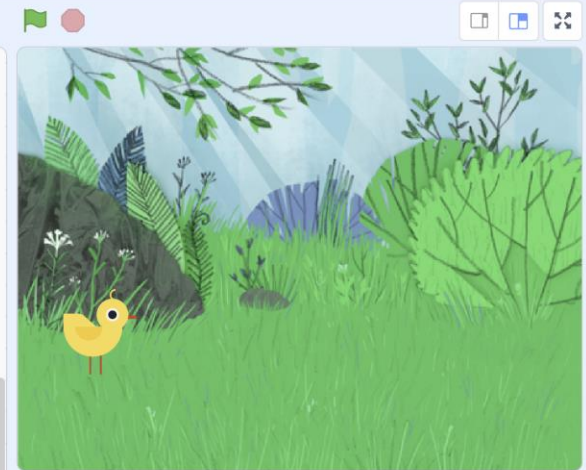


Looks

- say Hello! for 2 seconds
- say Hello!
- think Hmm... for 2 seconds
- think Hmm...
- switch costume to costume1
- next costume
- switch backdrop to School
- next backdrop
- change size by 10
- set size to 100 %
- change color effect by 25
- set color effect to 0
- clear graphic effects
- show
- hide

when clicked
show
wait 8 seconds
hide

Click and use
the code we
made before.



Sprite: Sprite1 x: 64 y: 61

Show: [Show] [Hide] Size: 100 Direction: 90

Chick Sprite1 Sprite2

Backdrops: 2

Scratch 3.29.1

Scratch

FileEditTutorialsScratch Project

CodeCostumesSounds

Motion

Looks

Sound

Events

Control

Sensing

Operators

Variables

My Blocks

Looks

say Hello! for 2 seconds

say Hello!

think Hmm... for 2 seconds

think Hmm...

switch costume to costume1

next costume

switch backdrop to School

next backdrop

change size by 10

set size to 100 %

change color effect by 25

set color effect to 0

clear graphic effects

show

hide

For Tree.

when clicked

show

wait 8 seconds

hide

Sprite2x179y5

ShowShowSize100Direction90

Chick

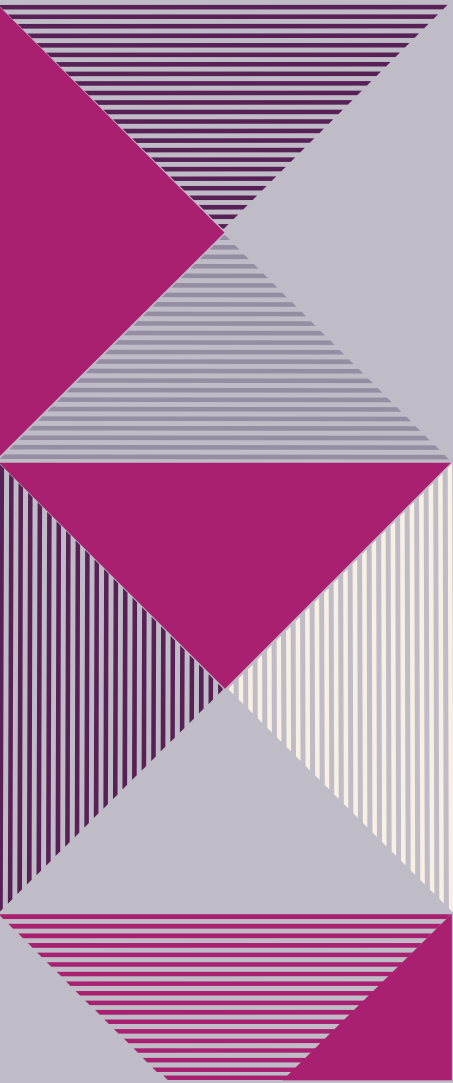
Sprite1

Sprite2

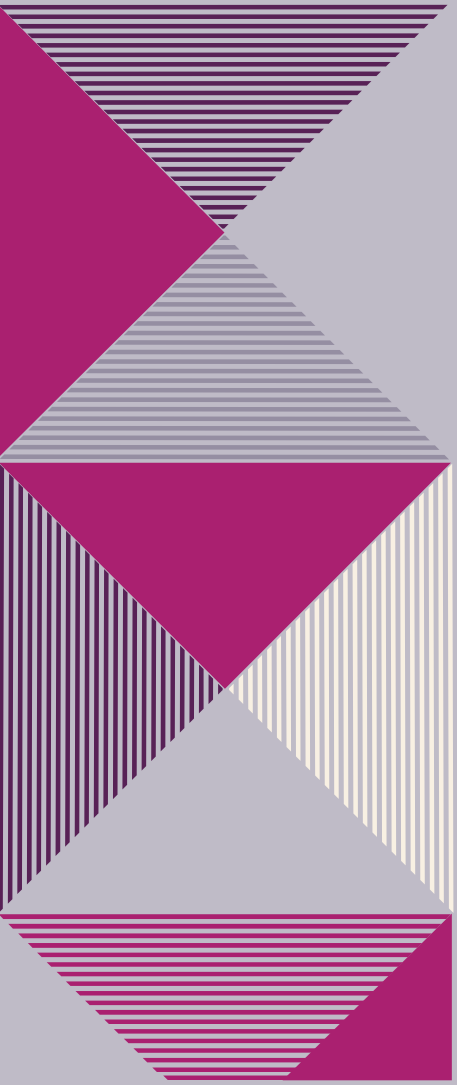
Stage

Backdrops2

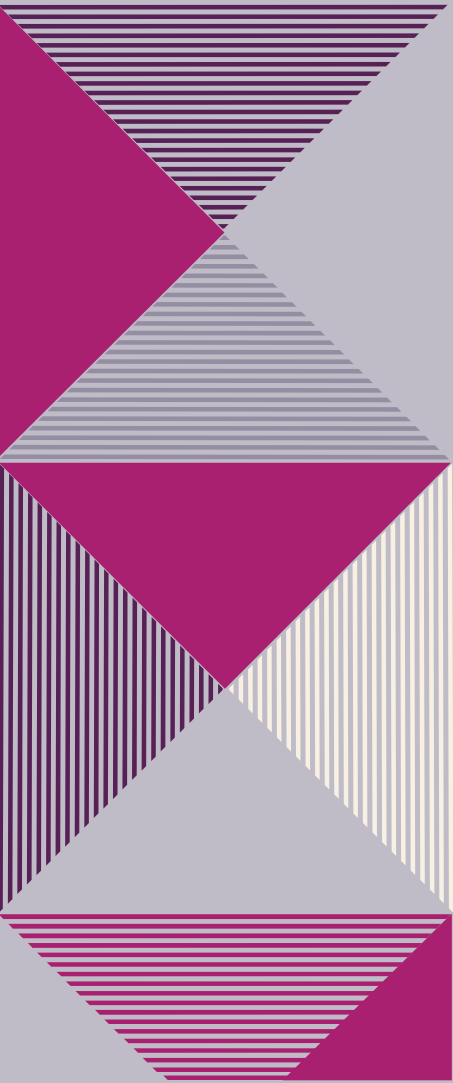
140



**NOW FOR
PLACE.**



WHERE SHOULD THE NEW PLACE BE?



SCHOOL!

Scratch 3.29.1

Scratch Project

Code Costumes Sounds

Motion: next costume

Looks: switch backdrop to Forest

Sound: next backdrop

Events: change size by 10

Control: set size to 100 %

Sensing: change color effect by 25

Operators: set color effect to 0

Variables: clear graphic effects

My Blocks: show, hide, go to front layer, go forward 1 layers, costume number, backdrop number

when clicked wait 8 seconds hide

Press the place

Sprite: Sprite2 x 179 y 5

Show: Size 100 Direction 90

Backdrops: 2

Chick Sprite1 Sprite2

Scratch 3.29.1

Scratch

FileEditTutorialsScratch Project

CodeBackdropsSounds

Motion

Stage selected: no motion blocks

Looks

switch backdrop to Forest

switch backdrop to Forest and wait

next backdrop

change color effect by 25

set color effect to 0

clear graphic effects

backdrop number

Sound

play sound pop until done

start sound pop

stop all sounds

change pitch effect by 10

set pitch effect to 100

clear sound effects

Press Backdrop

Sprite

Name

x

y

Show

Size

Direction

Chick

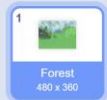
Sprite1

Sprite2

Stage

Backdrops

1

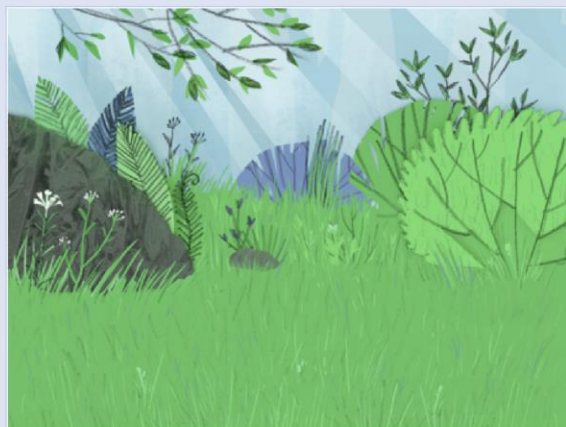


Costume Forest

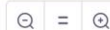
Fill



Press the
button :
add new
place



Convert to Vector



Sprite Name

Show

Size

Direction



Chick



Sprite1



Sprite2

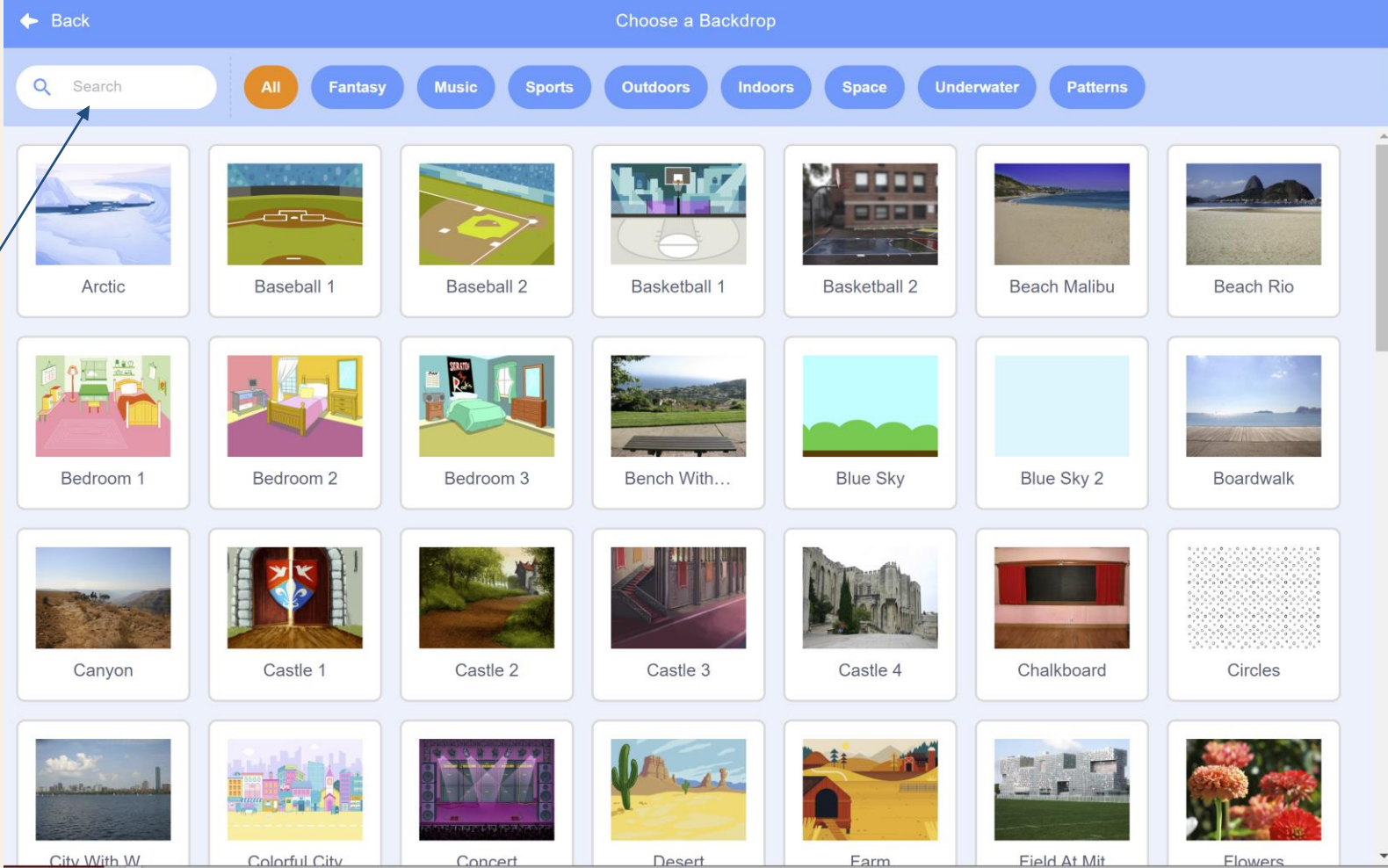
Stage



Backdrops

1





Let's
search
for
school

← Back

Choose a Backdrop



school



All

Fantasy

Music

Sports

Outdoors

Indoors

Space

Underwater

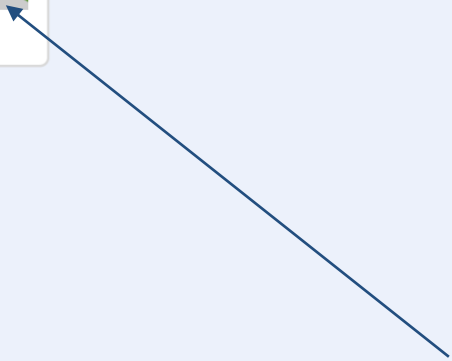
Patterns



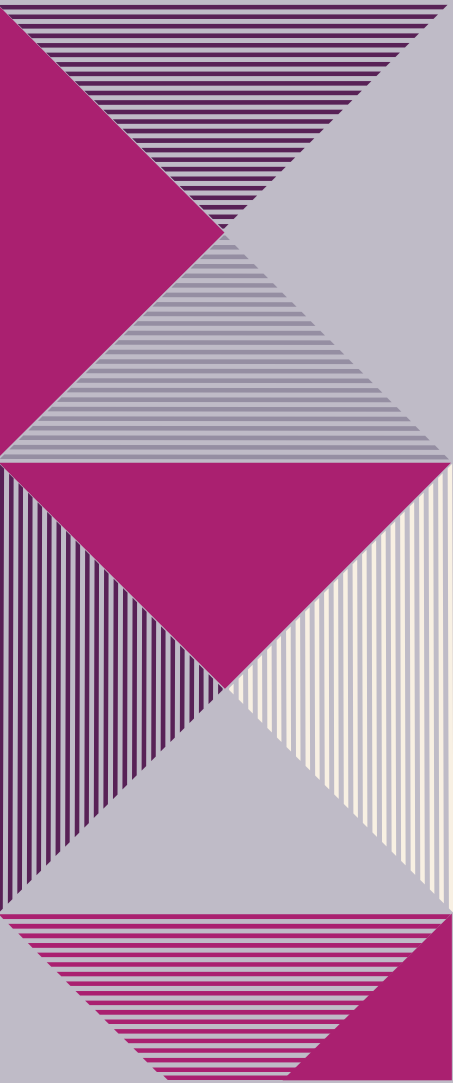
Chalkboard



School







**LET'S MAKE
COMPUTER
UNDERSTAND
WHEN TO GO TO
SCHOOL**

Scratch 3.29.1

Scratch Project

Code Backdrops Sounds

Costume School

Fill

Let's go back to code

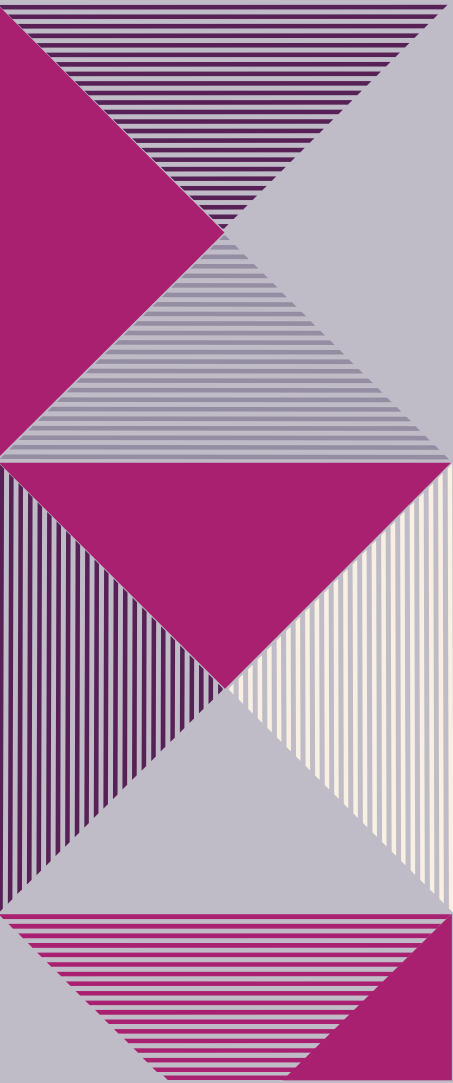
Convert to Vector

Sprite Name x x y y Show Size Direction

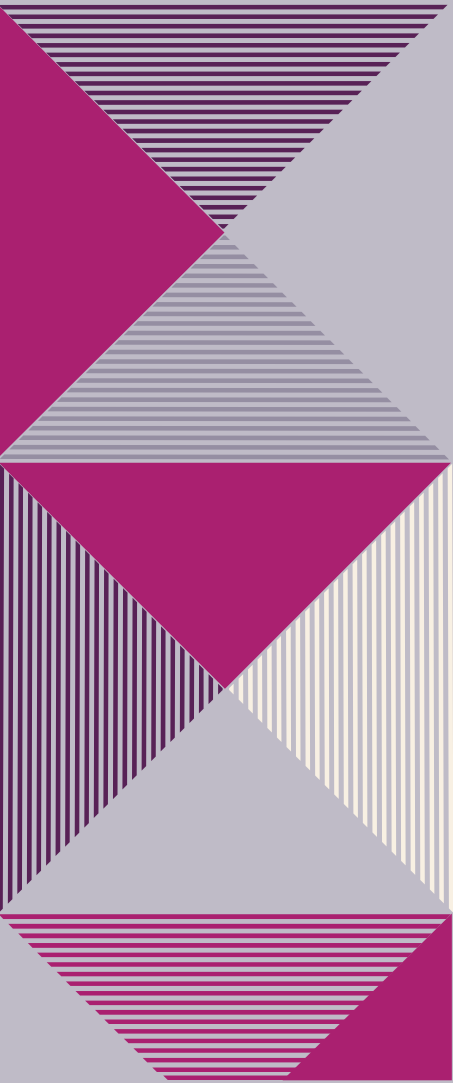
Chick Sprite1 Sprite2

Stage Backdrops 2

The image shows the Scratch 3.29.1 web interface. The top bar includes the Scratch logo, a globe icon, and menu options: File, Edit, Tutorials, and a 'Scratch Project' button. Below this is a tabbed interface with 'Code', 'Backdrops', and 'Sounds'. The 'Backdrops' tab is active, displaying a list of two backdrops: 'Forest' (480 x 360) and 'School' (480 x 360). The 'School' backdrop is selected and shown in a large preview window. To the left of the preview is a 'Costume' section with a 'School' dropdown and a 'Fill' color selector. Below the preview is a 'Convert to Vector' button. On the right side, the 'Stage' area shows the 'School' backdrop with a yellow chick sprite on it. Below the stage are controls for the selected sprite, including 'Name', 'Show' (eye icon), 'Size', 'Direction', and buttons for 'Chick', 'Sprite1', and 'Sprite2'. A sidebar on the far right shows 'Stage' and 'Backdrops' (2). The text 'Let's go back to code' is overlaid on the left side of the interface.



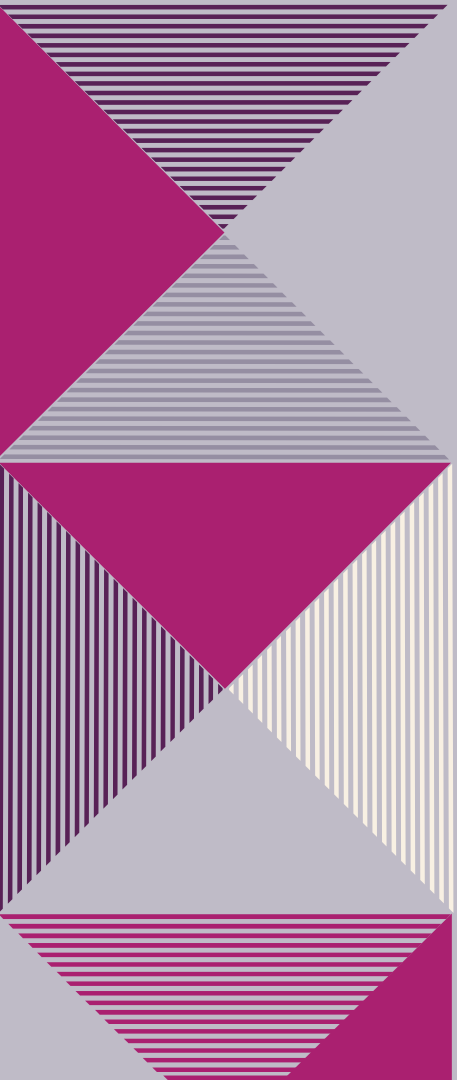
**THINK. WHAT
DO WE WANT
TO DO?**



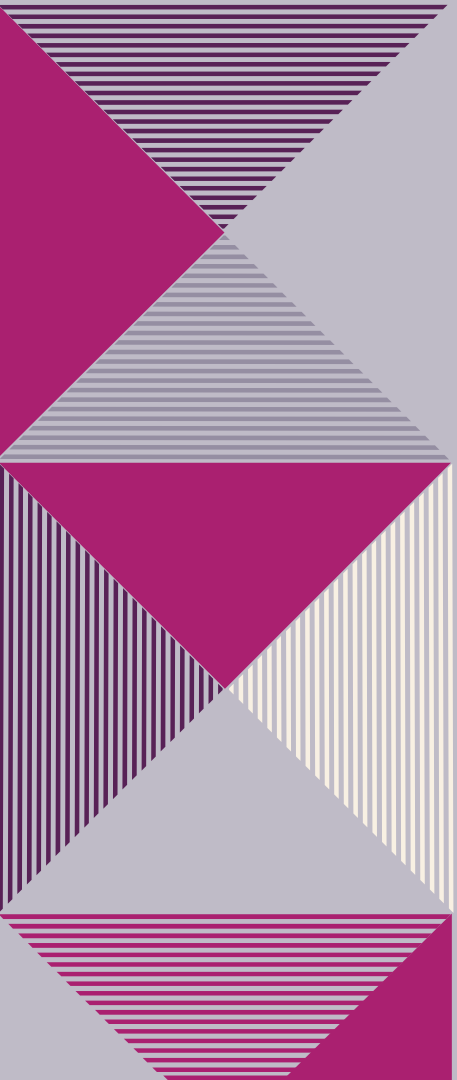
START IN FOREST.

WAIT.

**CHANGE TO
SCHOOL.**

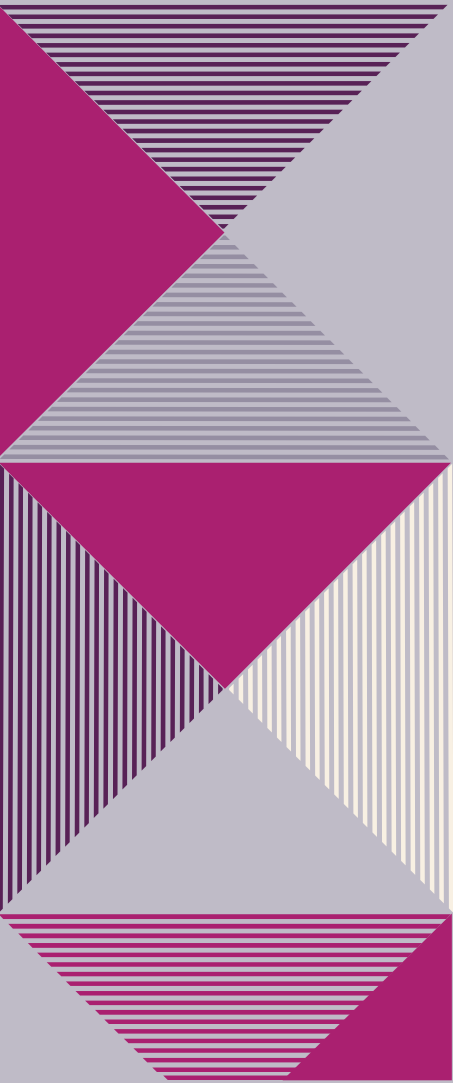


**HOW MANY
SECONDS DO
WE NEED TO
WAIT?**



**REMEMBER FOR
TREE AND RAIN.**

$$2 + 3 + 3 = 8$$



1. START IN FOREST.

2. WAIT 8 SECONDS

**3. CHANGE TO
SCHOOL.**

Scratch 3.29.1

Scratch

File

Edit

Tutorials

Scratch Project

Code

Backdrops

Sounds

Motion

Looks

Sound

Events

Control

Sensing

Operators

Variables

My Blocks

Looks

switch backdrop to School

switch backdrop to School and wait

next backdrop

change color effect by 25

set color effect to 0

clear graphic effects

backdrop number

Sound

play sound pop until done

start sound pop

stop all sounds

change pitch effect by 10

set pitch effect to 100

clear sound effects

change volume by -10

set volume to 100 %

Bring to code space

Stage

Sprite Name x y x y

Show Show Size Direction

Chick Sprite1 Sprite2

Backdrops

2

Scratch 3.29.1

Scratch

FileEditTutorialsScratch Project

CodeBackdropsSounds

Motion

Looks

Sound

Events

Control

Sensing

Operators

Variables

My Blocks

Looks

switch backdrop to School

switch backdrop to School and wait

next backdrop

change color effect by 25

set color effect to 0

clear graphic effects

backdrop number

Sound

play sound pop until done

start sound pop

stop all sounds

change pitch effect by 10

set pitch effect to 100

clear sound effects

change volume by -10

set volume to 100 %

switch backdrop to School

Forest

✓ School

next backdrop

previous backdrop

random backdrop

1. Start in Forest

Stage



Sprite

Name

↔ x x

↕ y y

Show

Size

Direction

Chick

Sprite1

Sprite2

Backdrops

2

Scratch 3.29.1

Scratch

File

Edit

Tutorials

Scratch Project

Code

Backdrops

Sounds

Motion

Looks

Sound

Events

Control

Sensing

Operators

Variables

My Blocks

Control

wait 1 seconds

repeat 10

forever

if then

if then else

wait until

repeat until

stop all

create clone of Chick

Sensing

ask What's your name? and wait

switch backdrop to Forest

Bring to code space

Stage

Sprite

Name

x

y

Show

Size

Direction

Chick

Sprite1

Sprite2

Backdrops

2

Scratch 3.29.1

Scratch Project

Code Backdrops Sounds

Control

wait 1 seconds

repeat 10

forever

if then

if then else

wait until

repeat until

stop all

create clone of Chick

Sensing

ask What's your name? and wait

switch backdrop to Forest

wait 8 seconds

2. Wait 8 seconds

Sprite Name x y Show Size Direction

Chick Sprite1 Sprite2

Stage

Backdrops 2

The image shows the Scratch 3.29.1 interface. The top bar includes the Scratch logo, version number, and tabs for File, Edit, Tutorials, and a project titled 'Scratch Project'. Below the top bar are three tabs: Code, Backdrops, and Sounds. The Code tab is active, showing a block palette on the left with categories: Motion, Looks, Sound, Events, Control, Sensing, Operators, Variables, and My Blocks. The Control category is selected, displaying various control blocks. A 'wait 8 seconds' block is highlighted with a blue arrow and the text '2. Wait 8 seconds'. The stage area on the right shows a forest backdrop with a yellow chick sprite. The stage panel includes a 'Sprite' section with a 'Name' field, 'x' and 'y' coordinates, 'Show' checkboxes, 'Size' and 'Direction' fields, and a 'Backdrops' section with a '2' indicator. The bottom of the interface features a 'Sensing' section with an 'ask' block.

Scratch 3.29.1

Scratch

FileEditTutorialsScratch Project

CodeBackdropsSounds

Motion

Looks

Sound

Events

Control

Sensing

Operators

Variables

My Blocks

Looks

switch backdrop to School

switch backdrop to School and wait

next backdrop

change color effect by 25

set color effect to 0

clear graphic effects

backdrop number

Sound

play sound pop until done

start sound pop

stop all sounds

change pitch effect by 10

set pitch effect to 100

clear sound effects

change volume by -10

set volume to 100 %

switch backdrop to Forest

wait 8 seconds

3. Change to school

Stage



Sprite

Name

↔ x x

↕ y y

Show

Size

Direction

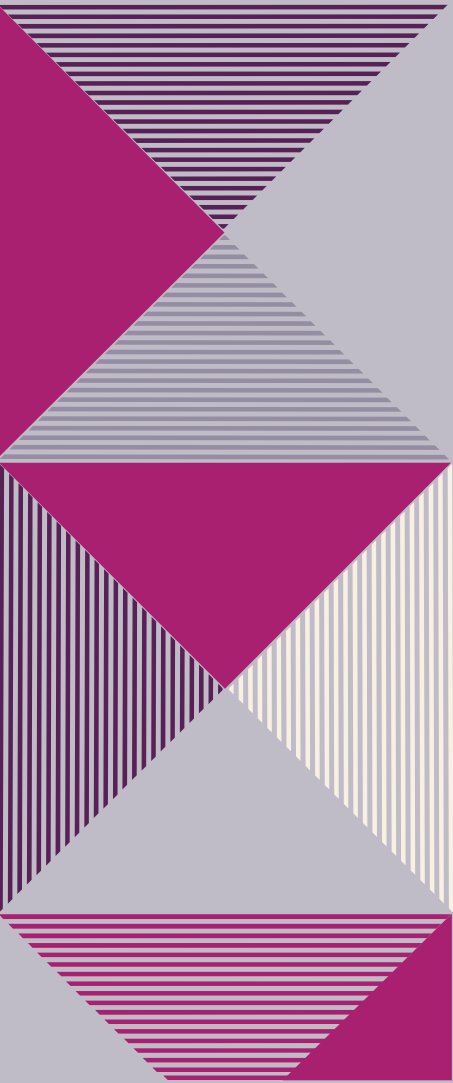
Chick

Sprite1

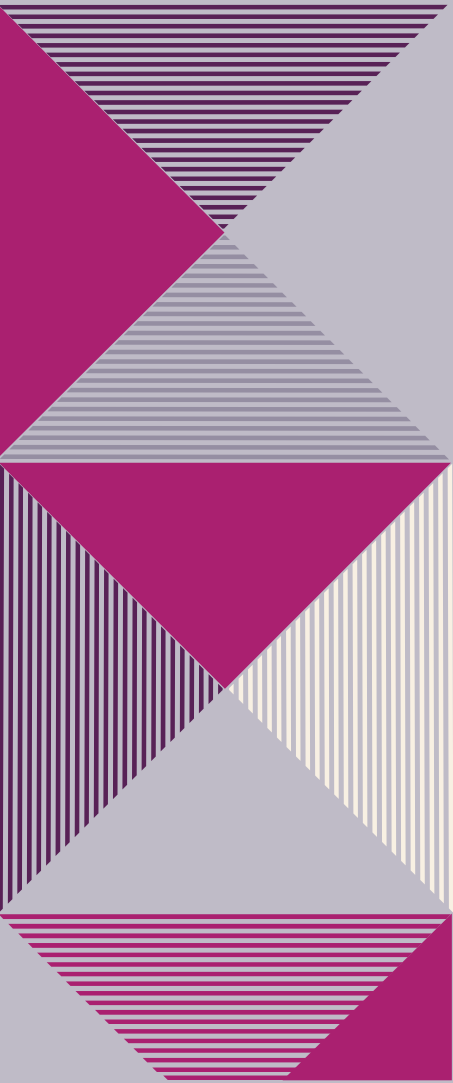
Sprite2

Backdrops

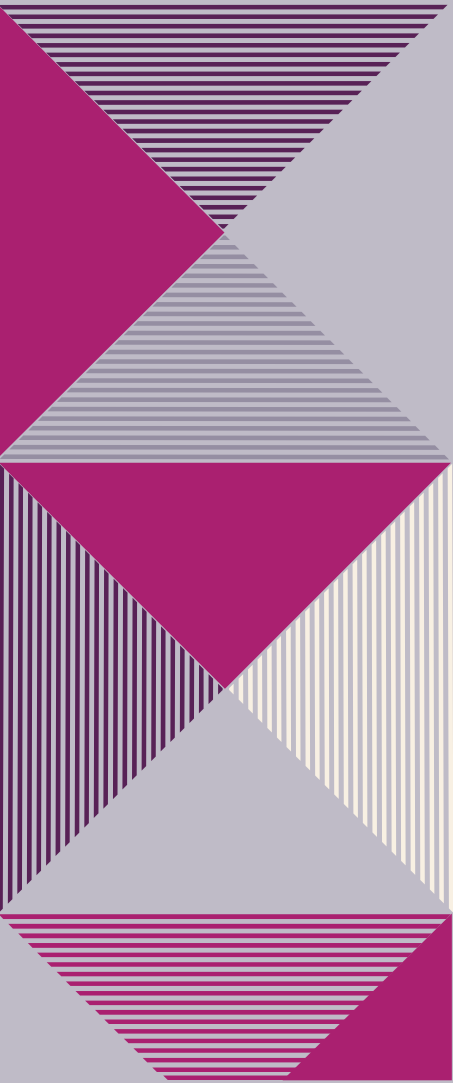
2



WHAT ARE WE FORGETTING?



HOW DO WE MAKE COMPUTER READ OUR CODE?



GREEN FLAG!

Scratch 3.29.1

Scratch

FileEditTutorialsScratch Project

CodeBackdropsSounds

Motion

Looks

Sound

Events

Control

Sensing

Operators

Variables

My Blocks

Looks

switch backdrop to School

switch backdrop to School and wait

next backdrop

change color effect by 25

set color effect to 0

clear graphic effects

backdrop number

Sound

play sound pop until done

start sound pop

stop all sounds

change pitch effect by 10

set pitch effect to 100

clear sound effects

change volume by -10

set volume to 100 %

switch backdrop to Forest

wait 8 seconds

3. Change to school

Stage

Sprite Name

↔ x x

↕ y y

Show

Size

Direction

Chick

Sprite1

Sprite2

Backdrops

2



Sprite

Name

↔ x x

↕ y y

Show

Size

Direction

Chick

Sprite1

Sprite2

Backdrops

2

Scratch 3.29.1

Scratch

FileEditTutorialsScratch Project

CodeBackdropsSounds

Motion

Looks

Sound

Events

Control

Sensing

Operators

Variables

My Blocks

Events

when green flag clicked

when space key pressed

when stage clicked

when backdrop switches to Forest

when loudness > 10

when I receive message1

broadcast message1

broadcast message1 and wait

Control

wait 1 seconds

repeat 10

forever

when green flag clicked

switch backdrop to Forest

wait 8 seconds

switch backdrop to School

Stage

Backdrops2

Sprite

Name

↔ x

x

↕ y

y

Show

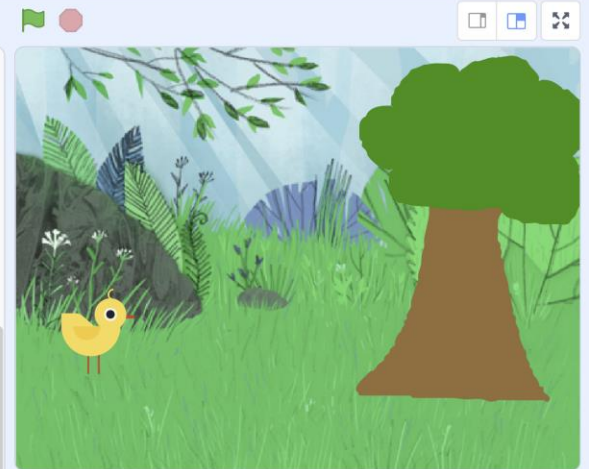
Size

Direction

Chick

Sprite1

Sprite2



Sprite

Name

↔ x

x

↕ y

y

Show

Size

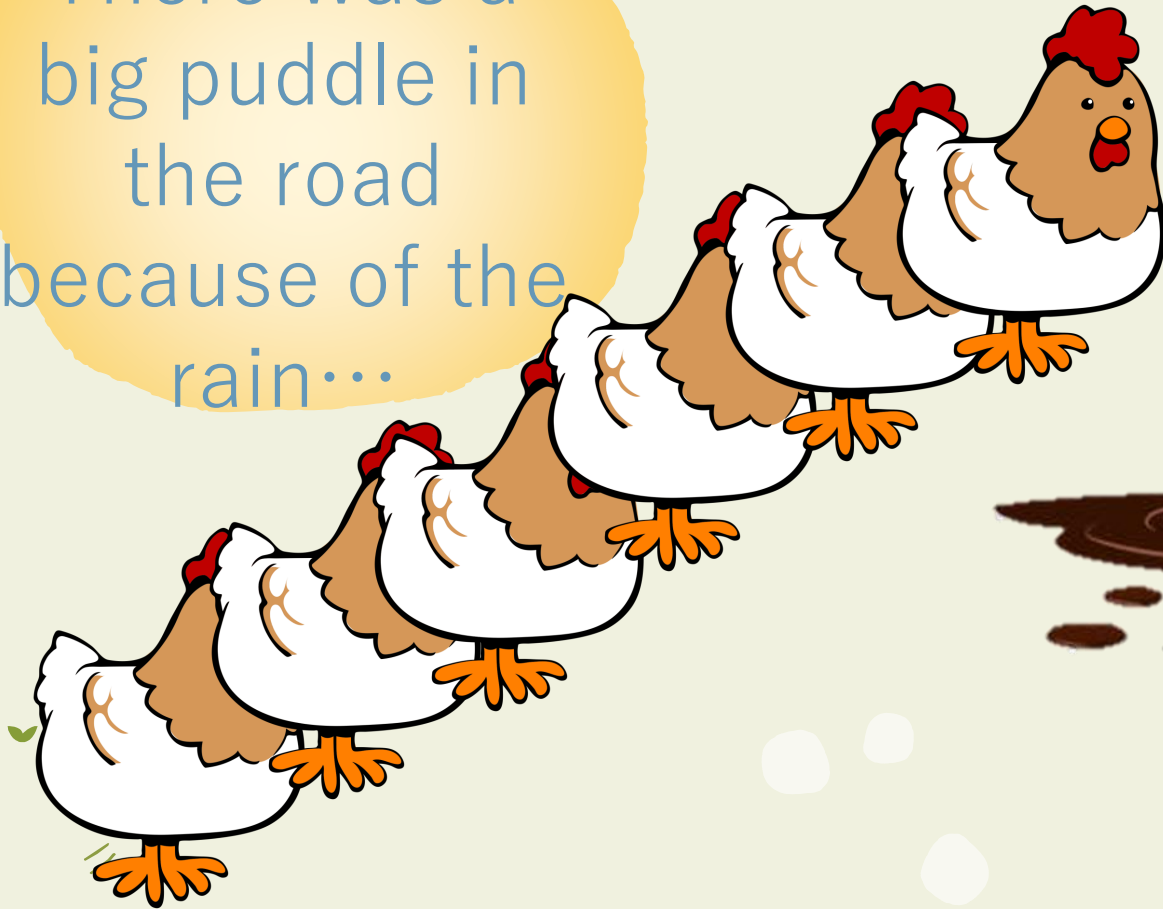
Direction

Chick

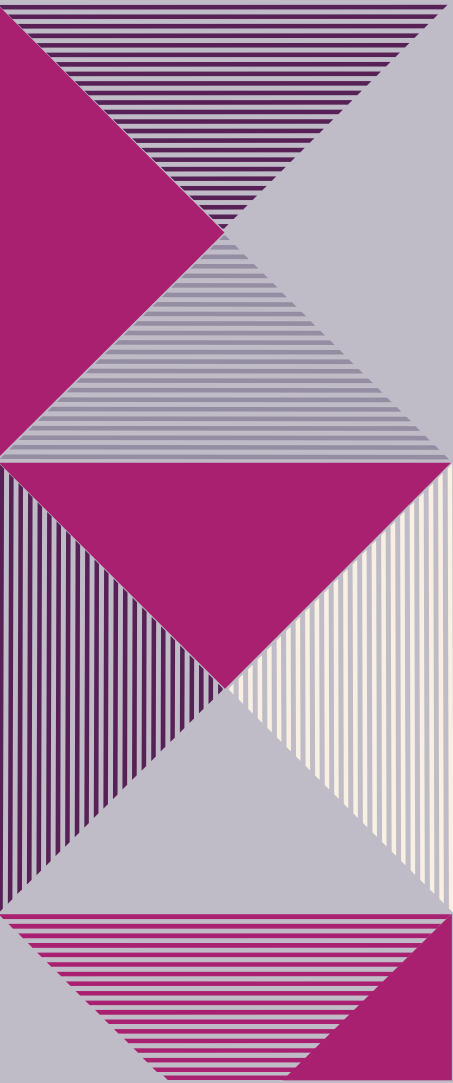
Sprite1

Sprite2

There was a
big puddle in
the road
because of the
rain...



Billy
walked
around
the
puddle.



**LET'S DRAW
PUDDLE.**

Scratch 3.29.1

Scratch

File

Edit

Tutorials

Scratch Project

Code

Costumes

Sounds

1

costume1

0 x 0

Costume

costume1

Group

Ungroup

Forward

Backward

Front

Back

Fill

Outline

4

Color

36

Saturation

100

Brightness

56

Convert to Bitmap

Stage

Chick

Sprite1

Sprite2

Sprite3

Sprite

Sprite3

x

36

y

28

Show

Size

100

Direction

90

Backdrops

2

Scratch 3.29.1

Scratch

File

Edit

Tutorials

Scratch Project

Code

Costumes

Sounds

1

costume1

171 x 125

costume1

Fill

Outline

4

10

Group

Ungroup

Forward

Backward

Front

Back

Paint bucket

Eraser

Text

Line

Rectangle

171 x 125

Convert to Bitmap

Q

=

Q

Stage

Backdrops

2

Sprite3

Chick

Sprite1

Sprite2

Sprite3

Sprite3

Size

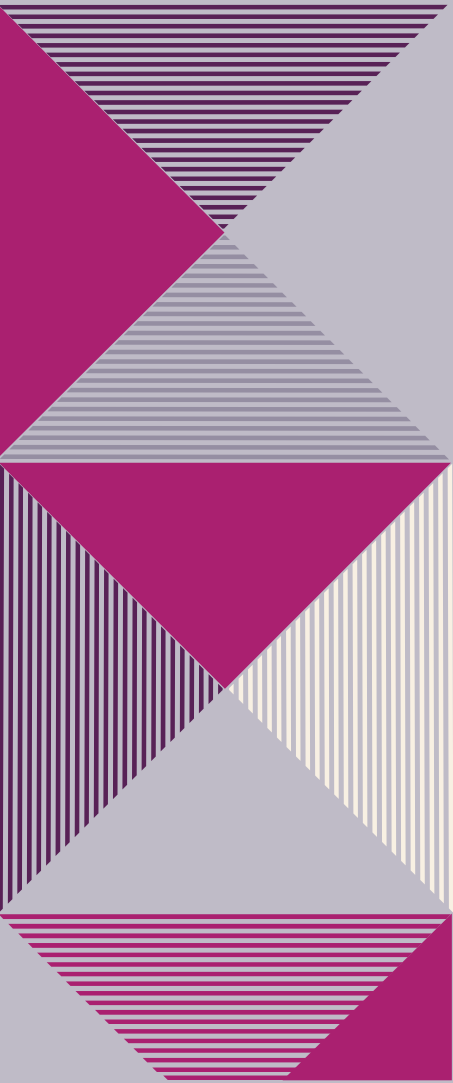
Direction

Chick

Sprite1

Sprite2

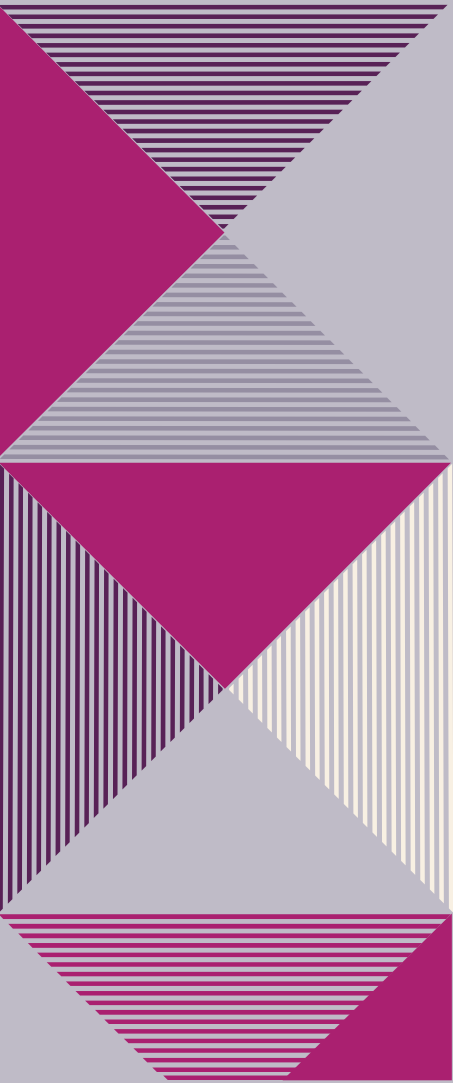
Sprite3



1. HIDE

2. WAIT 8 SECONDS

3. SHOW



YOU TRY

Scratch 3.29.1

Scratch

FileEditTutorialsScratch Project

CodeCostumesSounds

Motion

Looks

Sound

Events

Control

Sensing

Operators

Variables

My Blocks

Looks

say Hello! for 2 seconds

say Hello!

think Hmm... for 2 seconds

think Hmm...

switch costume to costume1

next costume

switch backdrop to School

next backdrop

change size by 10

set size to 100 %

change color effect by 25

set color effect to 0

clear graphic effects

show

hide

when clicked

hide

wait 8 seconds

show

Stage

Sprite3

x18y-70

Show

Size100Direction90

Chick

Sprite1

Sprite2

Sprite3

Backdrops

2



Sprite Sprite3

x18y-70

Show

Size100Direction90

Chick

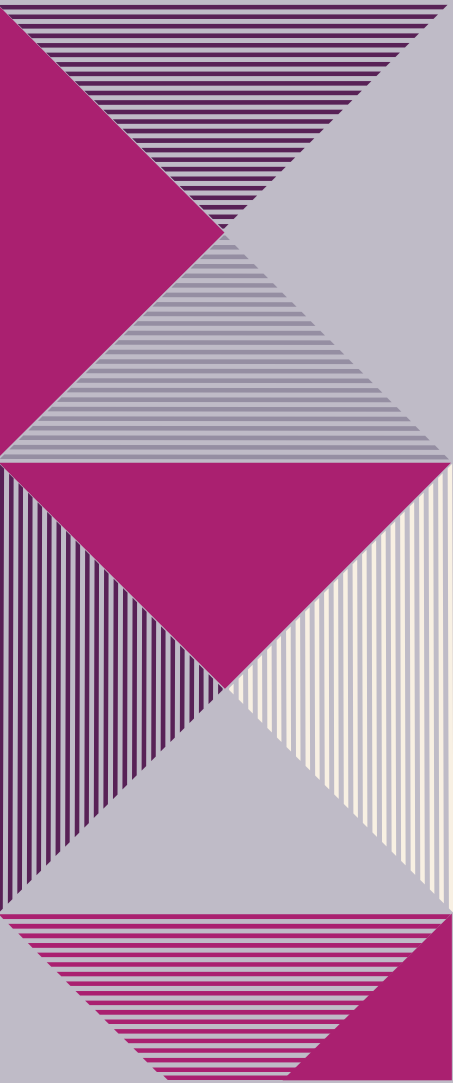
Sprite1

Sprite2

Sprite3

Backdrops

2



MAKE BILLY WALK AROUND

Scratch 3.29.1

ScratchFileEditTutorialsScratch Project

CodeCostumesSounds

Motion

Looks

Sound

Events

Control

Sensing

Operators

Variables

My Blocks

Looks

say Hello! for 2 seconds

say Hello!

think Hmm... for 2 seconds

think Hmm...

switch costume to chick-c

next costume

switch backdrop to School

next backdrop

change size by 10

set size to 100 %

change color effect by 25

set color effect to 0

clear graphic effects

show

hide

when clicked

think I go to school to learn science for 2 seconds

glide 3 secs to x: 165 y: -80

glide 3 secs to x: -170 y: -62

Chick

Sprite1

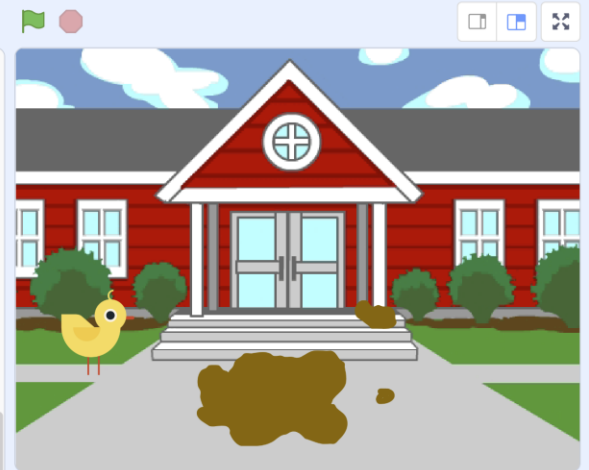
Sprite2

Sprite3

Stage

Backdrops

2



Sprite: Chick

Size: 100

Direction: 90

Chick

Sprite1

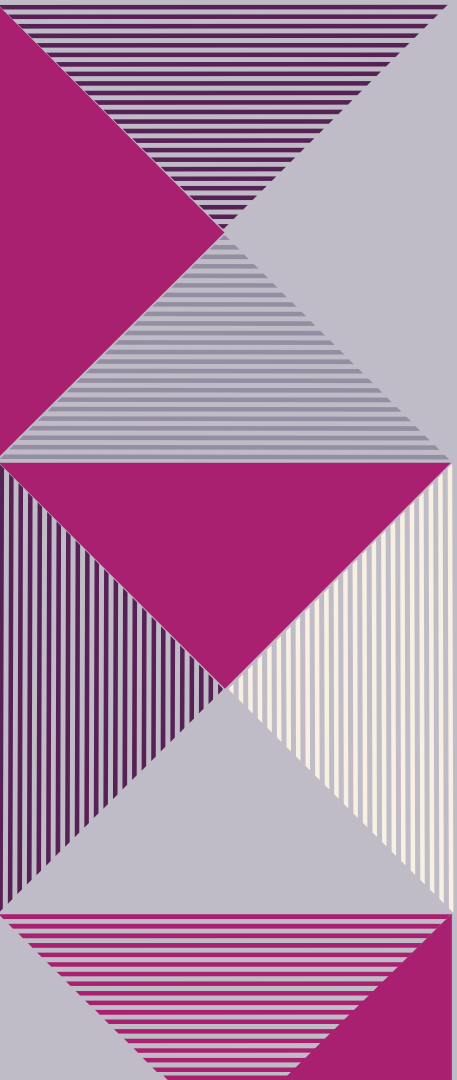
Sprite2

Sprite3

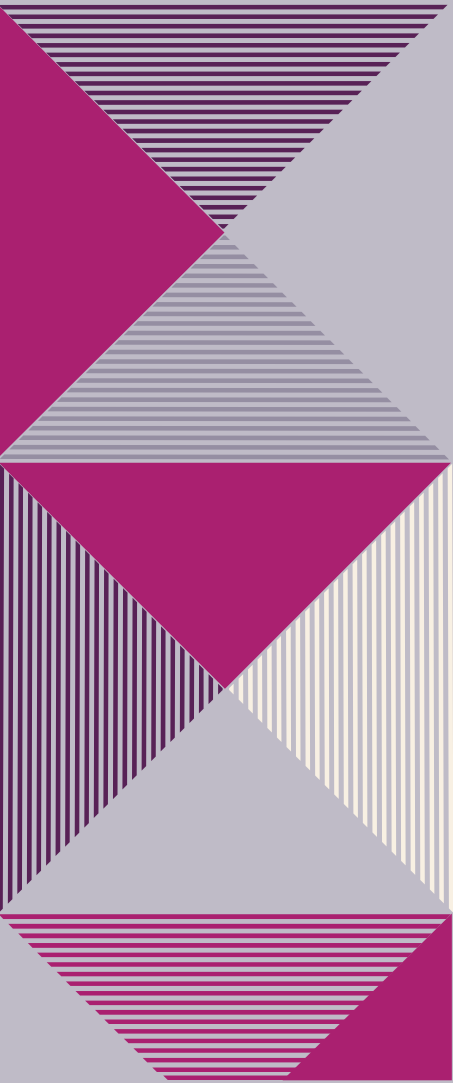
Stage

Backdrops

2



1. GLIDE AROUND PUDDLE



YOU TRY.

**(REMEMBER MOVE BILLY
AND COPY X AND Y FOR
POSITION)**

Code

Costumes

Sounds

Motion

Looks

Sound

Events

Control

Sensing

Operators

Variables

My Blocks

Motion

move 10 steps

turn 15 degrees

turn 15 degrees

go to random position

go to x: -11 y: -108

glide 1 secs to random position

glide 1 secs to x: -11 y: -108

point in direction 90

point towards mouse-pointer

change x by 10

set x to -11

change y by 10

set y to -108

when clicked

think I go to school to learn science for 2 seconds

glide 3 secs to x: 165 y: -80

glide 3 secs to x: -170 y: -62

glide 3 secs to x: -145 y: -47

glide 3 secs to x: -11 y: -108



Sprite Chick

x -11

y -108

Show

Size

100

Direction

90

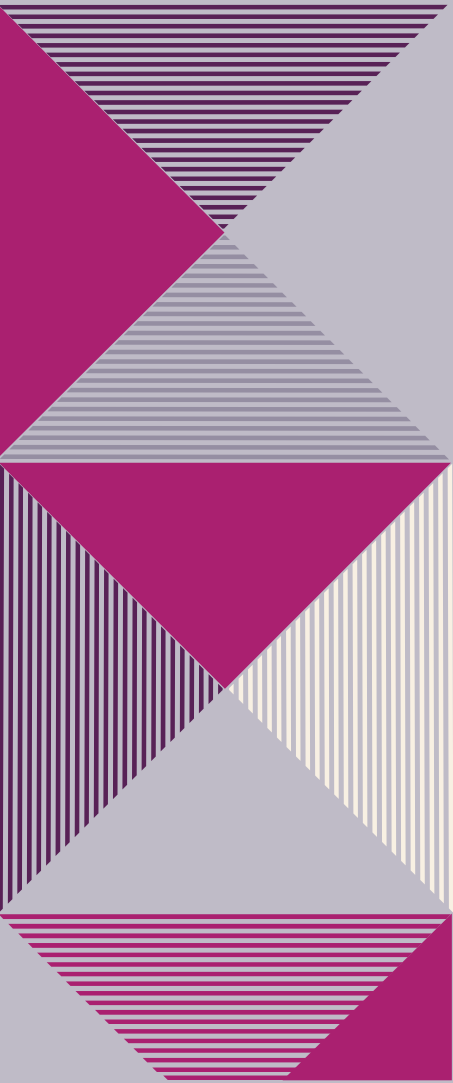


Stage

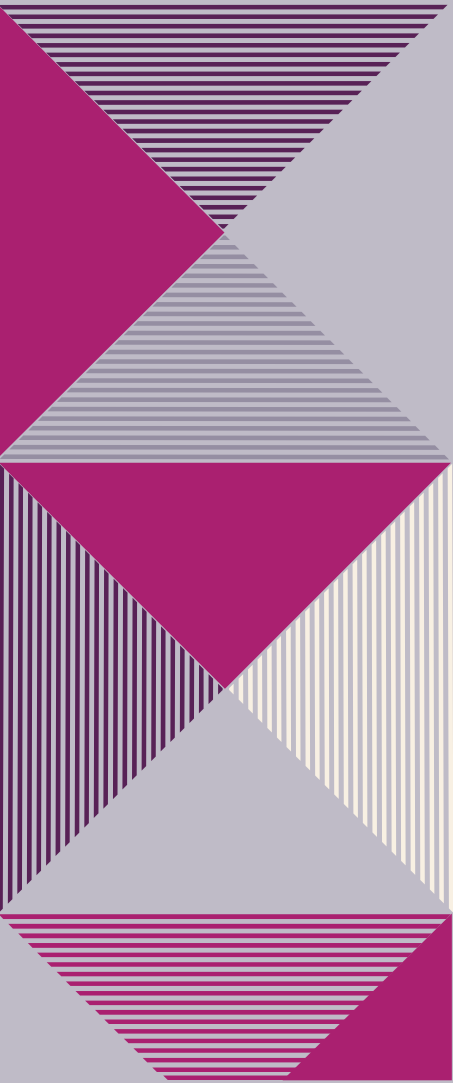


Backdrops

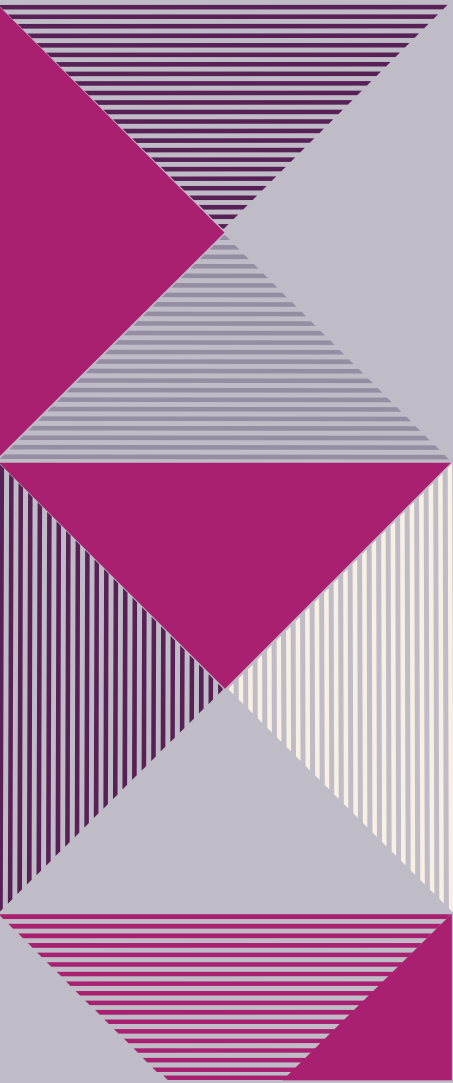
2



**PRESS GREEN FLAG.
DOES EVERYTHING
LOOK OK?**



**MAKE SURE BILLY
STARTS AT A SINGLE
POSITION**



HOW TO EDIT CODE.

Scratch 3.29.1

File Edit Tutorials Scratch Project

Code Costumes Sounds

Motion

- move 10 steps
- turn 15 degrees
- turn 15 degrees
- go to random position
- go to x: -11 y: -108
- glide 1 secs to random position
- glide 1 secs to x: -11 y: -108
- point in direction 90
- point towards mouse-pointer
- change x by 10
- set x to -11
- change y by 10
- set y to -108

when clicked

- think I go to school to learn science for 2 seconds
- glide 3 secs to x: 165 y: -80
- glide 3 secs to x: -170 y: -62
- glide 3 secs to x: -145 y: -47
- glide 3 secs to x: -11 y: -108

Let's start Billy at the position he is at now

Sprite: Chick

x: -11 y: -108

Show: [] Size: 180 Direction: 90

Backdrops: 2

Chick Sprite1 Sprite2 Sprite3

Code

Costumes

Sounds



Motion



Looks



Sound



Events



Control



Sensing



Operators



Variables



My Blocks

Motion

move 10 steps

turn 15 degrees

turn 15 degrees

go to random position

go to x: -11 y: -108

glide 1 secs to random position

glide 1 secs to x: -11 y: -108

point in direction 90

point towards mouse-pointer

change x by 10

set x to -11

change y by 10

set y to -108

when clicked

think I go to school to learn science for 2 seconds

glide 3 secs to x: 165 y: -80

glide 3 secs to x: -170 y: -62

glide 3 secs to x: -145 y: -47

glide 3 secs to x: -11 y: -108

go to x: -179 y: -84



Sprite Chick

x -11

y -108

Show



Size

100

Direction

90



Chick



Sprite1



Sprite2



Sprite3

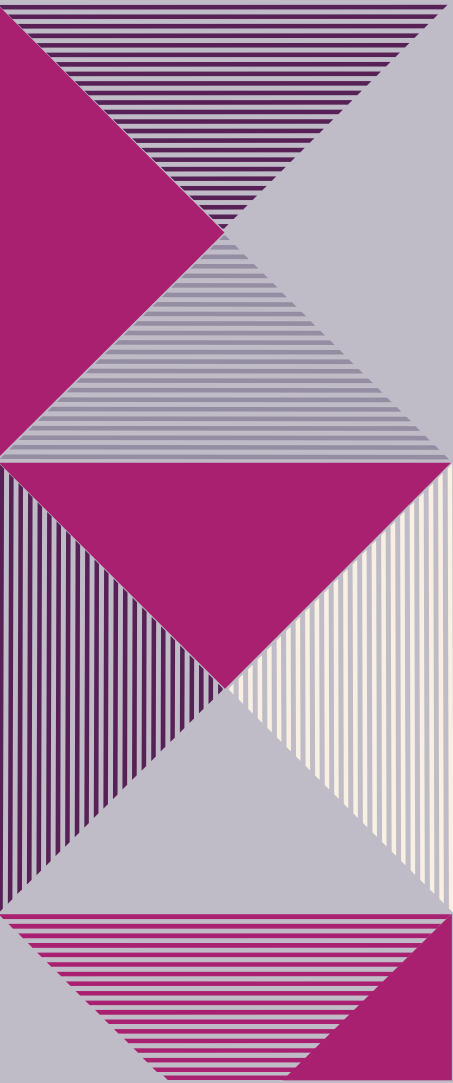
Stage



Backdrops

2





**WE WANT TO MAKE
IT AT THE
BEGINNING**

Code

Costumes

Sounds

Motion

Looks

Sound

Events

Control

Sensing

Operators

Variables

My Blocks

Motion

move 10 steps

turn 15 degrees

turn 15 degrees

go to random position

go to x: -11 y: -108

glide 1 secs to random position

glide 1 secs to x: -11 y: -108

point in direction 90

point towards mouse-pointer

change x by 10

set x to -11

change y by 10

set y to -108

when clicked

go to x: -179 y: -84

think I go to school to learn science for 2 seconds

glide 3 secs to x: 165 y: -80

glide 3 secs to x: -170 y: -62

glide 3 secs to x: -145 y: -47

glide 3 secs to x: -11 y: -108



Sprite Chick

x -11

y -108

Show

Size

100

Direction

90



Chick



Sprite1



Sprite2



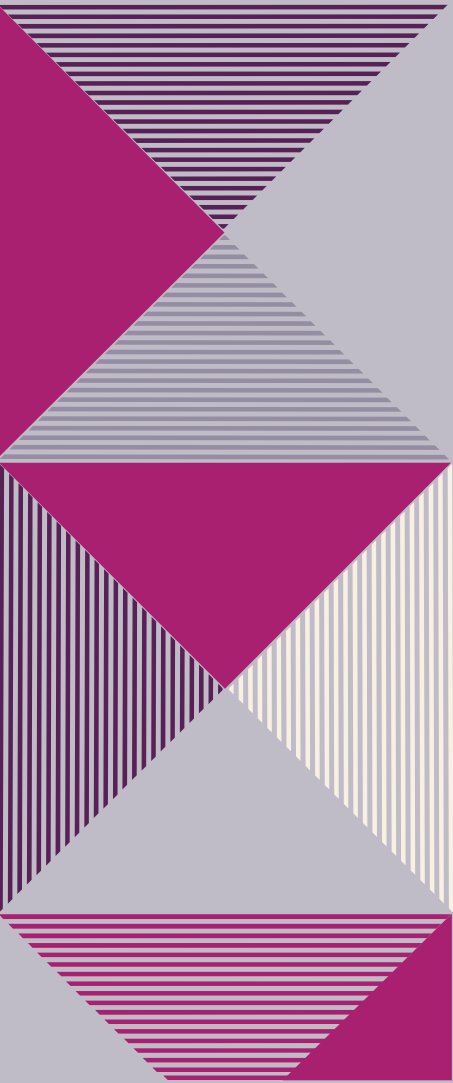
Sprite3

Stage



Backdrops

2



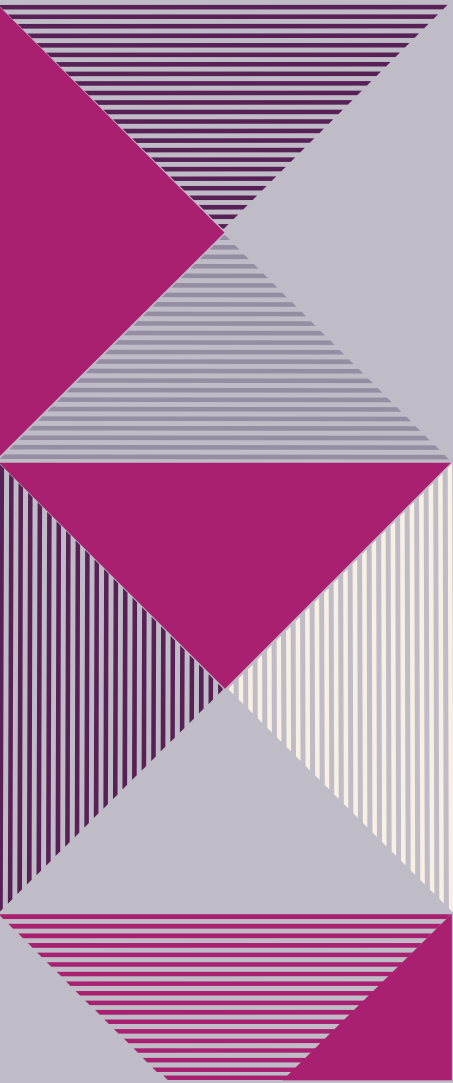
**PRESS GREEN FLAG.
DOES EVERYTHING
LOOK OK?**

An abstract geometric design on the left side of the slide. It features a diagonal line running from the top-left corner towards the bottom-right. To the left of this line, there are several overlapping geometric shapes: a dark purple triangle at the top-left, a light gray semi-circle, a blue square with concentric circles, a magenta square with diagonal lines, a magenta square with a grid of lines, a magenta square, a light gray triangle, and a magenta triangle. A small white circle is located at the top-left corner of the diagonal line.

**EDIT CODE AS
YOU NEED!**

Billy then
walked
straight to
school.





**MAKE BILLY WALK
STRAIGHT TO
SCHOOL!**

Scratch 3.29.1

Scratch Project

Code Costumes Sounds

Motion

- move 10 steps
- turn 15 degrees
- turn 15 degrees
- go to random position
- go to x: -11 y: -108
- glide 1 secs to random position
- glide 1 secs to x: -11 y: -108
- point in direction 90
- point towards mouse-pointer
- change x by 10
- set x to -11
- change y by 10
- set y to -108

when clicked

go to x: -179 y: -84

think I go to school to learn science for 2 seconds

glide 3 secs to x: 165 y: -80

glide 3 secs to x: -170 y: -62

glide 3 secs to x: -145 y: -47

glide 3 secs to x: -11 y: -108

You try!

Sprite Chick x: -11 y: -108

Show Size 100 Direction 90

Backdrops 2

Code

Costumes

Sounds



Motion



Looks



Sound



Events



Control



Sensing



Operators



Variables



My Blocks

Motion

move 10 steps

turn 15 degrees

turn 15 degrees

go to random position

go to x: 0 y: -8

glide 1 secs to random position

glide 1 secs to x: 0 y: -8

point in direction 90

point towards mouse-pointer

change x by 10

set x to 0

change y by 10

set y to -8

when clicked

go to x: -179 y: -84

think I go to school to learn science for 2 seconds

glide 3 secs to x: 165 y: -80

glide 3 secs to x: -170 y: -62

glide 3 secs to x: -145 y: -47

glide 3 secs to x: -11 y: -108

glide 1 secs to x: 0 y: -8



Sprite Chick



x

y

0

-8

Show



Size

100

Direction

90

Stage



Backdrops

2



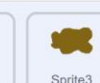
Chick



Sprite1

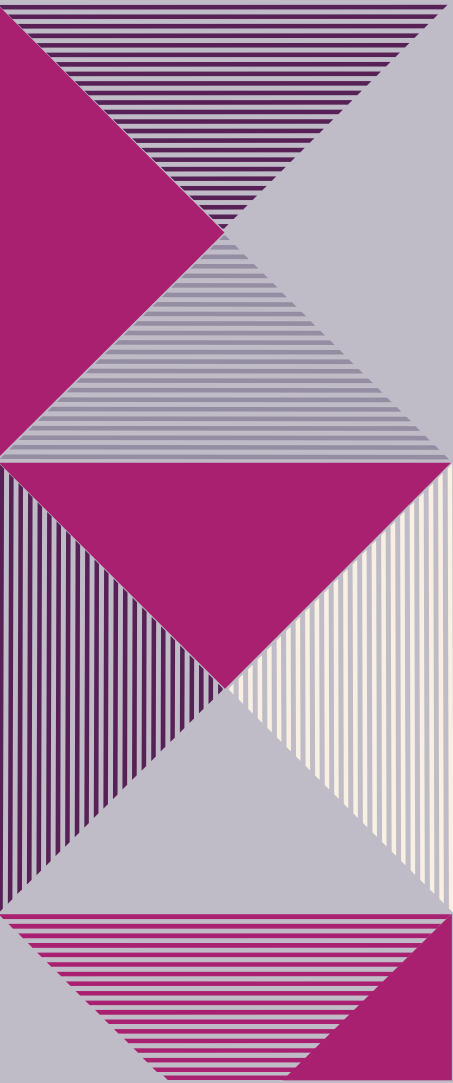


Sprite2

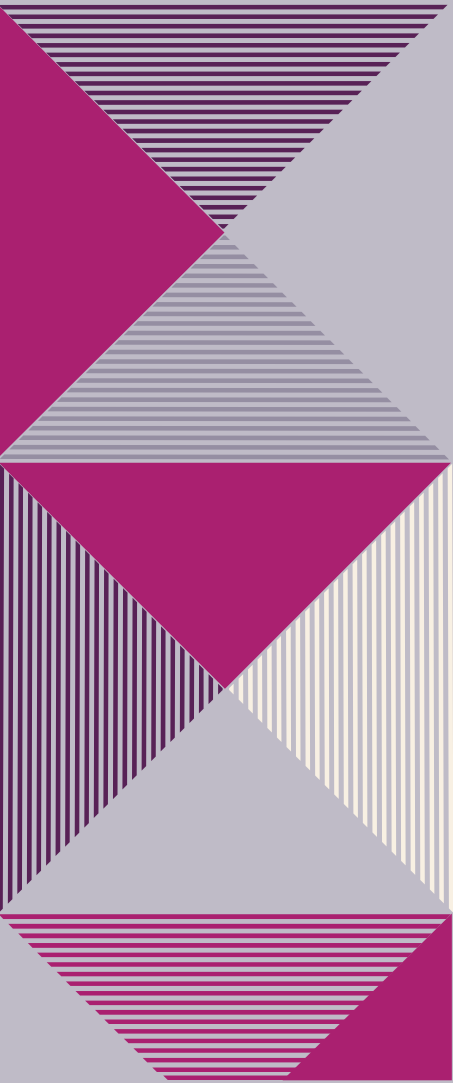


Sprite3





**PRESS GREEN FLAG.
DOES EVERYTHING
LOOK OK?**



THE END!